

Direct Operated **Pilot Operated**

2-Port Solenoid Valve



Refer to page 84 for details.



Refer to pages 85 to 89 for details.



**Improved environmental resistance
due to the stainless steel coil cover
[IP67 enclosure/NEMA4X*1]**

IP67

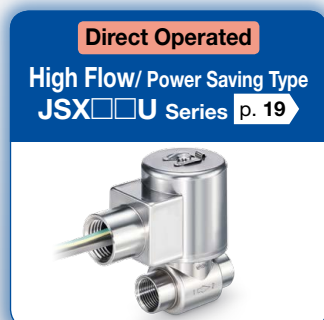
NEMA4X*1

*1 IP65 for models with a DIN terminal

**CO₂ emissions
(Power consumption) 71% reduction**

Series	CO ₂ emissions [kg-CO ₂ e/year]
Power Saving Type JSX31U Series	2.9 71% reduction
Existing model VX23 Series	10

Direct Operated
JSX Series p. 13

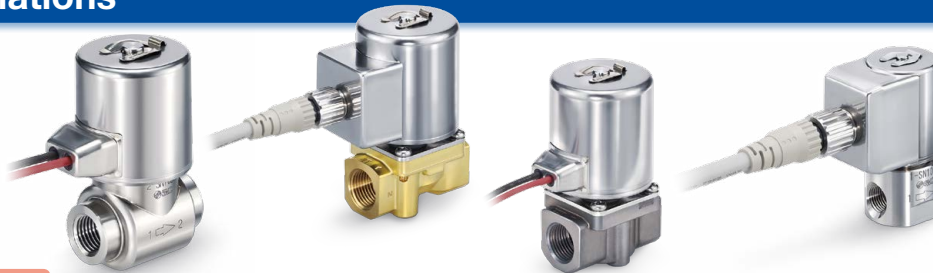


JSX/JSX□ Series



CAT.ES70-56F

Series Variations



Direct Operated **JSX Series** N.C. specification pp. 13, 15 N.O. specification p. 17

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]				Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			5	10	20	30						
JSX10 Series*2	1/8	1.6 2.4	5	(For orifice diameter ø2.4)			Air Water Oil	Stainless steel Brass Aluminum*2	N.C. N.O.	NBR FKM EPDM	Grommet DIN terminal Conduit M12 connector	CE UK CA UL US LISTED C UL US ®
JSX20 Series	1/8	3.2										
	1/4, 3/8	3.2, 4.0, 5.6, 7.1	15	(For orifice diameter ø5.6)								
JSX30 Series	1/4, 3/8	4.0, 5.6, 7.1	25	(For orifice diameters ø4.0 and ø5.6)								

* Refer to page 85 for details.

*1 At the max. operating pressure differential (Fluid: Water)

*2 Excludes N.O.

* Refer to page 85 for details.



Direct Operated **High Flow/ Power Saving Type JSX□□U Series** pp. 19, 21

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]				Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			5	10	20	30						
JSX10U Series	1/8	2.4										

*1 At the max. operating pressure differential (Fluid: Water)

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min] (ANR)				Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			500	1000	1500	2000						
JSX20U Series	1/4, 3/8	5.0		1000			Air	Aluminum	N.C.	NBR FKM EPDM	Grommet DIN terminal Conduit M12 connector	CE UK CA
JSX30U Series	1/4, 3/8	7.0				1700						

*1 At the max. operating pressure differential (Fluid: Air)

Series Variations



Direct Operated Vacuum Type JSX□□V Series p. 23

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]				Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			200	500	700	1000						
JSX10V Series	1/8	1.6 2.4	190				Air	Stainless steel Brass	N.C.	FKM	Grommet DIN terminal Conduit M12 connector	CE UK CA
JSX20V Series	1/8, 1/4, 3/8	3.2, 4 5.6, 7.1	470									
JSX30V Series	1/4, 3/8	4 5.6, 7.1				940						

*1 At the max. operating pressure differential (Fluid: Air)



Direct Operated High Pressure Type JSX□□H Series p. 25

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]						Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			500	750	1000	1500	2000	2250						
JSX30H Series	1/4, 3/8	3.2						2200	Air	Stainless steel Brass	N.C.	NBR FKM EPDM	Grommet DIN terminal Conduit M12 connector	CE UK CA

*1 At the max. operating pressure differential (Fluid: Air)



Direct Operated Steam Type JSX□□S Series p. 39

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]						Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			5	10	15	20	25	30						
JSX30S Series	1/4, 3/8	5.6, 7.1			15				Air (Steam) Heated water	Stainless steel Brass	N.C.	FKM	Conduit	CE UK CA

*1 At the max. operating pressure differential (Fluid: Steam)



Direct Operated Modular Mounting Type JSXM Series p. 77

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min] (ANR)		Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			500	1000						
JSXM20 Series	1/8, 1/4	3.2	650		Air	Aluminum	N.C.	NBR FKM	Grommet DIN terminal Conduit M12 connector	CE UK CA
JSXM30 Series	1/4, 3/8	4.0		1300						
JSXM40 Series	1/4, 3/8, 1/2	4.0		1300						

*1 At the max. operating pressure differential (Fluid: Air)

Series Variations



Pilot Operated

JSXD Series

N.C. specification p. 43

N.O. specification p. 47

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]			Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			200	400	1000						
JSXD30 Series	1/4, 3/8, 1/2*2	10	100			Air Water Oil	Stainless steel Brass Bronze Aluminum*2	N.C. N.O.	NBR FKM EPDM	Grommet DIN terminal Conduit M12 connector	
JSXD40 Series	3/8, 1/2	15	200								
JSXD50 Series	3/4	20	430								
JSXD60 Series	1	25	580								
JSXD70 Series	1 1/4, 32A	35			1000						
JSXD80 Series	1 1/2, 40A	40			1400						
JSXD90 Series	2, 50A	50			2200						

*1 At the max. operating pressure differential (Fluid: Water)

*2 Excludes N.O.



Pilot Operated

Steam Type

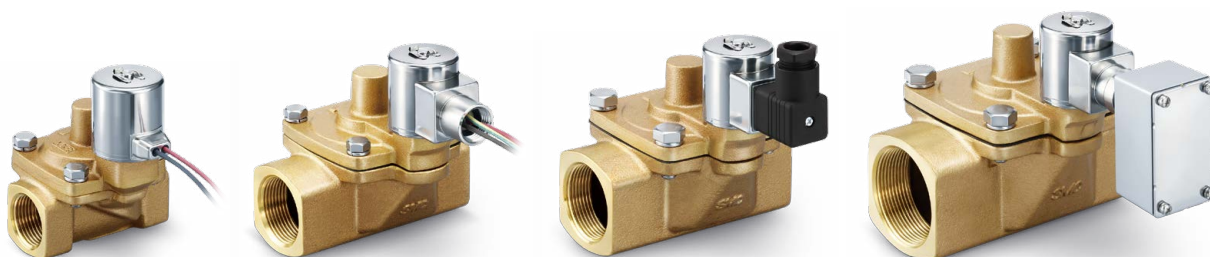
JSXP Series

N.C. specification p. 57

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]			Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			200	400	1000						
JSXP40 Series	3/8, 1/2	15	200			Steam Heated water	Stainless steel Brass Bronze	N.C.	FKM PTFE	Grommet Conduit	
JSXP50 Series	3/4	20	420								
JSXP60 Series	1	25	530								
JSXP70 Series	1 1/4, 32A	35			1000						
JSXP80 Series	1 1/2, 40A	40			1400						
JSXP90 Series	2, 50A	50			2200						

*1 At the max. operating pressure differential (Fluid: Steam)

Series Variations



Water Hammer Relief

JSXR Series p. 65

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]			Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			200	400	1000						
JSXR50 Series	1/2, 3/4	20	320			Water	Bronze	N.C.	NBR FKM	Grommet DIN terminal Conduit M12 connector Conduit terminal	CE UKCA
JSXR60 Series	1	25	540								
JSXR70 Series	1 1/4	35	1000								
JSXR80 Series	1 1/2	40	1400								
JSXR90 Series	2	50	2200								

*1 At the max. operating pressure differential (Fluid: Water)



Zero Differential Pressure Type Pilot Operated

JSXZ Series

N.C. specification p. 71

N.O. specification p. 73

Model	Port size	Orifice diameter [mmø]	Flow rate*1 [L/min]			Fluid	Body material	Valve type	Seal material	Electrical entry	Standards
			200	400	1000						
JSXZ30 Series	1/4, 3/8	10	100			Air Water Oil	Stainless steel Brass Aluminum*2	N.C. N.O.	NBR FKM EPDM	Grommet DIN terminal Conduit M12 connector	CE UKCA
JSXZ40 Series	1/2	15	200								
JSXZ50 Series	3/4	20	400								
JSXZ60 Series	1	25	460								

*1 At the max. operating pressure differential (Fluid: Water)

*2 Excludes N.O.

Space saving

Compact

Valve volume: **25% reduction**^{*1}

Lightweight

Weight: **30% reduction**^{*1}
*1 Compared with the existing model

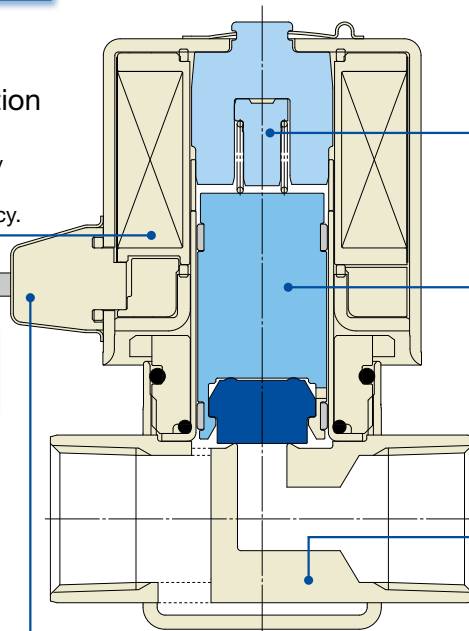
Energy saving^{*3}

Coil force: **10% increase**
 (Compared with the existing model)
 Power consumption: **14% reduction**
 (Compared with the existing model)
 The coil attraction force has been improved by 10% and the power consumption has been reduced by 14% for optimal magnetic efficiency.

*3 For JSX series N.C./DC specification valves

360° lead wire insertion and removal is possible.

As the coil rotates 360°, the lead wire is easy to handle.



Stopper construction

Metal noise reduced by the resin stopper
 Longer service life

Improved armature durability

IP67 enclosure

* IP65 for models with a DIN terminal

Choice of body material

· Stainless steel · Brass/Bronze^{*2}
 · Aluminum

*2 The bronze body is only selectable for the pilot operated type.

Power consumption * For DC voltages

Model	Size	10	20	30	40	50	60	70	80	90
Direct Operated JSX Series		4	6	8	—	—	—	—	—	—
Direct Operated High Flow/ Power Saving Type JSX□□U Series		2*1	3*1	3*1	—	—	—	—	—	—
Direct Operated Vacuum Type JSX□□V Series		4	6	8	—	—	—	—	—	—
Direct Operated Steam Type JSX□□S Series		—	—	13	—	—	—	—	—	—
Direct Operated High Pressure Type JSX□□H Series		—	—	13	—	—	—	—	—	—
Pilot Operated JSXD Series		—	—	6	6	6	8	8	8	8
Pilot Operated Steam Type JSXP Series		—	—	—	6	6	8	8	13	13
Pilot Operated Water Hammer Relief JSXR Series		—	—	—	—	6	8	8	13	13
Zero Differential Pressure Type Pilot Operated JSXZ Series		—	—	8	8	13	13	—	—	—
Modular Mounting Type JSXM Series		—	6	8	8	—	—	—	—	—

*1 When holding in an energized state

Full-wave rectifier type

Improved durability

Extended service life due to the special construction
 (Compared with the existing shading coil)

Reduced buzzing noise

Due to being rectified to DC
 by the full-wave rectifier

Reduced apparent power

* Class B, N.C. valve
 (Compared with the existing model)
 9.5 VA → **8 VA**
 (JSX20/JSXD60, 70 Series)
 12 VA → **9.5 VA**
 (JSX30/JSXD80, 90 Series)

Improved OFF response

Specially constructed to improve the OFF response when operated with high viscosity fluids such as oil

Low-noise construction

Specially constructed to reduce metal noise during operation

Improved weather resistance in outdoor environments*1

*1 Various tests for weather resistance have been passed, including the accelerated weathering test, combined cycle test, and ozone-proof exposure test. When using the product, refer to "Product Usage Precautions" in the **Web Catalog**.

Passed
1000 hours

Accelerated weathering test

ISO 4892-3 (JIS K 7350-3) compliant

Passed
960 hours

Combined cycle test

ISO 14993 (JIS H 8502:1999) compliant

Passed
1000 hours

Ozone-proof exposure test

ISO 1431 (JIS K 6259) compliant

Test details

Accelerated weathering test

UV exposure
8 hours

(Type 1A: UVA-340)

UV exposure and water spray
0.25 hours

(Type 1A: UVA-340)

Condensation/Dark
3.75 hours

Combined cycle test

Salt water spray
2 hours

(Ambient temperature 35°C, Salinity 5%)

Drying
4 hours

(Ambient temperature 60°C, Relative humidity 20 to 30%)

Humidity
2 hours

(Ambient temperature 50°C, Relative humidity 95% or more)

Ozone-proof exposure test

Exposure time
1000 hours

(Ozone-proof concentration 1 ppm, Ambient temperature 40°C)

Direct Operated



JSX Series

Product Usage Precautions

- Although this product features improved weather resistance in outdoor environments, outdoor use is not covered by the warranty.
- This product should be used within the specifications and should not be exposed to direct sunlight, rain, snow, etc.
- This product does not provide any corrosion resistance (anti-rust or antidiscoloration).

Improved weather resistance in outdoor environments*1

*1 Various tests for weather resistance have been passed, including the accelerated weathering test, combined cycle test, and ozone-proof exposure test. When using the product, refer to "Product Usage Precautions" in the **Web Catalog**.

Applicable series: JSX21/31□-S Series

Body size	2 types
Body size: 20, 30	
Body material	Stainless steel
Electrical entry	Conduit type
Thread type	3 types
Rc, NPT, G	
Valve type	N.C.



Ordering Examples

Enter the standard product number. **p. 13**

JSX Series **JSX** **2** **1** **□** **-S** **N** **403** **R** **-5** **CS** **-B**

Size

Symbol	Size
2	20
3	30

Body material

Symbol	Body material
S	Stainless steel

Valve type

Symbol	Valve type
1	N.C.

Type

Symbol	Type
Nil	Standard
U	High flow

Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant
CS	Conduit, NPT thread (With surge voltage suppressor)	 Refer to page 84.
FS	Conduit, G thread (With surge voltage suppressor)	

* Only the DC voltages are available for the high-flow type.

High Flow/ Power Saving Type **JSX□□U Series** p. 19

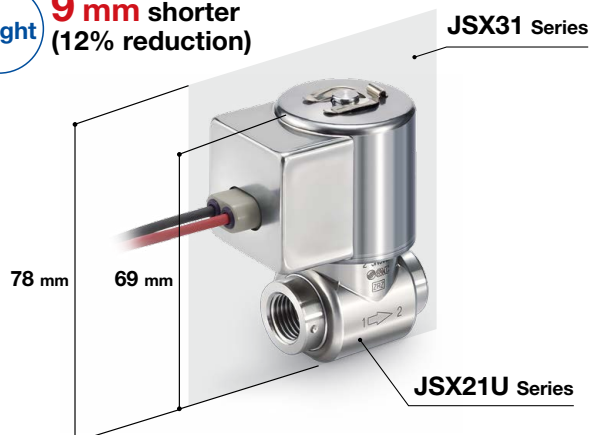
Flow rate increasable by up to 86%*1

*1 Operating pressure: 0.9 MPa

	Orifice diameter [mmø]	Flow rate [L/min]	
		5	10
High Flow/ Power Saving Type JSX11U Series	2.4	6.5	86% increase
JSX11 Series	1.6	3.5	

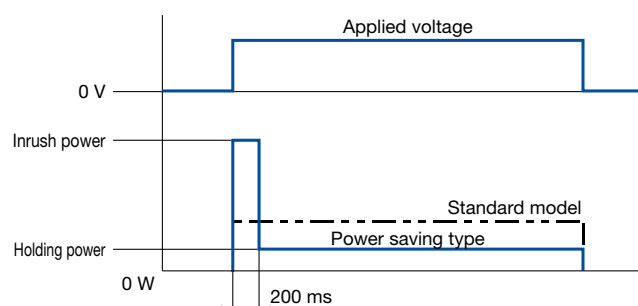
Reduced valve size*1

*1 Max. flow rate: 23.9 L/min, Orifice diameter: ø4 mm, Max. operating pressure differential: 1.0 MPa

JSX21U Series ← **JSX31 Series**
Height **9 mm** shorter
(12% reduction)

Weight **110 g** lighter (24% reduction)

	Height [mm]	Weight [g]
JSX21U Series	69	340
JSX31 Series	78	450

Substantial holding power consumption reduction

The overall power consumption amount can be reduced by up to **63%** by reducing the power consumption during holding.


* Effective after being energized for more than 200 ms

Power Consumption (Holding)		[W]		
		Size 10	Size 20	Size 30
JSX□□U Series	2	3	3	3
JSX□□ Series	4	6	8	8

Electrical entry



Grommet with PCB



Conduit

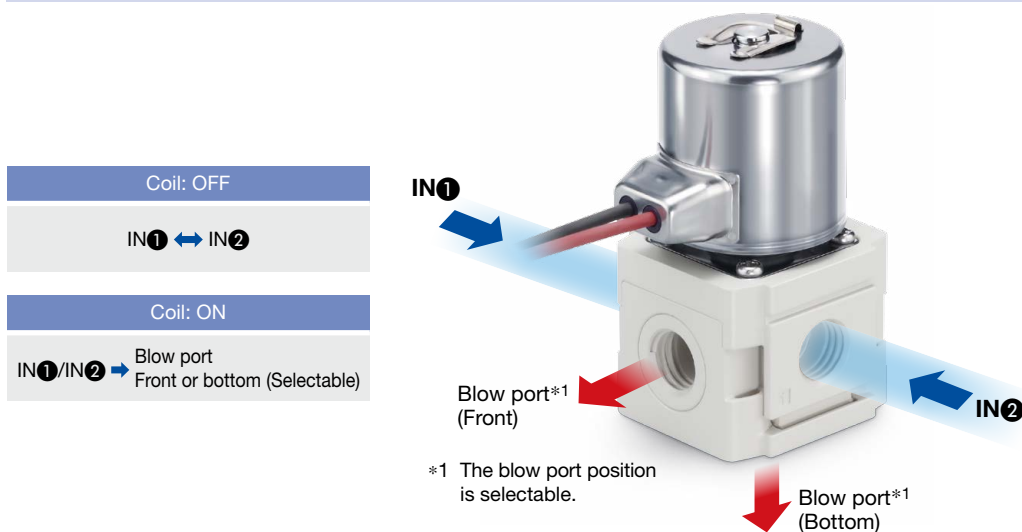


DIN terminal

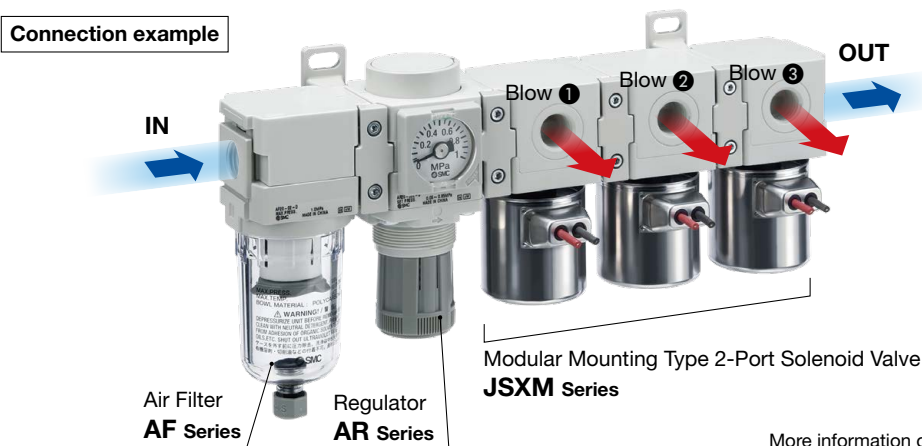

Conduit terminal
* JSXR only


M12 connector

Modular Mounting Type **JSXM** Series p. 77



Can be connected to modular type F.R.L. units



More information on F.R.L. units can be viewed here.



Simple Specials System

A system designed to respond quickly and easily to your special ordering needs

For modular connection units (shipped assembled), the simple specials system can be used.

Short lead times

This system enables us to respond to your special needs (additional machining, accessory assembly, or the designing of a modular unit) and deliver your personalized products as quickly as standard products.

Repeat orders

Once we receive a simple special part number from one of your previous orders, we will process the order, manufacture the product, and deliver it to you as quickly as possible.

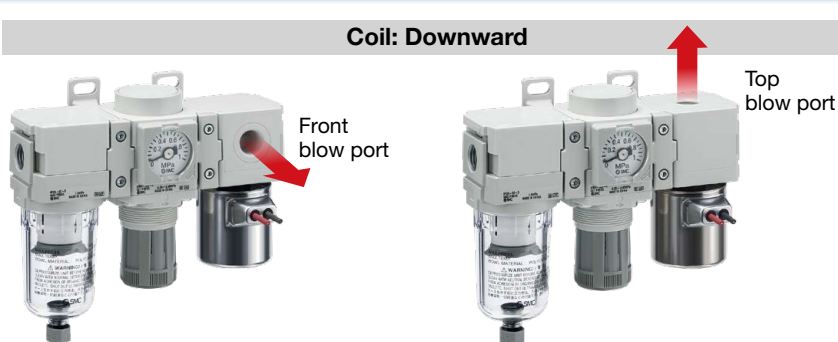
Please contact your local sales representative for more details.

The coil orientation and blow port position can be selected.

Coil: Upward



Coil: Downward



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JSX20, 30 **Port Size** 1/8, 1/4, 3/8 **Body Material** **Brass** **Body Material** **Stainless Steel, Brass** p. 33

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High Flow/ Power Saving Type

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For **Water** **Air** **Oil** **Body Material** **Stainless Steel, Brass**

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For **Vacuum** **Body Material** **Stainless Steel, Brass**

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JSX20V **Port Size** 1/8 **Body Material** **Stainless Steel** p. 29

JSX20V, 30V **Port Size** 1/4, 3/8 **Body Material** **Stainless Steel** p. 31

JSX20V, 30V **Port Size** 1/8, 1/4, 3/8 **Body Material** **Brass** **Body Material** **Stainless Steel, Brass** p. 33

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JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

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JSX30H Port Size 1/4, 3/8 Body Material **Stainless Steel** p. 31

JSX30H Port Size 1/4, 3/8 Body Material **Brass** p. 33

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Steam Type Direct Operated 2-Port Solenoid Valve JSX□□S Series p. 39

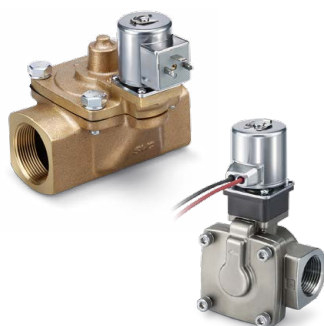
For **Steam** **Heated Water** Body Material **Stainless Steel, Brass**

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JSX

JSXD

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JSXZ

JSXM

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For **Water****Air****Oil**

Direct Operated 2-Port Solenoid Valve JSX Series



Differs depending on the voltage and electrical entry. For details, refer to page 84.



Refer to page 85 for details.

Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)		Normally Closed (N.C.)	Normally Open (N.O.)		High Flow/Power Saving Type		High Flow/Power Saving Type	Vacuum Type		High Pressure Type		Steam Type	
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How to Order

JSX **2** **1** - **S** **N** **302** **R** - **5** **G** - **D** - **B**

1 2 3 4 5 6 7 8 9 10

1 Size

Symbol	Size
1	10
2	20
3	30

2 Valve type

Symbol	Valve type
1	N.C.

* Refer to page 17 for N.O. type.

4 Seal material

Symbol	Seal material
N	NBR
F	FKM
E	EPDM

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

7 Rated voltage

Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC
2	200 VAC	8	48 VAC
3	120 (110) VAC	B	24 VAC
4	220 VAC	J	230 VAC

DC

Symbol	Rated voltage
5	24 VDC
6	12 VDC

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Option

Symbol	Option
Nil	None
B	With bracket*1 (Stainless steel)

*1 Refer to page 102 for bracket assembly part nos.

8 Electrical entry

Symbol	Electrical entry		Size			CE/UKCA-compliant	UL Standards
			10	20	30		
G	Grommet*1		●	●	●	Refer to page 84.	Refer to page 85.
GS	Grommet with PCB (With surge voltage suppressor)		●	●	●		
CS	Conduit (With surge voltage suppressor)		—	●	●		
DS	DIN terminal (With surge voltage suppressor)		●	●	●		
DZ	DIN terminal with light (With surge voltage suppressor)		●	●	●		
DN	DIN terminal without connector (With surge voltage suppressor)		●	●	●		
WN	M12 connector/Without connector cable (With surge voltage suppressor)*2		●	●	●		

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Size	Port size	Orifice diameter [mm]	Flow rate characteristics*1					Max. operating pressure differential [MPa]	Model	Weight*2	
			Air			Water, Oil				Stainless steel body*3	Brass body
			C [dm³/(s·bar)]	b	Cv	Kv	Conversion Cv				
10	1/8	1.6	0.36	0.58	0.08	0.07	0.08	0.9	JSX11- $\frac{1}{8}$ □101	160	160
		2.4	0.62	0.45	0.15	0.13	0.15	0.4	JSX11- $\frac{1}{8}$ □201	160	160
20	1/8	3.2	1.35	0.48	0.35	0.30	0.35	0.7	JSX21- $\frac{1}{8}$ □301	320	330
		3.2	1.35	0.48	0.35	0.30	0.35	0.7	JSX21- $\frac{1}{8}$ □302	320	330
	1/4	4.0	2.02	0.48	0.52	0.45	0.52	0.3	JSX21- $\frac{1}{4}$ □402	320	330
		5.6	2.62	0.43	0.73	0.63	0.73	0.2	JSX21- $\frac{1}{4}$ □502	320	330
		7.1	3.15	0.44	0.88	0.76	0.88	0.1	JSX21- $\frac{1}{4}$ □702	320	330
		3.2	1.35	0.48	0.35	0.30	0.35	0.7	JSX21- $\frac{1}{4}$ □303	320	360
	3/8	4.0	2.02	0.48	0.52	0.45	0.52	0.3	JSX21- $\frac{3}{8}$ □403	320	360
		5.6	2.62	0.43	0.73	0.63	0.73	0.2	JSX21- $\frac{3}{8}$ □503	320	360
		7.1	3.15	0.44	0.88	0.76	0.88	0.1	JSX21- $\frac{3}{8}$ □703	320	360
		30	1/4	4.0	2.02	0.48	0.52	0.45	0.52	1.0	JSX31- $\frac{1}{4}$ □402
5.6	2.62			0.43	0.73	0.63	0.73	0.5	JSX31- $\frac{1}{4}$ □502	450	490
7.1	3.15			0.44	0.88	0.76	0.88	0.2	JSX31- $\frac{1}{4}$ □702	450	490
3/8	4.0		2.02	0.48	0.52	0.45	0.52	1.0	JSX31- $\frac{3}{8}$ □403	450	520
	5.6		2.62	0.43	0.73	0.63	0.73	0.5	JSX31- $\frac{3}{8}$ □503	450	520
	7.1		3.15	0.44	0.88	0.76	0.88	0.2	JSX31- $\frac{3}{8}$ □703	450	520

Applicable Fluid Checklist

Applicable fluid	Seal material		
	NBR	FKM	EPDM
Air	●	●	●
Water	●	●	●
Oil	—	●	—

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use. If something is not clear, please contact SMC.

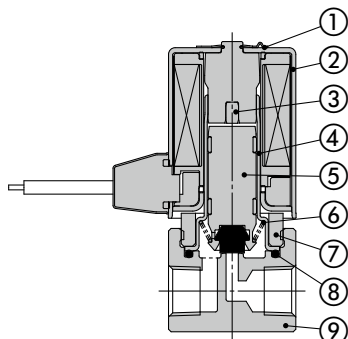
*1 The flow rate characteristics of this product vary.
*2 The values were calculated based on the combination of an Rc or NPT thread and a grommet. Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.

*3 Add 30 g for the G thread (port size 3/8) type.

Construction

JSX10

Body material: Stainless steel, Brass

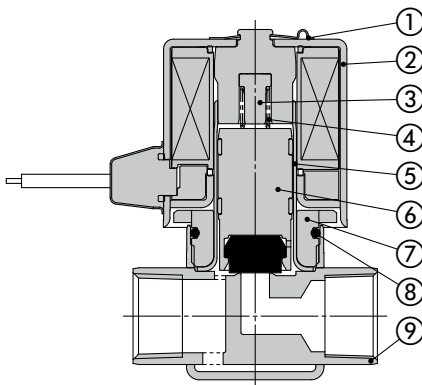


Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Tube assembly	Stainless steel
5	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
6	Spring	Stainless steel
7	Set nut	Stainless steel
8	Gasket	NBR, (FKM, EPDM)
9	Body	Stainless steel Brass

JSX20, 30

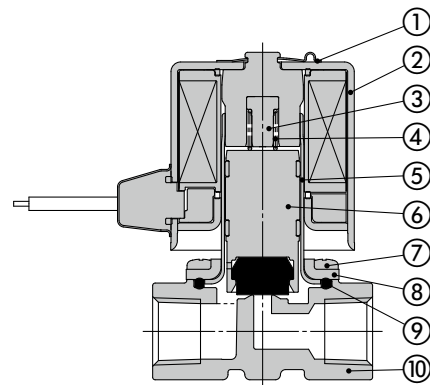
Body material: Stainless steel



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
7	Nut	Stainless steel
8	Gasket	NBR (FKM, EPDM)
9	Body	Stainless steel

Body material: Brass



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	NBR (FKM, EPDM)
10	Body	Brass

Common Specifications

Size			10	20	30
Valve specifications	Valve construction		Direct operated poppet		
	Valve type		Normally closed (N.C.)		
	Fluid and fluid temperature		Air: -10 to 60°C (Dew point temperature: -10°C or less) Water: 1 to 60°C (No freezing) Oil: -5 to 60°C (Kinematic viscosity: 50 mm²/s or less)		
	Withstand pressure		2.0 MPa		
	Max. system pressure		1.0 MPa		
	Ambient temperature		-20 to 60°C		
	Valve leakage*1 / External leakage*1	Air	1 cm³/min (ANR) or less		
		Water, Oil	0.1 cm³/min or less		
	Mounting orientation		Unrestricted		
	Enclosure*2		IP67 (IP65 for the DIN terminal)		
	Standards*3		CE/UKCA, UL Recognized, UL Listed		
	Operating environment		Location without the presence of corrosive gases or explosive gases		
	Body material		Stainless steel, Brass		
	Seal material		NBR, FKM, EPDM		
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V		
		DC	12 V, 24 V		
	Allowable voltage fluctuation		±10% of the rated voltage		
	Allowable leakage voltage	AC	5% or less of the rated voltage		
		DC	2% or less of the rated voltage		
	Apparent power*4, *5	AC	4.5 VA	8 VA	9.5 VA
	Power consumption*4	DC	4 W	6 W	8 W
	Temperature rise*6	AC/DC	70/65°C		

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to pages 84, 85.

*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

Direct Operated 2-Port Solenoid Valve JSX Series

CE UK
CA

Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

For **Air**

Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)		Normally Closed (N.C.)	Normally Open (N.O.)		High Flow/Power Saving Type		High Flow/Power Saving Type	Vacuum Type		High Pressure Type		Steam Type	
▶ p. 13		▶ p. 15	▶ p. 17		▶ p. 19		▶ p. 21	▶ p. 23		▶ p. 25		▶ p. 39	

How to Order

JSX **2** **1** - **A** **N** **302** **R** - **5** **G** - **D** - **B**

1 2 3 4 5 6 7 8 9 10



1 Size

Symbol	Size
2	20
3	30

2 Valve type

Symbol	Valve type
1	N.C.

* Refer to page 17 for N.O. type.

3 Body material

Symbol	Body material
A	Aluminum

8 Electrical entry

Symbol	Electrical entry		Size		CE/UKCA-compliant
			20	30	
G	Grommet*1		●	●	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)		●	●	
CS	Conduit (With surge voltage suppressor)		●	●	
DS	DIN terminal (With surge voltage suppressor)		●	●	
DZ	DIN terminal with light (With surge voltage suppressor)		●	●	
DN	DIN terminal without connector (With surge voltage suppressor)		●	●	
WN	M12 connector/Without connector cable (With surge voltage suppressor)*2		●	●	

4 Seal material

Symbol	Seal material
N	NBR
F	FKM

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

5 Orifice diameter and port size

Symbol	Orifice diameter [mmø]	Port size	Size	
			20 Aluminum body	30 Aluminum body
301	3	1/8	●	—
302		1/4	●	—
402	4	1/4	—	●
403		3/8	—	●
501	5	1/8	●	—
502		1/4	●	—
702	7	1/4	—	●
703		3/8	—	●

7 Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Option

Symbol	Option
Nil	None
B	With bracket*1

*1 Refer to page 102 for bracket assembly part nos.

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Aluminum Body Type

Size	Port size	Orifice diameter [mmø]	Flow rate characteristics*1			Max. operating pressure differential [MPa]	Model	Weight*2 [g]
			C [dm ³ /(s·bar)]	b	Cv			
20	1/8, 1/4	3	1.41	0.54	0.35	0.7	JSX21-A□30□	240
		5	1.66	0.54	0.52	0.2	JSX21-A□50□	240
30	1/4, 3/8	4	1.57	0.59	0.52	1.0	JSX31-A□40□	400
		7	3.02	0.53	0.88	0.2	JSX31-A□70□	400

*1 The flow rate characteristics of this product vary.

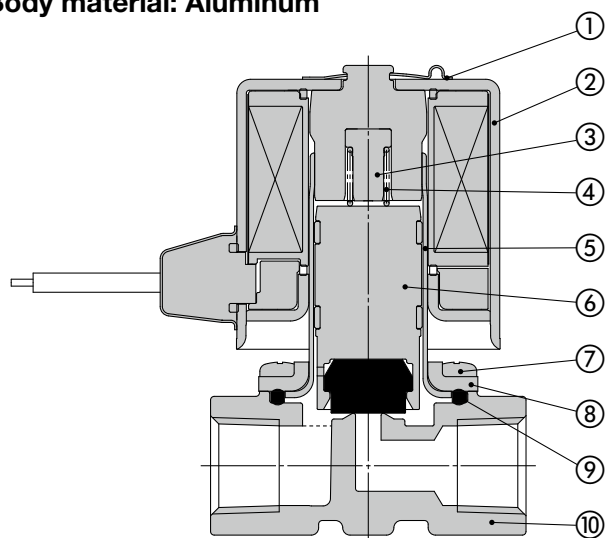
*2 Indicates case of grommet type

Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.

Construction

JSX20, 30

Body material: Aluminum



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR, (FKM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	NBR, (FKM)
10	Body	Aluminum

Common Specifications

Size			10	20	30
Valve specifications	Valve construction		Direct operated poppet		
	Valve type		Normally closed (N.C.)		
	Fluid and fluid temperature		Air: -10 to 60°C (Dew point temperature: -10°C or less)		
	Withstand pressure		2.0 MPa		
	Max. system pressure		1.0 MPa		
	Ambient temperature		-20 to 60°C		
	Valve leakage ^{*1} /External leakage ^{*1}	Air	1 cm ³ /min (ANR) or less		
	Mounting orientation		Unrestricted		
	Enclosure ^{*2}		IP67 (IP65 for the DIN terminal)		
	Standards ^{*3}		CE/UKCA		
	Operating environment		Location without the presence of corrosive gases or explosive gases		
	Body material		Aluminum		
	Seal material		NBR, FKM		
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V		
		DC	12 V, 24 V		
	Allowable voltage fluctuation		±10% of the rated voltage		
	Allowable leakage voltage	AC	5% or less of the rated voltage		
		DC	2% or less of the rated voltage		
	Apparent power ^{*4, *5}	AC	4.5 VA	8 VA	9.5 VA
	Power consumption ^{*4}	DC	4 W	6 W	8 W
	Temperature rise ^{*6}	AC/DC	70/65°C		

^{*1} Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

^{*2} This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

^{*3} Standards compliance varies depending on the model. For details, refer to page 84.

^{*4} Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

^{*5} There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

^{*6} Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

For **Water**
Air
Oil

Direct Operated 2-Port Solenoid Valve JSX Series

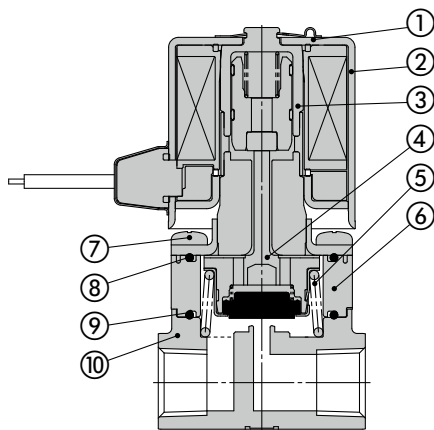
CE UK
CA
Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)		Normally Closed (N.C.)	Normally Open (N.O.)		High Flow/Power Saving Type		High Flow/Power Saving Type	Vacuum Type		High Pressure Type		Steam Type	
▶ p. 13		▶ p. 15	▶ p. 17		▶ p. 19		▶ p. 21	▶ p. 23		▶ p. 25		▶ p. 39	

Construction

JSX20, 30 series Normally open (N.O.)
Body material: Stainless steel, Brass



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Sleeve assembly	Stainless steel, PPS
4	Push rod assembly	Stainless steel, PPS, NBR (FKM, EPDM)
5	Spring	Stainless steel
6	Adapter	PPS
7	Mounting screw	Stainless steel
8	O-ring	NBR (FKM, EPDM)
9	O-ring	NBR (FKM, EPDM)
10	Body	Stainless steel, Brass

Specifications

Size			20	30
Valve specifications	Valve construction		Direct operated poppet	
	Valve type		Normally open (N.O.)	
	Fluid and fluid temperature		Air: -10 to 60°C (Dew point temperature: -10°C or less) Water: 1 to 60°C (No freezing) Oil: -5 to 60°C (Kinematic viscosity: 50 mm ² /s or less)	
	Withstand pressure		2.0 MPa	
	Max. system pressure		1.0 MPa	
	Ambient temperature		-20 to 60°C	
	Valve leakage*1/External leakage*1	Air	1 cm ³ /min (ANR) or less	
		Water, Oil	0.1 cm ³ /min or less	
	Mounting orientation		Unrestricted	
	Enclosure*2		IP67 (IP65 for the DIN terminal)	
	Standards*3		CE/UKCA	
	Operating environment		Location without the presence of corrosive gases or explosive gases	
Coil specifications	Body material		Stainless steel, Brass	
	Seal material		NBR, FKM, EPDM	
	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V	
		DC	12 V, 24 V	
	Allowable voltage fluctuation		±10% of the rated voltage	
	Allowable leakage voltage	AC	5% or less of the rated voltage	
		DC	2% or less of the rated voltage	
	Apparent power*4, *5	AC	8 VA	9.5 VA
	Power consumption*4	DC	6 W	8 W
	Temperature rise*6	AC/DC	70/65°C	

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

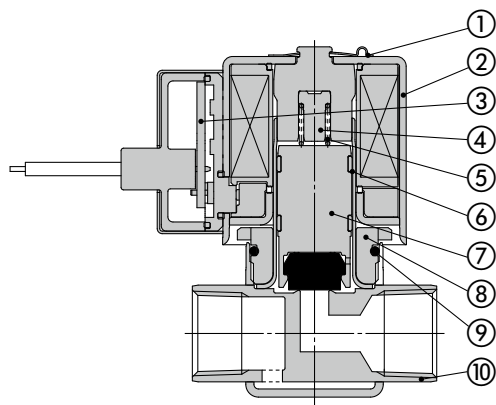
*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

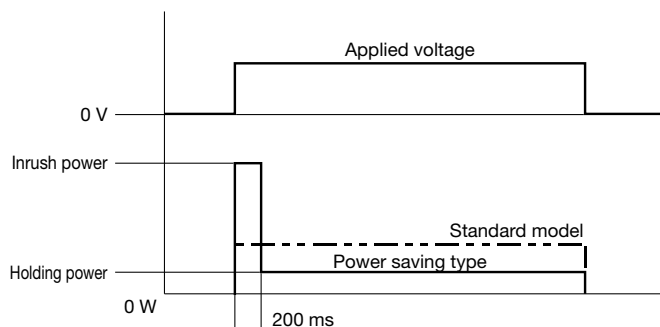
Construction



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Board assembly	—
4	Stopper	PPS
5	Spring	Stainless steel
6	Tube assembly	Stainless steel
7	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
8	Nut	Stainless steel
9	Gasket	NBR (FKM, EPDM)
10	Body	Stainless steel

Power Saving Specification



Power is saved by reducing the wattage required to hold the valve.
Effective after being energized for more than 200 ms
* The valve has polarity. Refer to the “Electrical Circuits” on page 106 and be careful not to reverse the polarity.

Common Specifications

Size		10	20	30
Valve specifications	Valve construction	Direct operated poppet		
	Valve type	Normally closed (N.C.)		
	Fluid and fluid temperature	Air: -10 to 60°C (Dew point temperature: -10°C or less) Water: 1 to 60°C (No freezing) Oil: -5 to 60°C (Kinematic viscosity: 50 mm ² /s or less)		
	Withstand pressure	2.0 MPa		
	Max. system pressure	1.0 MPa		
	Ambient temperature	-20 to 60°C		
	Valve leakage/ External leakage*1	1 cm ³ /min (ANR) or less		
		0.1 cm ³ /min or less		
	Mounting orientation	Unrestricted		
	Enclosure*2	IP67 (IP65 for the DIN terminal)		
	Standards*3, *4	CE/UKCA		
	Operating environment	Location without the presence of corrosive gases or explosive gases		
	Body material	Stainless steel, Brass		
	Seal material	NBR, FKM, EPDM		
Coil specifications	Vibration/Impact resistance*7	30/100 m/s ²		
	Rated voltage	12 V, 24 V		
	Allowable voltage fluctuation	±10% of the rated voltage		
	Allowable leakage voltage	2% or less of the rated voltage		
	Power consumption (Holding)*5	2 W	3 W	3 W
	Inrush current	1.25 A	2 A	2 A
	Temperature rise*6	25°C	25°C	25°C

- *1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C
 *2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage. Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.
 *3 Standards compliance varies depending on the model. For details, refer to page 84.
 *4 The high flow type is not in compliance with UL standards.
 *5 Power consumption: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)
 *6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.
 *7 Vibration resistance: No malfunction occurred in a one-sweep test between 45 and 2000 Hz. The test was performed in both an energized and de-energized state in the axial direction and at a right angle to the armature.
 Impact resistance: No malfunction occurred when tested with a drop tester in the axial direction and at a right angle to the armature in both an energized and de-energized state, once in each condition. (Value in the initial state)
 Do not use in an environment subject to constant vibration and/or impact.

Be sure to read the “Specific Product Precautions” before handling the product.

High Flow/ Power Saving Type

Direct Operated 2-Port Solenoid Valve



Differs depending on the voltage and electrical entry. For details, refer to page 84.

For **Air**

JSX□□U Series

RoHS

Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)	Normally Closed (N.C.)	Normally Closed (N.C.)	Normally Open (N.O.)	Normally Open (N.O.)	High Flow/ Power Saving Type	High Flow/ Power Saving Type	High Flow/ Power Saving Type	Vacuum Type	Vacuum Type	High Pressure Type	High Pressure Type	Steam Type	Steam Type
▶ p. 13	▶ p. 13	▶ p. 15	▶ p. 15	▶ p. 17	▶ p. 19	▶ p. 19	▶ p. 21	▶ p. 23	▶ p. 23	▶ p. 25	▶ p. 25	▶ p. 39	▶ p. 39

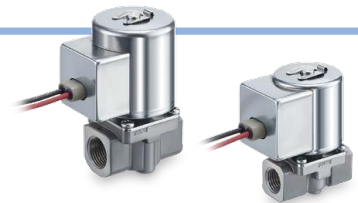
The dimensions are the same as those of the standard JSX series model. Refer to pages 27 to 38 for details.

How to Order

JSX **2** **1** **U** - **A** **N** **501** **R** - **5** **GS** - **D** - **B**

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩

U High flow type



① Size

Symbol	Size
2	20
3	30

② Valve type

Symbol	Valve type
1	N.C.

③ Body material

Symbol	Body material
A	Aluminum

⑧ Electrical entry

Symbol	Electrical entry	Size	CE/UKCA-compliant
		20	30
GS	Grommet with PCB (With surge voltage suppressor)	●	●
CS	Conduit (With surge voltage suppressor)	●	●
DS	DIN terminal (With surge voltage suppressor)	●	●
DZ	DIN terminal with light (With surge voltage suppressor)	●	●
DN	DIN terminal without connector (With surge voltage suppressor)	●	●
WN	M12 connector/Without connector cable (With surge voltage suppressor)*1	●	●

Refer to page 84.

*1 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

④ Seal material

Symbol	Seal material
N	NBR
F	FKM

⑤ Orifice diameter and port size

Symbol	Orifice diameter [mmø]	Port size	Size	
			20	30
501	5.0	1/8	●	—
502		1/4	●	—
702	7.0	1/4	—	●
703		3/8	—	●

⑥ Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

⑦ Rated voltage

Symbol	Rated voltage
5	24 VDC
6	12 VDC

⑨ Oil-free option

Symbol	Option
Nil	None
D	Oil-free

⑩ Option

Symbol	Option
Nil	None
B	With bracket*1 (Stainless steel)

*1 Refer to page 102 for bracket assembly part nos.

Flow Rate Characteristics

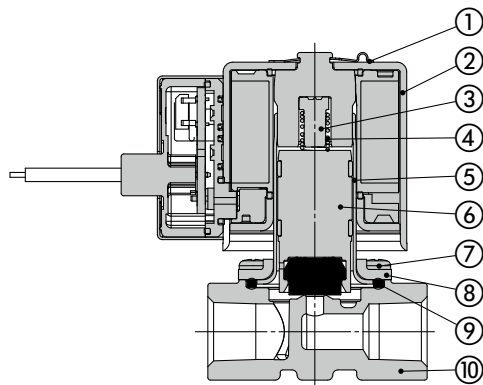
Size	Port size	Orifice diameter [mmø]	Flow rate characteristics*1			Max. operating pressure differential [MPa]	Model	Weight*2 [g]
			Air					
			C	b	Cv			
20	1/8	5.0	1.66	0.54	0.52	0.9	JSX21U-A-□501	260
	1/4	5.0	1.66	0.54	0.52	0.9	JSX21U-A-□502	260
30	1/4	7.0	3.02	0.53	0.88	0.8	JSX31U-A-□702	420
	3/8	7.0	3.02	0.53	0.88	0.8	JSX31U-A-□703	420

*1 The flow rate characteristics of this product vary.

*2 Add 50 g for the conduit type, 30 g for the DIN terminal type, and -5 g for the M12 connector type.

Construction

Body material: Aluminum



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	NBR (FKM, EPDM)
10	Body	Aluminum

Common Specifications

Size		20	30
Valve specifications	Valve construction	Direct operated poppet	
	Valve type	Normally closed (N.C.)	
	Fluid and fluid temperature	Air: -10 to 60°C (Dew point temperature: -10°C or less)	
	Withstand pressure	2.0 MPa	
	Max. system pressure	1.0 MPa	
	Ambient temperature	-20 to 60°C	
	Valve leakage/External leakage*1	1 cm ³ /min (ANR) or less	
	Mounting orientation	Unrestricted	
	Enclosure*2	IP67 (IP65 for the DIN terminal)	
	Standards*3	CE/UKCA	
	Operating environment	Location without the presence of corrosive gases or explosive gases	
	Body material	Aluminum	
Coil specifications	Seal material	NBR, FKM, EPDM	
	Vibration/Impact resistance*6	30/100 m/s ²	
	Rated voltage	12 V, 24 V	
	Allowable voltage fluctuation	±10% of the rated voltage	
	Allowable leakage voltage	2% or less of the rated voltage	
	Power consumption (Holding)*4	3 W	3 W
	Inrush current	2 A	2 A
	Temperature rise*5	25°C	25°C

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

The high flow type is not in compliance with UL standards.

*4 Power consumption: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

*6 Vibration resistance: No malfunction occurred in a one-sweep test between 8.3 and 2000 Hz. The test was performed in both an energized and de-energized state in the axial direction and at a right angle to the armature.

Impact resistance: No malfunction occurred when tested with a drop tester in the axial direction and at a right angle to the armature in both an energized and de-energized state, once in each condition. (Value in the initial state)

Do not use in an environment subject to constant vibration and/or impact.

Be sure to read the "Specific Product Precautions" before handling the product.

Vacuum Type



Differs depending on the voltage and electrical entry. For details, refer to page 84.

For Vacuum

Direct Operated 2-Port Solenoid Valve

JSX□□V Series

RoHS

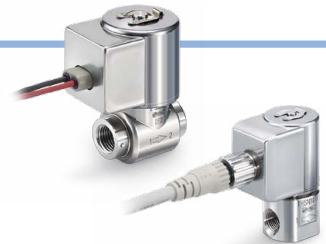
Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)		Normally Closed (N.C.)		Normally Open (N.O.)		High Flow/Power Saving Type		High Flow/Power Saving Type		Vacuum Type		High Pressure Type	
▶ p. 13		▶ p. 15		▶ p. 17		▶ p. 19		▶ p. 21		▶ p. 23		▶ p. 25	
												Steam Type	
												▶ p. 39	

How to Order

JSX **2** **1** V - **S** **F** **302** **R** - **5** **GS** - **D** - **B**

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩

V Vacuum type



① Size

Symbol	Size
1	10
2	20
3	30

② Valve type

Symbol	Valve type
1	N.C.

③ Body material

Symbol	Body material
S	Stainless steel
C	Brass

④ Seal material

Symbol	Seal material
F	FKM

⑧ Electrical entry

Symbol	Electrical entry	Size			CE/UKCA-compliant	
		10	20	30		
G	Grommet*1		●	●	●	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)		●	●	●	
CS	Conduit (With surge voltage suppressor)		—	●	●	
DS	DIN terminal (With surge voltage suppressor)		●	●	●	
DZ	DIN terminal with light (With surge voltage suppressor)		●	●	●	
DN	DIN terminal without connector (With surge voltage suppressor)		●	●	●	
WN	M12 connector/Without connector cable (With surge voltage suppressor)*2		●	●	●	

⑤ Orifice diameter and port size

Symbol	Orifice diameter [mmø]	Port size	Size		
			10	20	30
101	1.6	1/8	●	—	—
201	2.4	1/8	●	—	—
301	3.2	1/8	—	●	—
302		1/4	—	●	—
303		3/8	—	●	—
402	4.0	1/4	—	●	●
403		3/8	—	●	●
502	5.6	1/4	—	●	●
503		3/8	—	●	●
702	7.1	1/4	—	●	●
703		3/8	—	●	●

⑥ Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

* Only thread type "F" (G thread) can be selected for the JSX10.

⑦ Rated voltage

Symbol	Rated voltage
1	100 VAC
2	200 VAC
3	120 (110) VAC
4	220 VAC
7	240 VAC
8	48 VAC
B	24 VAC
J	230 VAC

DC

Symbol	Rated voltage
5	24 VDC
6	12 VDC

⑨ Oil-free option

Symbol	Option
Nil	None
D	Oil-free

⑩ Option

Symbol	Option
Nil	None
B	With bracket*1 (Stainless steel)

*1 Refer to page 102 for bracket assembly part nos.

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Size	Port size	Orifice diameter [mmø]	Flow rate characteristics*1			Operating pressure range [Pa abs]	Model	Weight*2 [g]	
			Air					Stainless steel body*3	Brass body
			C	b	Cv				
10	1/8	1.6	0.36	0.58	0.08	0.1 to atmospheric pressure	JSX11V- $\frac{S}{C}$ F101	160	160
		2.4	0.62	0.45	0.15		JSX11V- $\frac{S}{C}$ F201	160	160
20	1/8	3.2	1.35	0.48	0.35		JSX21V- $\frac{S}{C}$ □301	320	330
		3.2	1.35	0.48	0.35		JSX21V- $\frac{S}{C}$ □302	320	330
	1/4	4.0	2.02	0.48	0.52		JSX21V- $\frac{S}{C}$ □402	320	330
		5.6	2.62	0.43	0.73		JSX21V- $\frac{S}{C}$ □502	320	330
		7.1	3.15	0.44	0.88		JSX21V- $\frac{S}{C}$ □702	320	330
		3/8	3.2	1.35	0.48		0.35	JSX21V- $\frac{S}{C}$ □303	320
	4.0		2.02	0.48	0.52		JSX21V- $\frac{S}{C}$ □403	320	360
	5.6		2.62	0.43	0.73		JSX21V- $\frac{S}{C}$ □503	320	360
7.1	3.15		0.44	0.88	JSX21V- $\frac{S}{C}$ □703		320	360	
30	1/4	4.0	2.02	0.48	0.52		JSX31V- $\frac{S}{C}$ □402	450	490
		5.6	2.62	0.43	0.73		JSX31V- $\frac{S}{C}$ □502	450	490
		7.1	3.15	0.44	0.88		JSX31V- $\frac{S}{C}$ □702	450	490
		4.0	2.02	0.48	0.52		JSX31V- $\frac{S}{C}$ □403	450	520
	3/8	5.6	2.62	0.43	0.73		JSX31V- $\frac{S}{C}$ □503	450	520
		7.1	3.15	0.44	0.88		JSX31V- $\frac{S}{C}$ □703	450	520

*1 The flow rate characteristics of this product vary.

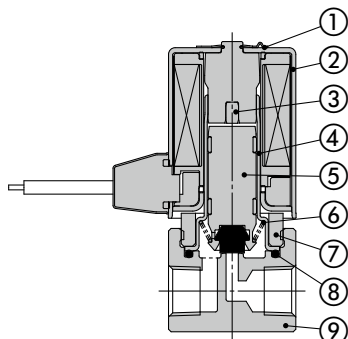
*2 Add 50 g for the conduit type, 30 g for the DIN terminal type, and -5 g for the M12 connector type.

*3 The values were calculated based on the combination of an Rc or NPT thread and a grommet with PCB. Add 30 g for the G thread (port size 3/8) type.

Construction

JSX10V

Body material: Stainless steel, Brass

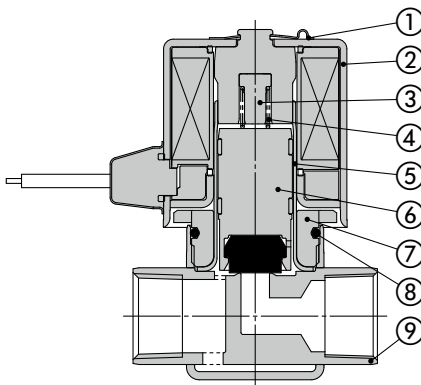


Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Tube assembly	Stainless steel
5	Armature assembly	Stainless steel, PPS (FKM)
6	Spring	Stainless steel
7	Set nut	Stainless steel
8	Gasket	FKM
9	Body	Stainless steel Brass

JSX20V, 30V

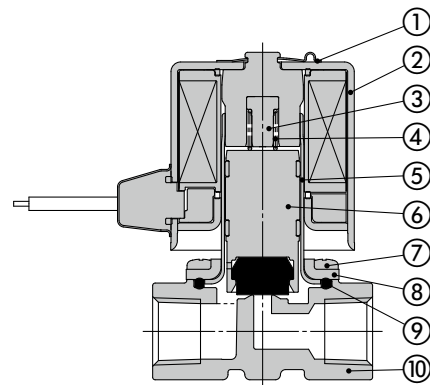
Body material: Stainless steel



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS (FKM)
7	Nut	Stainless steel
8	Gasket	FKM
9	Body	Stainless steel

Body material: Brass



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS (FKM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	FKM
10	Body	Brass

Common Specifications

Size			10	20	30
Valve specifications	Valve construction		Direct operated poppet		
	Valve type		Normally closed (N.C.)		
	Fluid and fluid temperature		Vacuum: -10 to 60°C (Dew point temperature: -10°C or less)		
	Withstand pressure		2.0 MPa		
	Max. system pressure		1.0 MPa		
	Ambient temperature		-20 to 60°C		
	Valve leakage/External leakage ^{*1}	Vacuum	10 ⁻⁶ Pa·m ³ /s or less		
	Mounting orientation		Unrestricted		
	Enclosure ^{*2}		IP67 (IP65 for the DIN terminal)		
	Standards ^{*3}		CE/UKCA		
	Operating environment		Location without the presence of corrosive gases or explosive gases		
	Body material		Stainless steel, Brass		
	Seal material		FKM		
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V		
		DC	12 V, 24 V		
	Allowable voltage fluctuation		±10% of the rated voltage		
	Allowable leakage voltage	AC	5% or less of the rated voltage		
		DC	2% or less of the rated voltage		
	Apparent power (Holding) ^{*4, *5}	AC	4.5 VA	8 VA	9.5 VA
	Power consumption (Holding) ^{*4}	DC	4 W	6 W	8 W
		AC/DC	70/65°C		

^{*1} Leakage (10⁻⁶ Pa·m³/s): The value at 0.1 Pa-abs and an ambient temperature of 20°C

^{*2} This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

^{*3} Standards compliance varies depending on the model. For details, refer to page 84.

^{*4} Power consumption: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

^{*5} There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

^{*6} Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

High Pressure Type



Differs depending on the voltage and electrical entry. For details, refer to page 84.

For **Air**

Direct Operated 2-Port Solenoid Valve

JSX□□H Series

RoHS

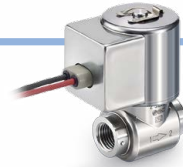
Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass		
Normally Closed (N.C.)		Normally Closed (N.C.)		Normally Open (N.O.)		High Flow/Power Saving Type		High Flow/Power Saving Type		Vacuum Type		High Pressure Type		Steam Type	
▶ p. 13		▶ p. 15		▶ p. 17		▶ p. 19		▶ p. 21		▶ p. 23		▶ p. 25		▶ p. 39	

How to Order

JSX 3 1 H - S N 302 R - 5 G - D - B

1 2 3 4 5 6 7 8 9 10

H High pressure type



1 Size

Symbol	Size
3	30

2 Valve type

Symbol	Valve type
1	N.C.

3 Body material

Symbol	Body material
S	Stainless Steel
C	Brass

8 Electrical entry

Symbol	Electrical entry	Size	CE/UKCA-compliant
G	Grommet* ¹	30	●
GS	Grommet with PCB (With surge voltage suppressor)	30	●
CS	Conduit (With surge voltage suppressor)	30	●
DS	DIN terminal (With surge voltage suppressor)	30	●
DZ	DIN terminal with light (With surge voltage suppressor)	30	●
DN	DIN terminal without connector (With surge voltage suppressor)	30	●
WN	M12 connector/Without connector cable (With surge voltage suppressor)* ²	30	●

4 Seal material

Symbol	Seal material
N	NBR
F	FKM
E	EPDM

5 Orifice diameter and port size

Symbol	Orifice diameter [mmø]	Port size	Size
302	3.2	1/4	●
303	3.2	3/8	●

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

7 Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Option

Symbol	Option
Nil	None
B	With bracket* ¹ (Stainless steel)

*1 Refer to page 102 for bracket assembly part nos.

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Size	Port size	Orifice diameter [mmø]	Flow rate characteristics* ¹			Max. operating pressure differential [MPa]	Model	Weight* ² [g]	
			C	b	Cv			Stainless steel body* ³	Brass body
30	1/4	3.2	1.2	0.43	0.33	3.0	JSX31H-502	450	490
	3/8	3.2	1.2	0.43	0.33	3.0	JSX31H-503	450	520

*1 The flow rate characteristics of this product vary.

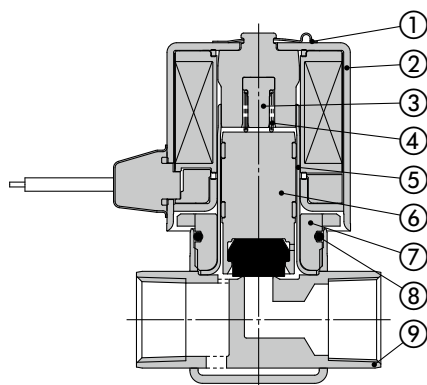
*2 Add 50 g for the conduit type, 30 g for the DIN terminal type, and -5 g for the M12 connector type.

*3 The values were calculated based on the combination of an Rc or NPT thread and a grommet with PCB. Add 30 g for the G thread (port size 3/8) type.

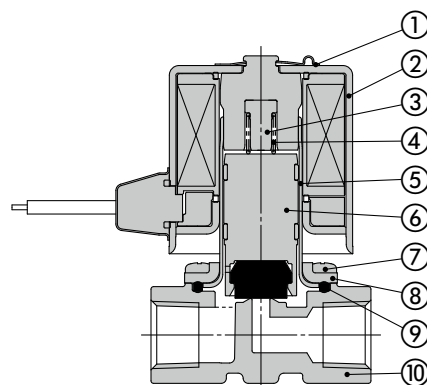
Construction

JSX30H

Body material: Stainless steel



Body material: Brass



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
7	Nut	Stainless steel
8	Gasket	NBR (FKM, EPDM)
9	Body	Stainless steel

Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM, EPDM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	NBR (FKM, EPDM)
10	Body	Brass

Common Specifications

Size			30
Valve specifications	Valve construction		Direct operated poppet
	Valve type		Normally closed (N.C.)
	Fluid and fluid temperature		Air: -10 to 60°C (Dew point temperature: -10°C or less)
	Withstand pressure		4.5 MPa
	Max. system pressure		3.0 MPa
	Ambient temperature		-20 to 60°C
	Valve leakage/External leakage*1	Air	1 cm ³ /min (ANR) or less
	Mounting orientation		Unrestricted
	Enclosure*2		IP67 (IP65 for the DIN terminal)
	Standards*3		CE/UKCA
	Operating environment		Location without the presence of corrosive gases or explosive gases
	Body material		Stainless steel, Brass
Coil specifications	Seal material		NBR, FKM, EPDM
	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V
		DC	12 V, 24 V
	Allowable voltage fluctuation		±10% of the rated voltage
	Allowable leakage voltage	AC	5% or less of the rated voltage
		DC	2% or less of the rated voltage
	Apparent power (Holding)*4, *5		16 VA
	Power consumption (Holding)*4		13 W
	Temperature rise*6		70/65°C

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

*4 Power consumption: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

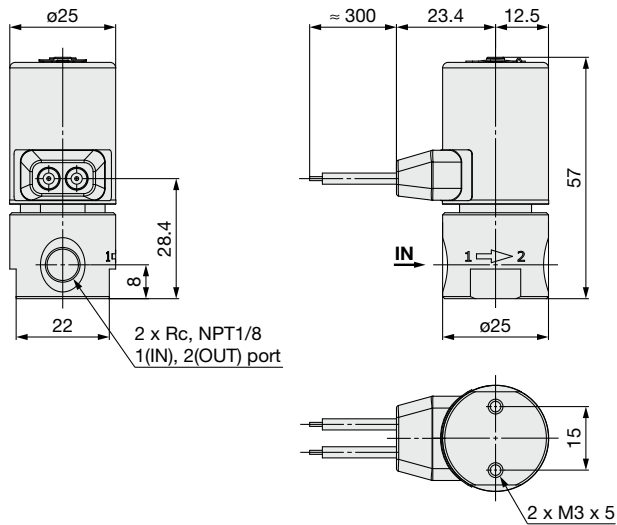
*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

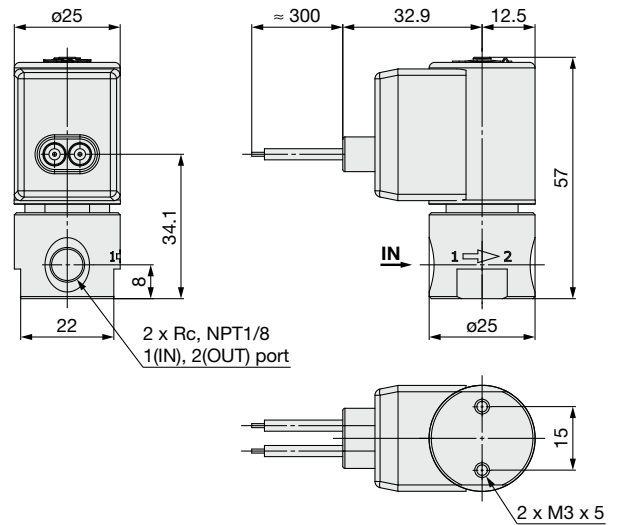
Dimensions: JSX **10, 10U, 10V** Port Size **1/8** Body Material **Stainless Steel, Brass**

G: Grommet

* JSX10 only

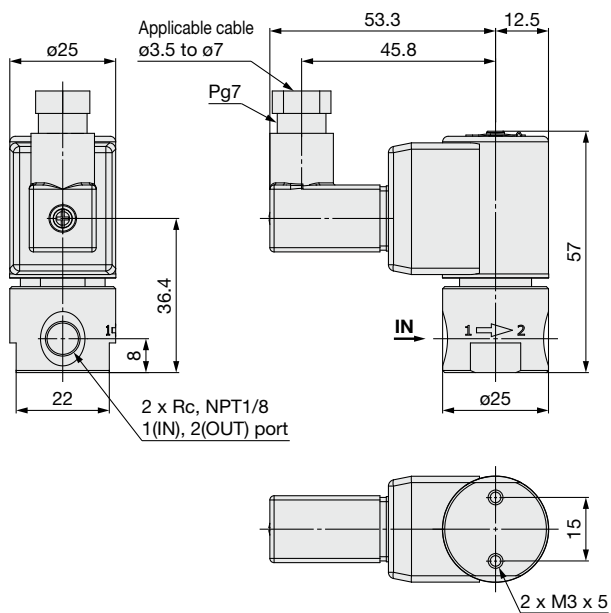


GS: Grommet with PCB



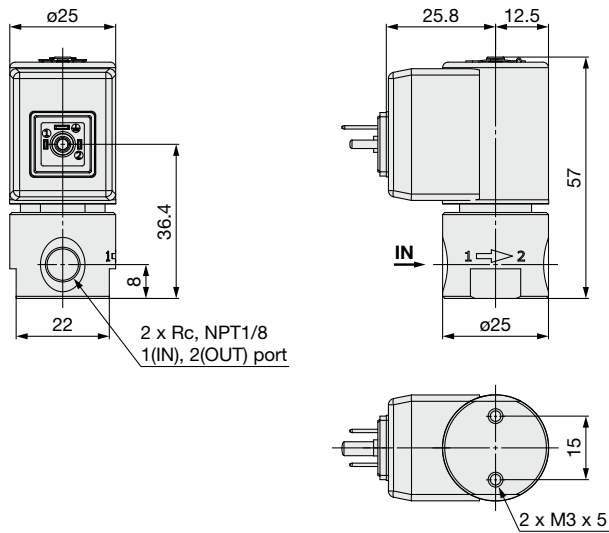
DS: DIN terminal

DZ: DIN terminal with light

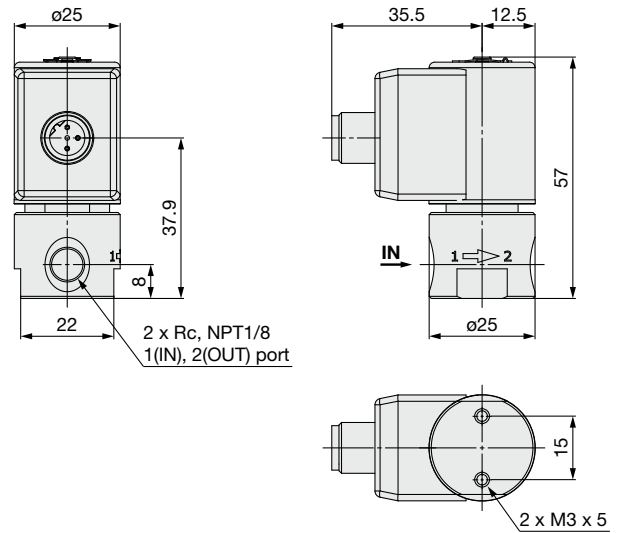


Dimensions: JSX **10, 10U, 10V** Port Size **1/8** Body Material **Stainless Steel, Brass**

DN: DIN terminal without connector

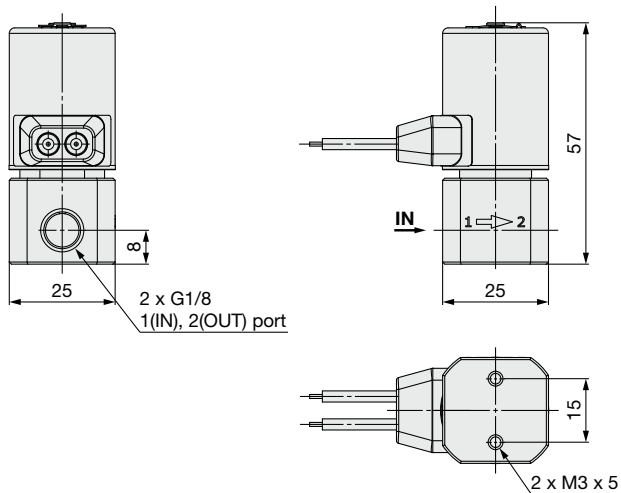


WN: M12 connector



G thread type

- * The dimensions other than those below are the same as those of the Rc type.
- * The grommet type is only available for the JSX10.



JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance Table

UL-compliance Table

Option

Replacement Parts

Glossary of Terms

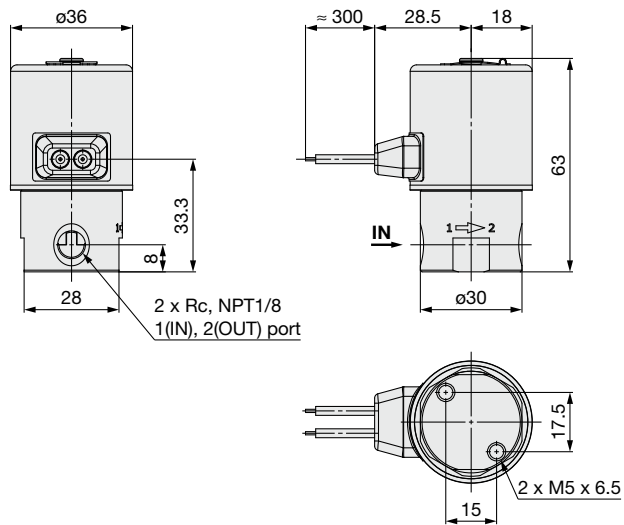
Flow Rate Characteristics

Specific Product Precautions

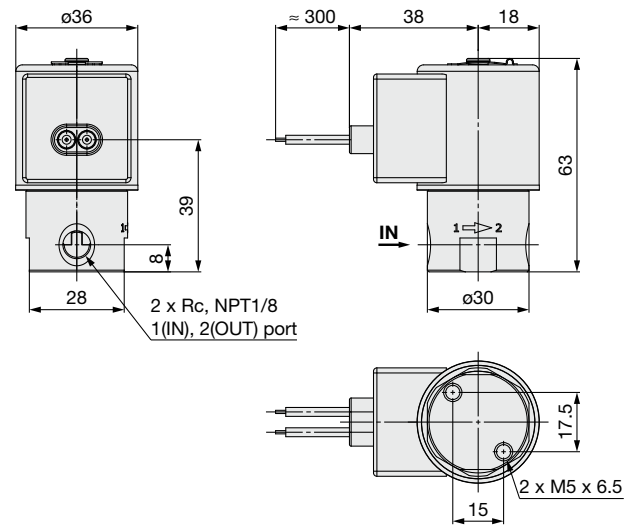
Dimensions: JSX**20, 20U, 20V** Port Size **1/8** Body Material **Stainless Steel**

G: Grommet

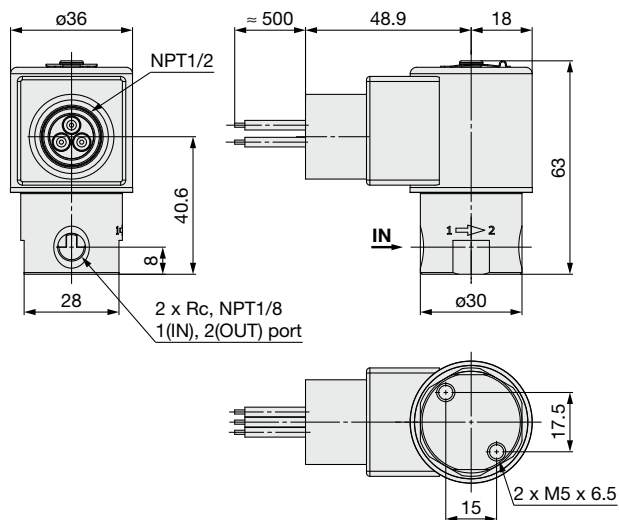
* JSX20 only



GS: Grommet with PCB

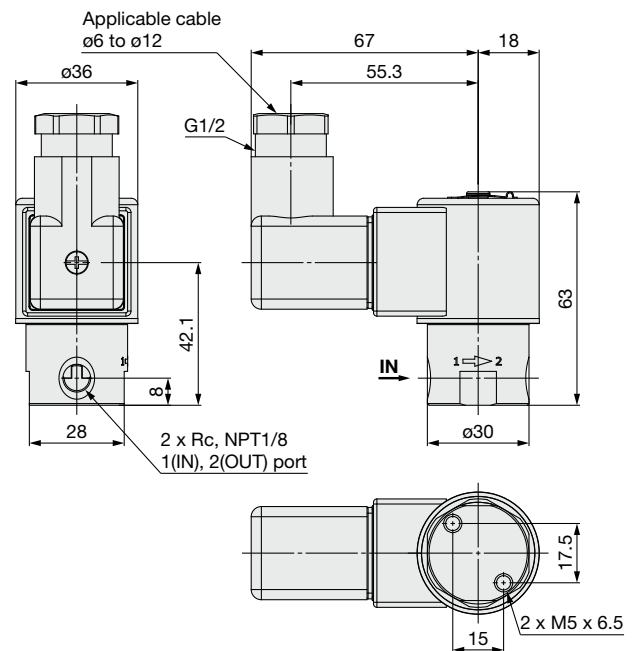


CS: Conduit



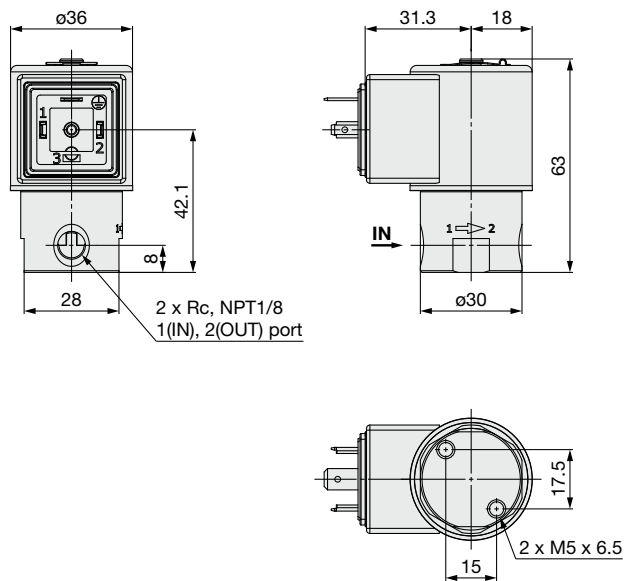
DS: DIN terminal

DZ: DIN terminal with light

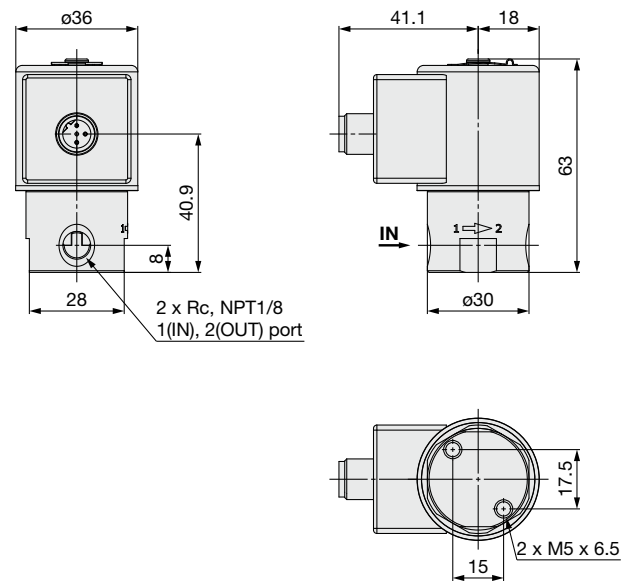


Dimensions: **JSX20, 20U, 20V** Port Size **1/8** Body Material **Stainless Steel**

DN: DIN terminal without connector

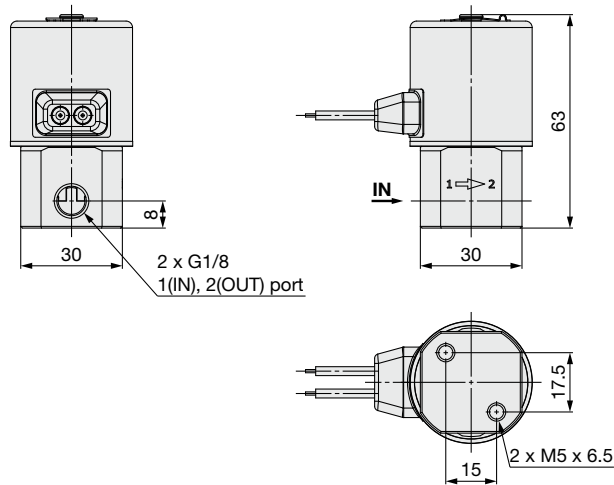


WN: M12 connector



G thread type

- * The dimensions other than those below are the same as those of the Rc type.
- * The grommet type is only available for the JSX20.



JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

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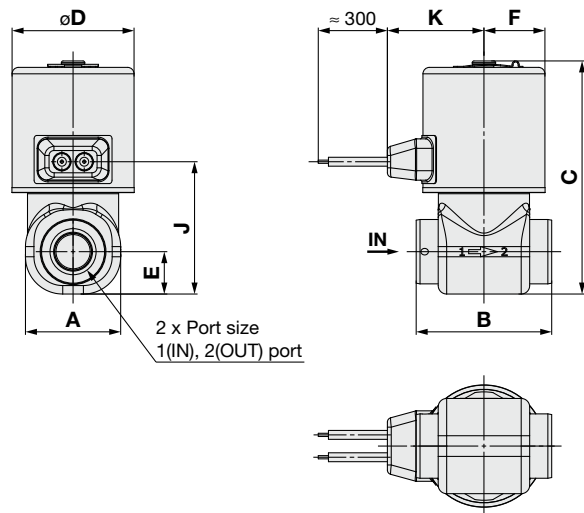
JSX Series

JSX20, 30, 20U, 30U

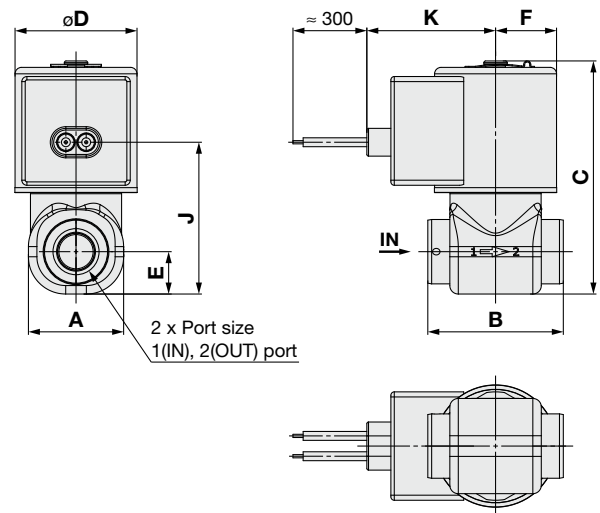
Dimensions: JSX20V, 30V, 30H Port Size 1/4, 3/8 Body Material Stainless Steel

G: Grommet

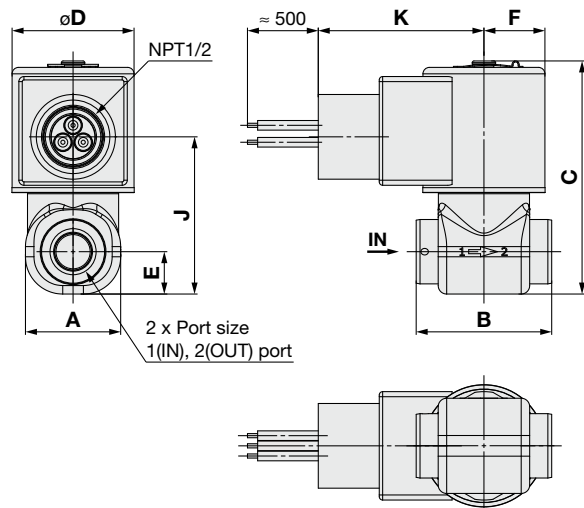
* JSX20 and 30 only



GS: Grommet with PCB

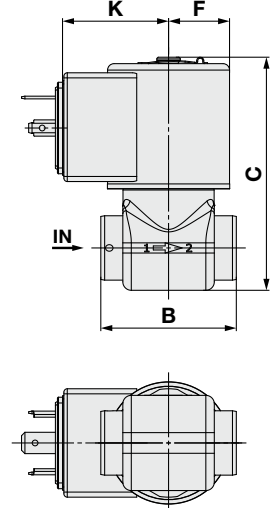
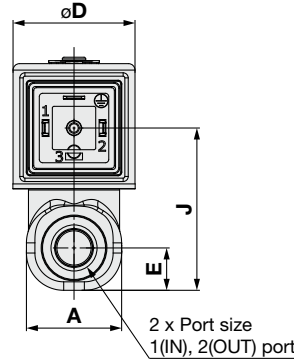
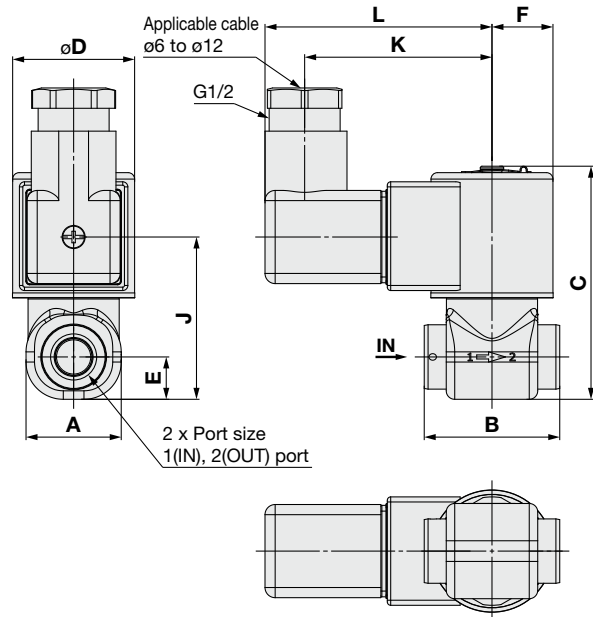
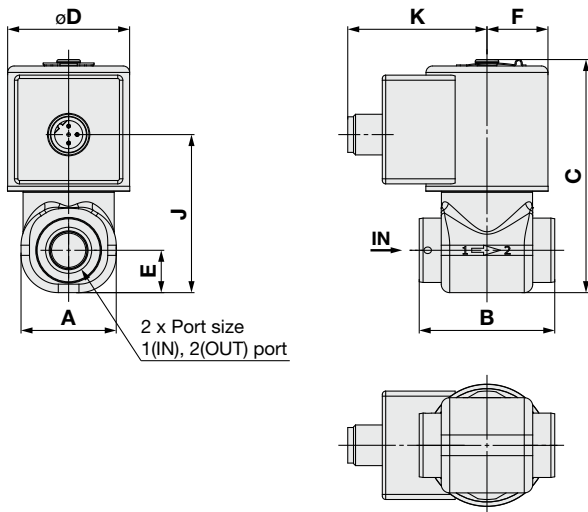


CS: Conduit



							[mm]
Size	Port size	A	B	C	D	E	F
20	1/4	28.1	40	69	36	12.5	18
	3/8		48			14	
	G3/8			72			
30	1/4	28.1	40	78	42	12.5	21
	3/8		48			14	
	G3/8			81			

Size	Port size	Grommet		Grommet with PCB		Conduit	
		J	K	J	K	J	K
20	1/4	39	28.5	44.8	38	46.4	48.9
	3/8						
	G3/8			47.8		49.4	
30	1/4	40	31.1	45.8	41	47.4	51.9
	3/8						
	G3/8	43		48.8		50.4	

JSX20, 30, 20U, 30U
Dimensions: JSX20V, 30V, 30H Port Size **1/4, 3/8** Body Material **Stainless Steel**
DS: DIN terminal
DZ: DIN terminal with light
DN: DIN terminal without connector

WN: M12 connector


[mm]							
Size	Port size	A	B	C	D	E	F
20	1/4	28.1	40	69	36	12.5	18
	3/8		48	72		14	
	G3/8						
30	1/4	28.1	40	78	42	12.5	21
	3/8		48	81		14	
	G3/8						

Size	Port size	DIN terminal			DIN terminal without connector		M12 connector	
		J	K	L	J	K	J	K
20	1/4	47.9	55.3	67	47.9	31.3	46.7	41.1
	3/8				50.9		49.7	
	G3/8							
30	1/4	48.9	58.3	70	48.9	34.3	47.7	44.1
	3/8				51.9		50.7	
	G3/8							

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JSX Series

Port Size Normally Closed (N.C.)
Normally Open (N.O.)

1/8, 1/4, 3/8

Body Material **Brass**

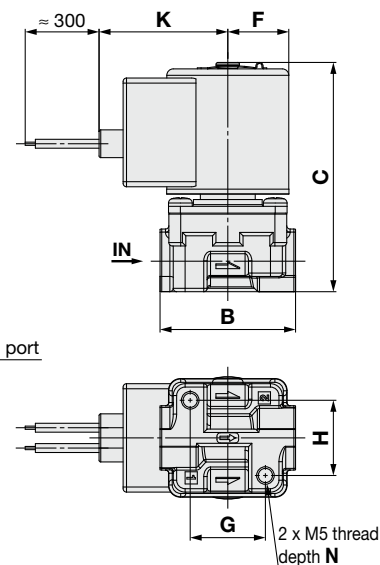
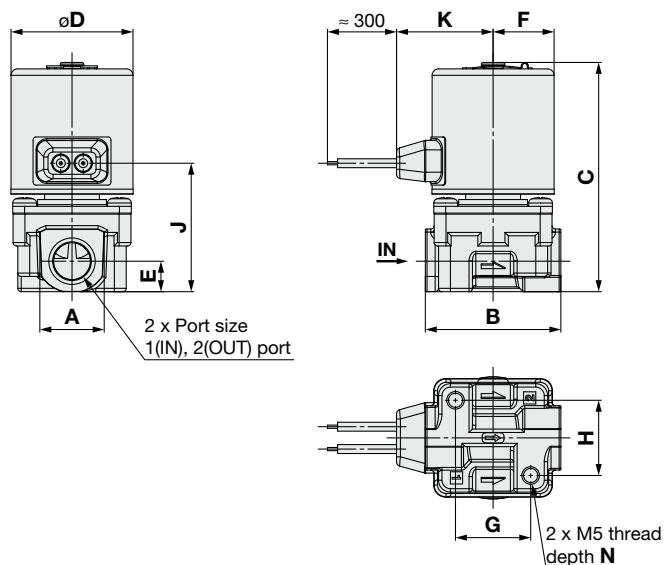
Body Material **Stainless Steel, Brass**

Dimensions: **JSX20, 30, 20U, 30U, 20V, 30V, 30H**

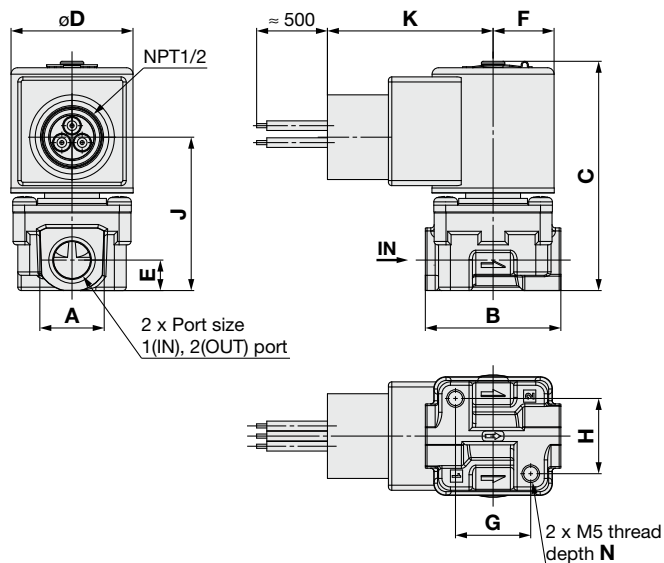
G: Grommet

* JSX20 and 30 only

GS: Grommet with PCB



CS: Conduit



[mm]										
Size	Port size	A	B	C	D	E	F	G	H	N
20	1/8	14	30	69.2 (79.1)	36	9	18	15	17.5	6.4
	1/4	19	40	67.7 (77.6)		11		22.2	22.2	7.6
	3/8	22	48	70.7 (80.6)				19	20.6	6
30	1/8	14	30	— (87.6)	42	9	21	15	17.5	6.4
	1/4	19	40	76.7 (86.1)		11		22.2	22.2	7.6
	3/8	22	48	79.7 (89.1)				19	20.6	6

Size	Port size	Grommet		Grommet with PCB		Conduit	
		J	K	J	K	J	K
20	1/8	39.4 (49.4)	28.5	45.2 (55.1)	38	46.8 (56.7)	48.9
	1/4	37.9 (47.9)		43.7 (53.6)		45.3 (55.2)	
	3/8	40.9 (50.9)		46.7 (56.6)		48.3 (58.2)	
30	1/8	— (49.9)	31.1	— (55.6)	41	— (57.2)	51.9
	1/4	39 (48.4)		44.7 (54.1)		46.3 (55.7)	
	3/8	42 (51.4)		47.7 (57.1)		49.3 (58.7)	

* (): Denotes the Normally Open (N.O.) dimensions

Direct Operated 2-Port Solenoid Valve **JSX Series**

Port Size Normally Closed (N.C.)
Normally Open (N.O.)

1/8, 1/4, 3/8

Body Material Brass

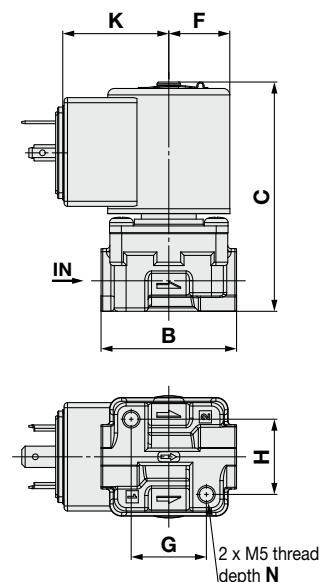
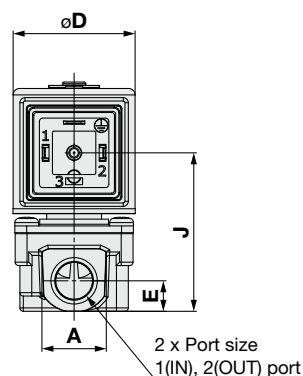
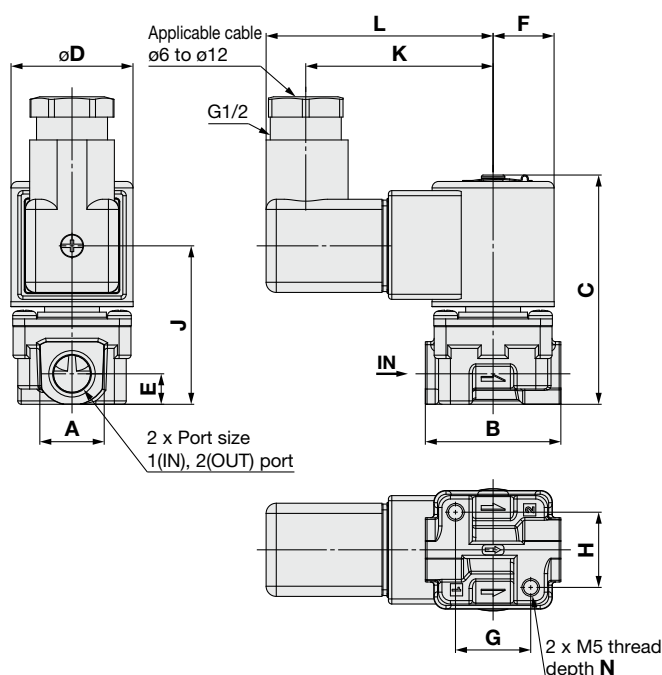
Body Material Stainless Steel, Brass

Dimensions: JSX20, 30, 20U, 30U, 20V, 30V, 30H

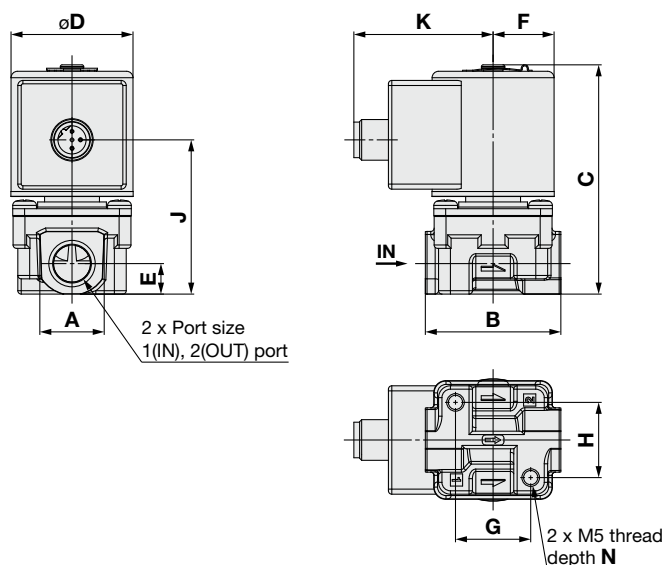
DS: DIN terminal

DZ: DIN terminal with light

DN: DIN terminal without connector



WN: M12 connector



[mm]										
Size	Port size	A	B	C	D	E	F	G	H	N
20	1/8	14	30	69.2 (79.1)	36	9	18	15	17.5	6.4
	1/4	19	40	67.7 (77.6)		11		22.2	22.2	7.6
	3/8	22	48	70.7 (80.6)				19	20.6	6
30	1/8	14	30	— (87.6)	42	9	21	15	17.5	6.4
	1/4	19	40	76.7 (86.1)		11		22.2	22.2	7.6
	3/8	22	48	79.7 (89.1)				19	20.6	6

Size	Port size	DIN terminal			DIN terminal without connector		M12 connector	
		J	K	L	J	K	J	K
20	1/8	48.3 (58.2)	55.3	67	48.3 (58.2)	31.3	47 (57)	41.1
	1/4	46.8 (56.7)			46.8 (56.7)		45.5 (55.5)	
	3/8	49.8 (59.7)			49.8 (59.7)		48.5 (58.5)	
30	1/8	— (58.7)	58.3	70	— (58.7)	34.3	— (57.5)	44.1
	1/4	47.8 (57.2)			47.8 (57.2)		46.6 (56)	
	3/8	50.8 (60.2)			50.8 (60.2)		49.6 (59)	

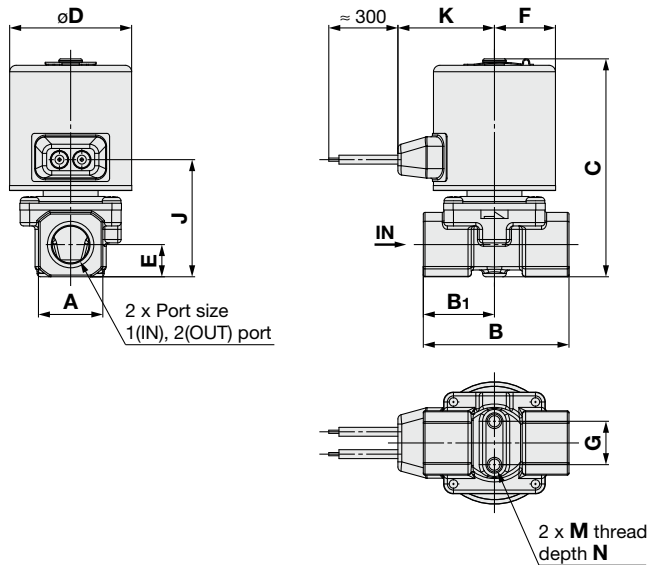
* (): Denotes the Normally Open (N.O.) dimensions

JSX Series

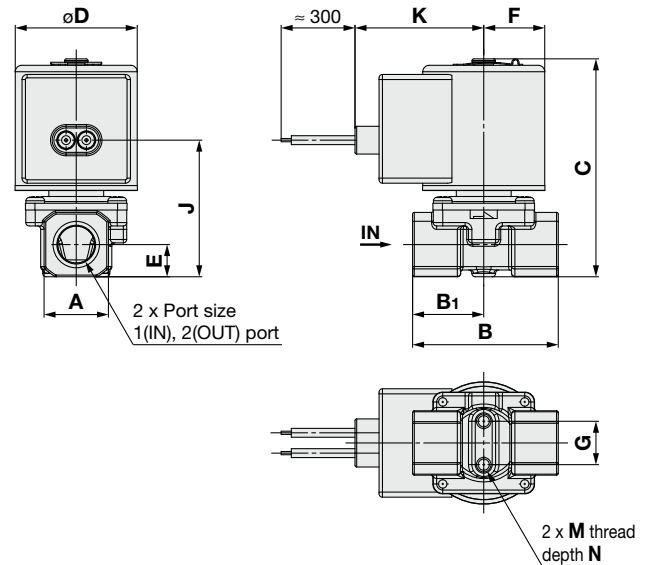
JSX20, 30

Dimensions: JSX20U, 30U Port Size 1/8, 1/4, 3/8 Body Material Aluminum

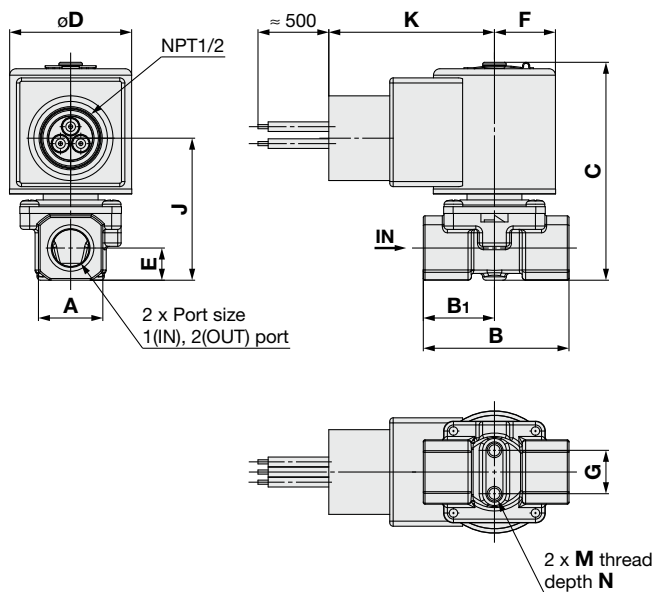
G: Grommet



GS: Grommet with PCB



CS: Conduit

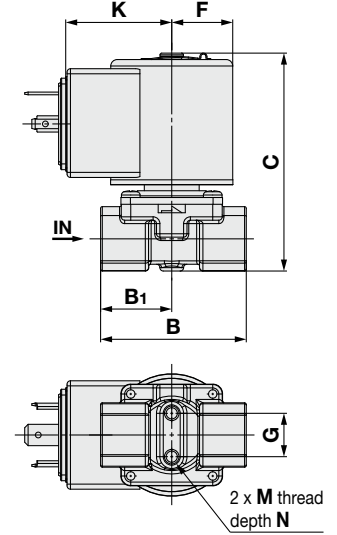
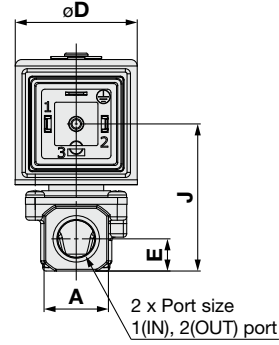
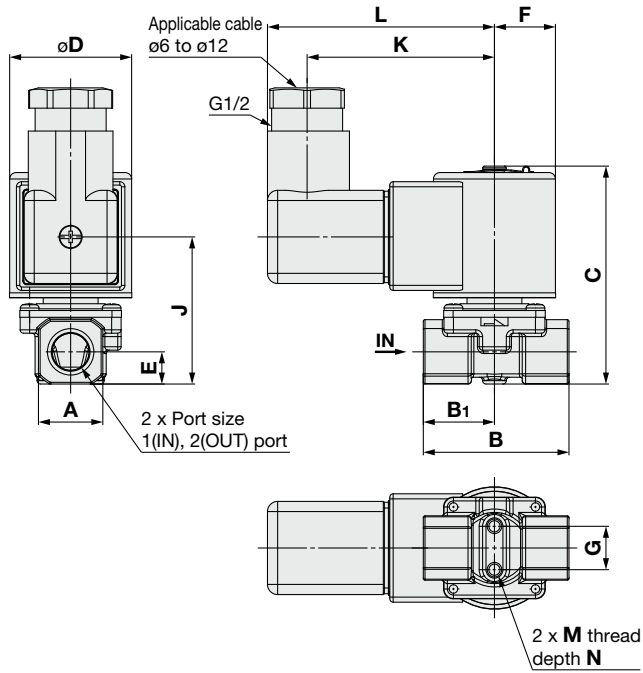
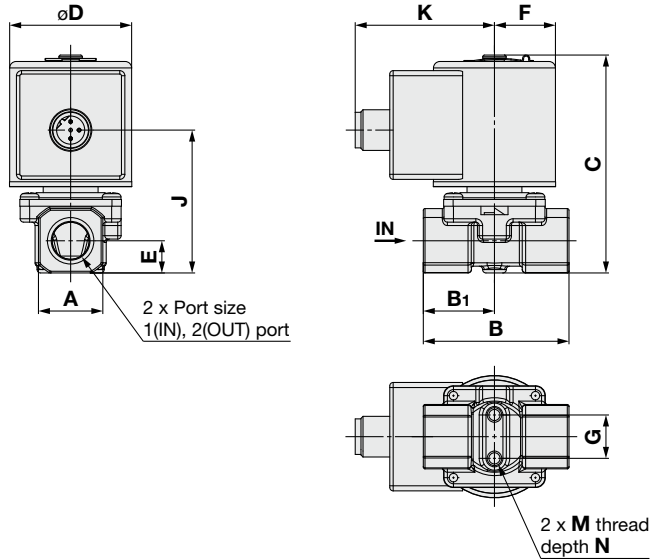


[mm]											
Size	Port size	A	B	B ₁	C	D	E	F	G	M	N
20	1/8, 1/4	19	43	21	64.3	36	9.5	18	12.8	M4	6
30	1/4, 3/8	24	45	22.5	80.7	42	12	21	19	M5	8

Size	Port size	Grommet		Grommet with PCB		Conduit	
		J	K	J	K	J	K
20	1/8, 1/4	34.6	28.5	40.3	38	41.9	48.9
30	1/4, 3/8	43	31.1	48.7	41	50.3	51.9

JSX20, 30
Dimensions: JSX20U, 30U Port Size **1/8, 1/4, 3/8** Body Material **Aluminum**
DS: DIN terminal

DZ: DIN terminal with light

DN: DIN terminal without connector

WN: M12 connector


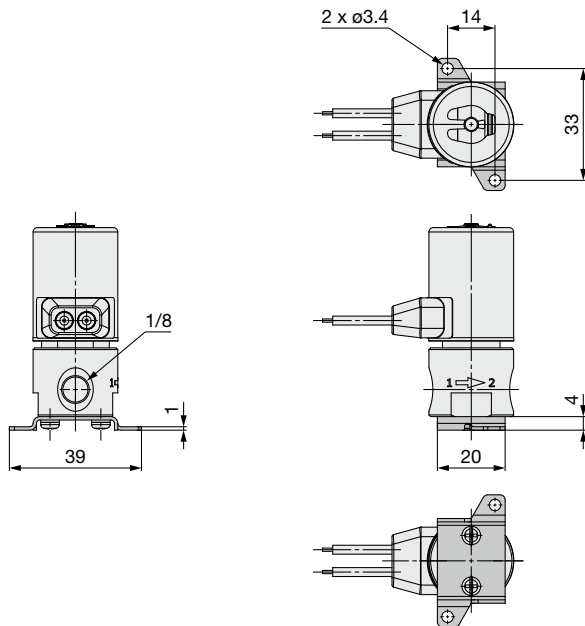
Size	Port size	A	B	B ₁	C	D	E	F	G	M	N
20	1/8, 1/4	19	43	21	64.3	36	9.5	18	12.8	M4	6
30	1/4, 3/8	24	45	22.5	80.7	42	12	21	19	M5	8

Size	Port size	DIN terminal			DIN terminal without connector		M12 connector	
		J	K	L	J	K	J	K
20	1/8, 1/4	43.4	55.3	67	43.4	31.3	42.2	41.1
30	1/4, 3/8	51.8	58.3	70	51.8	34.3	50.6	44.1

Dimensions: Bracket Options

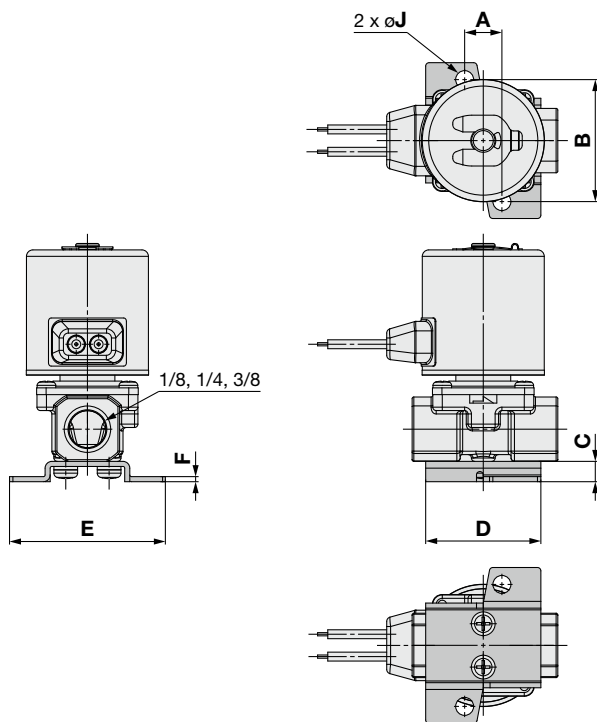
JSX10, 10U, 10V Body Material Stainless Steel, Brass

* The grommet type is only available for the JSX10.



JSX20, 30 JSX20U, 30U Body Material Aluminum

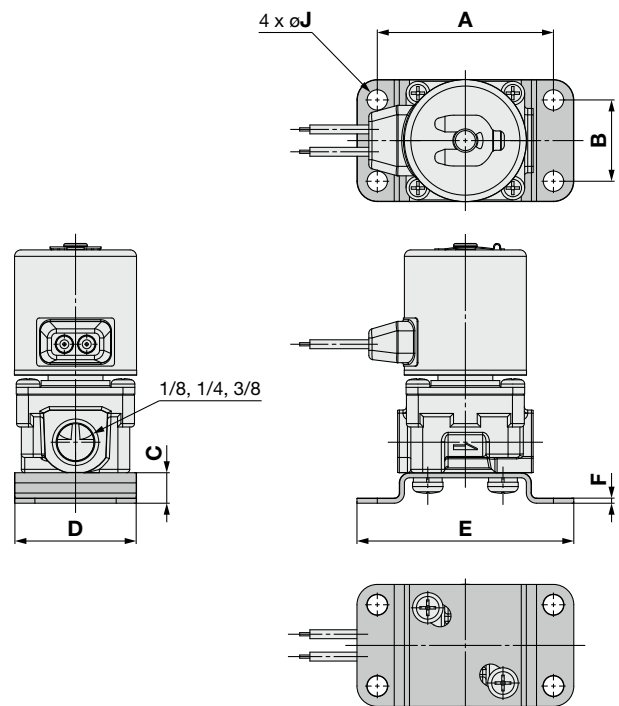
* The grommet type is only available for the JSX20 and 30.



Body Material: Aluminum							[mm]
Size	Port size	A	B	C	D	E	F
20	1/8, 1/4	11	36	6	34	46	1.5
30	1/4, 3/8	13	46	7	40	56	1.5
							øJ
							5.3

JSX20, 30, 20U, 30U JSX20V, 30V, 30H Body Material Brass

* The grommet type is only available for the JSX20 and 30.



Body Material: Brass							[mm]
Size	Port size	A	B	C	D	E	F
20	1/8	52	24	9	36	64	1.5
20, 30	1/4, 3/8	52	24	9	36	64	1.5
							øJ
							6

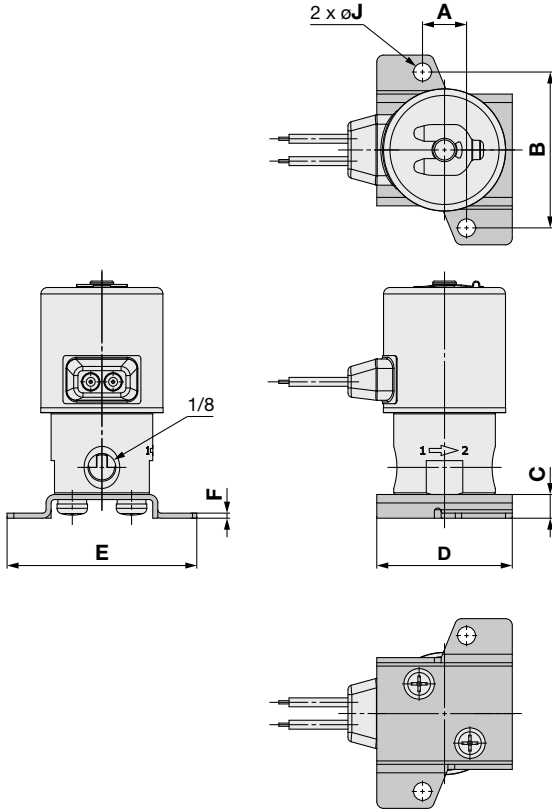
Dimensions: Bracket Options

JSx20, 20V

Body Material **Stainless Steel**

* The grommet type is only available for the JSX20 and 30.

(Port size 1/8 type)



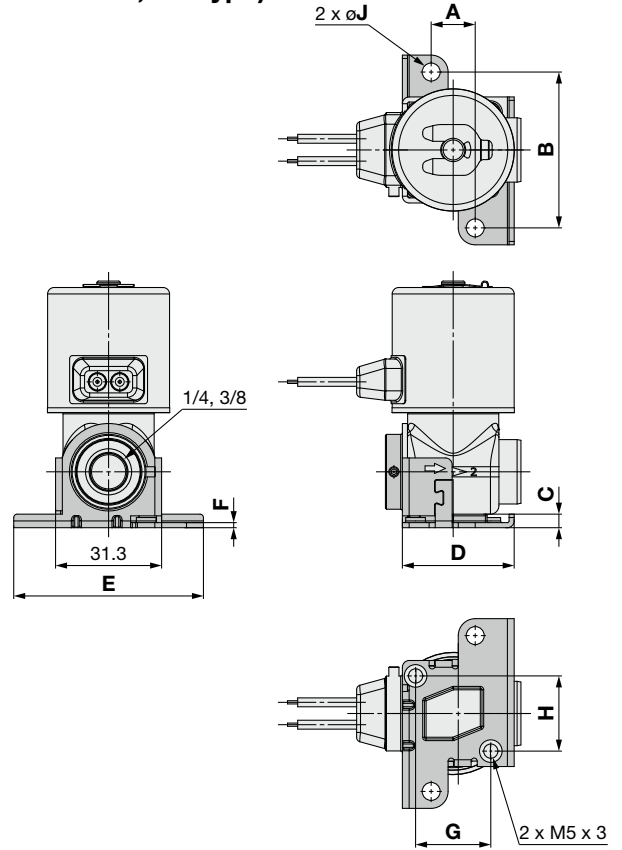
JSx20, 30, 20U, 30U

JSx20V, 30V, 30H

Body Material **Stainless Steel**

* The grommet type is only available for the JSX20 and 30.

(Port size 1/4, 3/8 type)



[mm]										
Size	Port size	A	B	C	D	E	F	G	H	øJ
20	1/8	13	46	7	40	56	1.5	—	—	5.3
20, 30	1/4, 3/8	13	46	4	33	56	1.5	22.2	22.2	5.3
	G3/8							19	20.6	

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

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Steam Type



Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

For **Steam**
Heated Water

Direct Operated 2-Port Solenoid Valve

JSX□□S Series

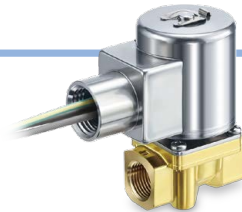
Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Aluminum	Stainless Steel	Brass	Stainless Steel	Brass	Stainless Steel	Brass
Normally Closed (N.C.)	Normally Closed (N.C.)	Normally Closed (N.C.)	Normally Open (N.O.)	High Flow/Power Saving Type	High Flow/Power Saving Type	Vacuum Type	High Pressure Type	Steam Type					
▶ p. 13	▶ p. 15	▶ p. 17	▶ p. 19	▶ p. 21	▶ p. 23	▶ p. 25	▶ p. 39						

How to Order

JSX 3 1 S - S F 502 R - 5 CS - D - B

1 2 3 4 5 6 7 8 9 10

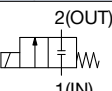
S Steam type



1 Size

Symbol	Size
3	30

2 Valve type

Symbol	Valve type
1	N.C. 

3 Body material

Symbol	Body material
S	Stainless steel
C	Brass

4 Seal material

Symbol	Seal material
F	FKM

5 Orifice diameter and port size

Symbol	Orifice diameter [mm]	Port size	Size
502	5.6	1/4	●
503	5.6	3/8	●
702	7.1	1/4	●
703	7.1	3/8	●

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

7 Rated voltage

AC			
Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC
2	200 VAC	8	48 VAC
3	120 (110) VAC	B	24 VAC
4	220 VAC	J	230 VAC

DC

Symbol	Rated voltage
5	24 VDC
6	12 VDC

8 Electrical entry

Symbol	Electrical entry	Size	CE/UKCA-compliant
CS	Conduit (With surge voltage suppressor) 	30	Refer to page 84.

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Option

Symbol	Option
Nil	None
B	With bracket*1 (Stainless steel)

*1 Refer to page 102 for bracket assembly part nos.

Flow Rate Characteristics

Size	Port size	Orifice diameter [mm]	Flow rate characteristics*1					Max. operating pressure differential [MPa]	Model	Weight [g]	
			Air			Water, Oil				Stainless steel body*2	Brass body
			C	b	Cv	Kv	Conversion Cv				
30	1/4	5.6	2.62	0.43	0.73	0.63	0.73	1.0	JSX31S-  502	500	540
		7.1	3.15	0.44	0.88	0.76	0.88	0.5	JSX31S-  702	500	540
	3/8	5.6	2.62	0.43	0.73	0.63	0.73	1.0	JSX31S-  503	500	570
		7.1	3.15	0.44	0.88	0.76	0.88	0.5	JSX31S-  703	500	570

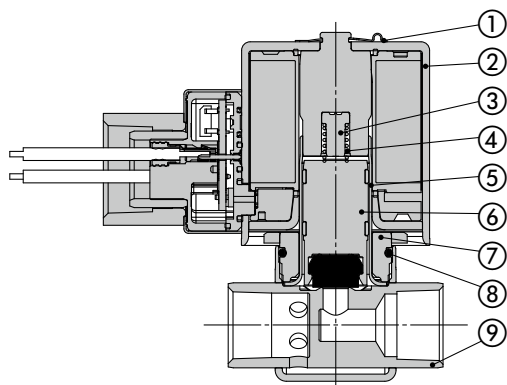
*1 The flow rate characteristics of this product vary.

*2 The values were calculated based on the combination of an Rc or NPT thread and a grommet with PCB. Add 30 g for the G thread (port size 3/8) type.

Construction

JSX30S

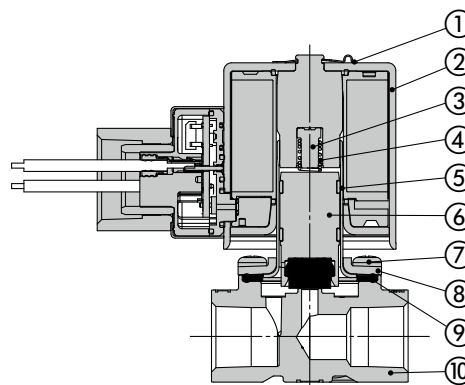
Body material: Stainless steel



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS (FKM)
7	Nut	Stainless steel
8	Gasket	FKM
9	Body	Stainless steel

Body material: Brass



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS (FKM)
7	Mounting screw	Fe
8	Bonnet	Stainless steel
9	Gasket	FKM
10	Body	Brass

Common Specifications

Size			30
Valve specifications	Valve construction		Direct operated poppet
	Valve type		Normally closed (N.C.)
	Fluid and fluid temperature		Steam: 183°C or less
			Heated water: 99°C or less
	Withstand pressure		2.0 MPa
	Max. system pressure		1.0 MPa
	Ambient temperature		-20 to 60°C
	Valve leakage/ External leakage*1	Steam	1.0 cm³/min or less
		Heated water	0.1 cm³/min or less
	Mounting orientation		Unrestricted
	Enclosure*2		IP67
	Standards*4		CE/UKCA
	Operating environment		Location without the presence of corrosive gases or explosive gases
Coil specifications	Body material		Stainless steel, Brass
	Seal material		FKM
	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V
		DC	12 V, 24 V
	Allowable voltage fluctuation		±10% of the rated voltage
	Allowable leakage voltage	AC	5% or less of the rated voltage
		DC	2% or less of the rated voltage
	Apparent power (Holding)*5, *6		16 VA
	Power consumption (Holding)*5		13 W
	Temperature rise*7		100°C

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Protect the lead wire part with a wiring conduit.

*4 Standards compliance varies depending on the model. For details, refer to page 84.

*5 Power consumption: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*6 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*7 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

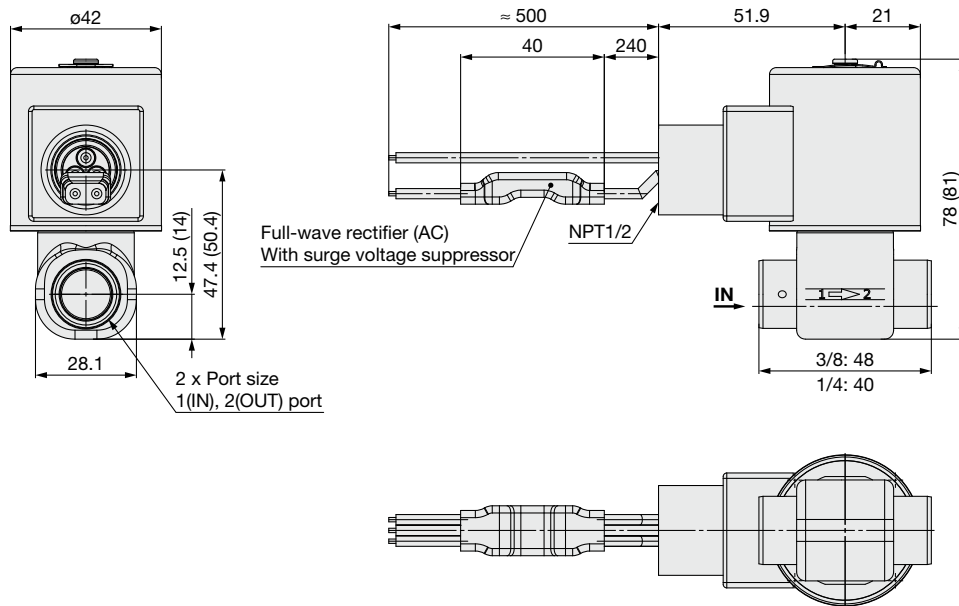
Be sure to read the "Specific Product Precautions" before handling the product.

JSX Series

Dimensions: JSX**30S** Port Size **1/4, 3/8** Body Material **Stainless Steel, Brass**

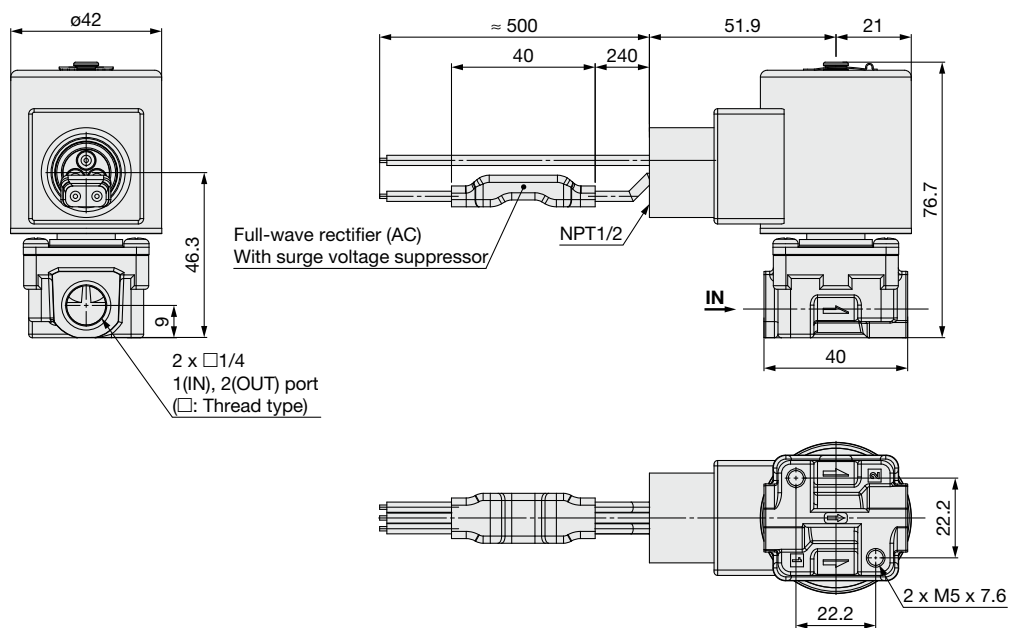
JSX**30S** Body Material **Stainless Steel**

CS: Conduit



JSX**30S** Body Material **Brass**

CS: Conduit



Specific Product
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UL-compliance
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CE/UKCA-compliance
Table

JSXM

JSXZ

JSXR

JSXP

JSXD

JSX

Pilot Operated 2-Port Solenoid Valve *JSXD Series*

CE UK
CA
Differs depending on the
voltage and electrical entry. For
details, refer to page 84.

UL US
LISTED
Refer to pages 86 to 89 for details.

RoHS

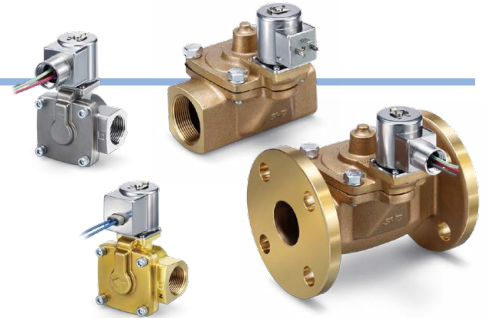
Stainless Steel	Brass	Bronze	Aluminum
Normally Closed (N.C.)			
▶ p. 43			

Stainless Steel	Brass	Bronze
Normally Open (N.O.)		
▶ p. 47		

How to Order

JSXD **3** **1** - **C** **N** **02** **R** - **5** **G** - **D** - **B**

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩



① Size

Symbol	Size
3	30
4	40
5	50
6	60
7	70
8	80
9	90

② Valve type

Symbol	Valve type
1	N.C.

* Refer to page 47 for N.O. type.

③ Body material

Symbol	Body material	Size		
		30	40, 50, 60	70, 80, 90
C	Brass	●	●	—
S	Stainless steel	●	●	—
B	Bronze	—	—	●
A	Aluminum	●	—	—

④ Seal material

Symbol	Seal material
N	NBR
F	FKM
E *1	EPDM

*1 Cannot be used in combination with the aluminum body

⑤ Port size

Symbol	Connection	Port size	Size						
			30	40	50	60	70	80	90
02	Thread	1/4	●	—	—	—	—	—	—
03		3/8	●	●	—	—	—	—	—
04		1/2	●	●	—	—	—	—	—
06		3/4	—	—	●	—	—	—	—
10		1	—	—	—	●	—	—	—
12		1 1/4	—	—	—	—	●	—	—
14	Flange	1 1/2	—	—	—	—	—	●	—
20		2	—	—	—	—	—	—	●
32		32A	—	—	—	—	●	—	—
40		40A	—	—	—	—	—	●	—
50		50A	—	—	—	—	—	—	●

⑧ Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant	UL Standards
G	Grommet*1	Refer to page 84.	Refer to pages 86 to 89.
GS	Grommet with PCB (With surge voltage suppressor)		
CS	Conduit (With surge voltage suppressor)		
DS	DIN terminal (With surge voltage suppressor)		
DZ	DIN terminal with light (With surge voltage suppressor)		
DN	DIN terminal without connector (With surge voltage suppressor)		
WN	M12 connector without cable (With surge voltage suppressor)*2		

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

⑥ Thread type

Symbol	Thread type	Connection
R	Rc	Thread
N	NPT	
F	G	
Nil	—	Flange

⑦ Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

⑨ Oil-free option

Symbol	Option
Nil	None
D	Oil-free

⑩ Bracket

Symbol	With bracket	Size		
		30	40, 50, 60	70, 80, 90
Nil	None	●	●	●
B	With bracket	●	●	—*1

*1 Sizes 70 to 90 are not available with a bracket.

Flow Rate Characteristics

Size	Body material	Port size	Orifice diameter [mmø]	Flow rate characteristics*1						Min. operating pressure differential [MPa]	Max. operating pressure differential [MPa]	Model	Weight*2 [g]
				Air				Water, Oil					
				C [dm³/(s·bar)]	b	Cv	Effective area [mm²]	Kv	Conversion Cv				
30	Aluminum	1/4	10	8.5	0.35	2.0	—	—	—	0.02	1.0	JSXD31-A□02	410
		3/8		9.2		2.4						JSXD31-A□03	410
		1/2		9.2		2.4						JSXD31-A□04	410
	Brass Stainless steel	1/4		8.5	0.35	2.0		1.6	1.9			JSXD31-S□02	500
		3/8		9.2		2.4		2.0	2.4			JSXD31-S□03	500
		1/2		9.2		2.4		2.0	2.4			JSXD31-S□04	500
				Brass		3/8		18	0.35			5.0	3.9
Stainless steel	1/2	20	5.5	4.6	5.5	JSXD41-S□04	720						
40	Brass	3/8	15	38	0.30	9.5	8.2	9.5	0.03	1.0	JSXD51-S□06	880	
Stainless steel	1/2	20		5.5		4.6	5.5	JSXD41-S□04			720		
50	Brass/Stainless steel	3/4	20	38	—		8.2	9.5	0.03	1.0	JSXD51-S□06	880	
60	Brass/Stainless steel	1	25	225			11.0	13.0			JSXD61-S□10	1460	
70	Bronze	1 1/4, 32A	35	415			19.6	23.0			JSXD71-B□(12, 32)	5500/3000	
80	Bronze	1 1/2, 40A	40	560			26.4	31.0			JSXD81-B□(14, 40)	6900/4100	
90	Bronze	2, 50A	50	880			42.8	49.0			JSXD91-B□(20, 50)	8500/5500	

*1 The flow rate characteristics of this product vary.

*2 Indicates rate of grommet type

Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.
For sizes 70, 80, and 90, the weight on the left is for the flange type, and the weight on the right is for the thread type.

Applicable Fluid Checklist

Applicable fluid	Seal material		
	NBR	FKM	EPDM
Air	●	●	●
Water	●	●	●
Oil	—	●	—

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use. If something is not clear, please contact SMC.

Common Specifications

Size		30		40	50	60	70	80	90
Valve specifications	Body material		Aluminum	Brass, Stainless steel	Brass, Stainless steel			Bronze	
	Valve construction		Pilot operated diaphragm						
	Valve type		Normally closed (N.C.)						
	Fluid and fluid temperature	Air*1	-10 to 60°C						
		Water, Oil	—	Water: 1 to 60°C (No freezing), Oil: -5 to 60°C (Kinematic viscosity: 50 mm²/s or less)					
	Withstand pressure		2 MPa						
	Max. system pressure		1 MPa						
	Ambient temperature		-20 to 60°C						
	Valve leakage*2	Air	15 cm³/min (ANR) or less	2 cm³/min (ANR) or less			10 cm³/min (ANR) or less		
		Water, Oil	—	0.2 cm³/min or less			1 cm³/min or less		
	External leakage*2	Air	15 cm³/min (ANR) or less	1 cm³/min (ANR) or less					
		Water, Oil	—	0.1 cm³/min or less					
	Mounting orientation		Unrestricted						
	Enclosure*3		IP67 (IP65 for the DIN terminal)						
Standards*4		CE/UKCA, UL Recognized, UL Listed							
Operating environment		Location without the presence of corrosive gases or explosive gases							
Seal material		NBR, FKM, EPDM							
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V						
		DC	12 V, 24 V						
	Allowable voltage fluctuation		±10% of the rated voltage						
	Allowable leakage voltage	AC	5% or less of the rated voltage						
		DC	2% or less of the rated voltage						
	Apparent power*5, *6	AC	8 VA				9.5 VA		
	Power consumption*5	DC	6 W				8 W		
	Temperature rise*7		AC/DC	70/65°C					

*1 Dew point temperature: -10°C or less

*2 Leakage: The value at a differential pressure the same as or higher than the min. operating pressure differential, and an ambient temperature of 20°C

*3 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*4 Standards compliance varies depending on the model. For details, refer to pages 84, 86 through to 89.

*5 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*6 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

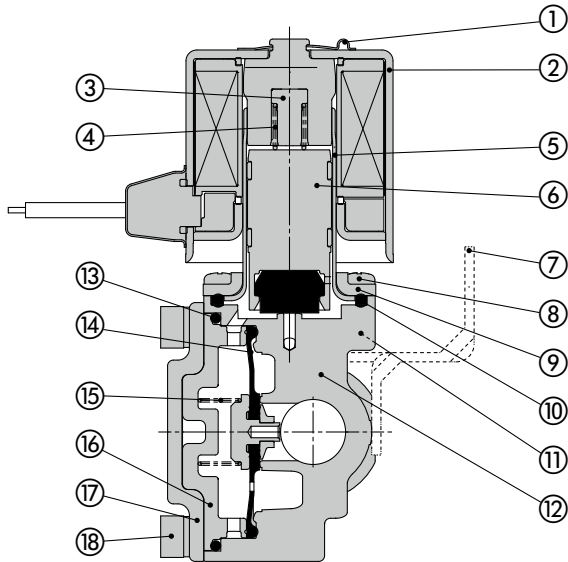
*7 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

Construction

JSXD30, Normally closed (N.C.)

Body material: Brass, Stainless steel, Aluminum

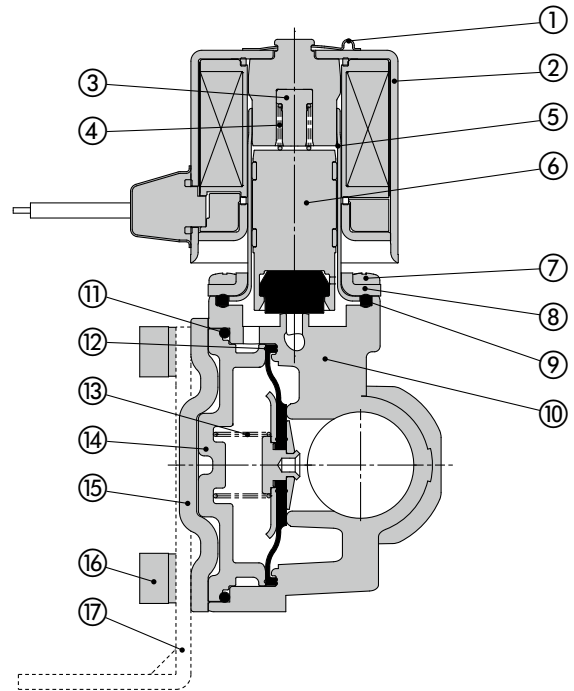


Component Parts

No.	Description	Material		
		Brass	Stainless steel	Aluminum
1	Clip		Stainless steel	
2	Solenoid coil		Stainless steel, Cu, Resin	
3	Stopper		PPS	
4	Spring		Stainless steel	
5	Tube assembly		Stainless steel	
6	Armature assembly	Stainless steel, PPS, NBR, (FKM, EPDM)		Stainless steel, PPS, NBR, (FKM)
7	Bracket		Fe	
8	Mounting screw		Fe	
9	Bonnet		Stainless steel	
10	Gasket	NBR, (FKM, EPDM)		NBR, (FKM)
11	Bolt		Fe	
12	Body	Brass	Stainless steel	Aluminum
13	O-ring	NBR, (FKM, EPDM)		NBR, (FKM)
14	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)		Stainless steel, NBR, (FKM)
15	Valve spring		Stainless steel	
16	Buffer		PPS	
17	Bonnet		Stainless steel	
18	Bolt		Fe	

JSXD40, Normally closed (N.C.)

Body material: Brass, Stainless steel

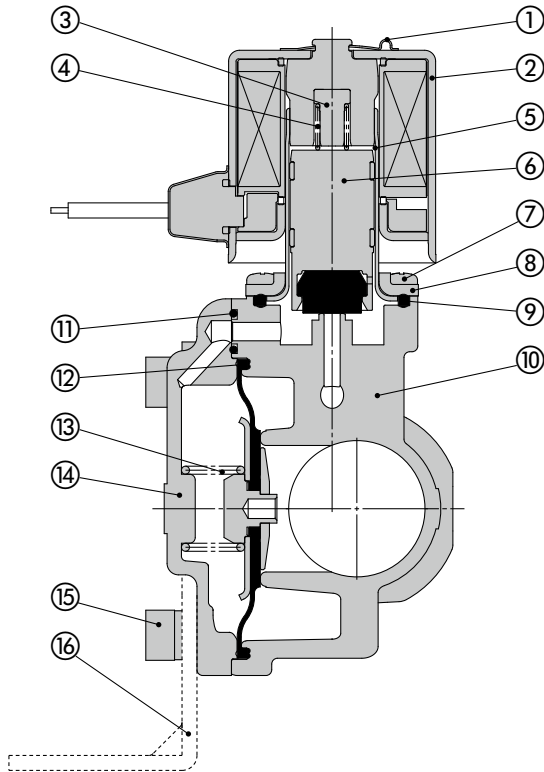


Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip		Stainless steel
2	Solenoid coil		Stainless steel, Cu, Resin
3	Stopper		PPS
4	Spring		Stainless steel
5	Tube assembly		Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
7	Mounting screw		Fe
8	Bonnet		Stainless steel
9	Gasket		NBR, (FKM, EPDM)
10	Body	Brass	Stainless steel
11	O-ring		NBR, (FKM, EPDM)
12	Diaphragm assembly		Stainless steel, NBR, (FKM, EPDM)
13	Valve spring		Stainless steel
14	Buffer		PPS
15	Bonnet		Stainless steel
16	Bolt		Fe
17	Bracket		Fe

Construction

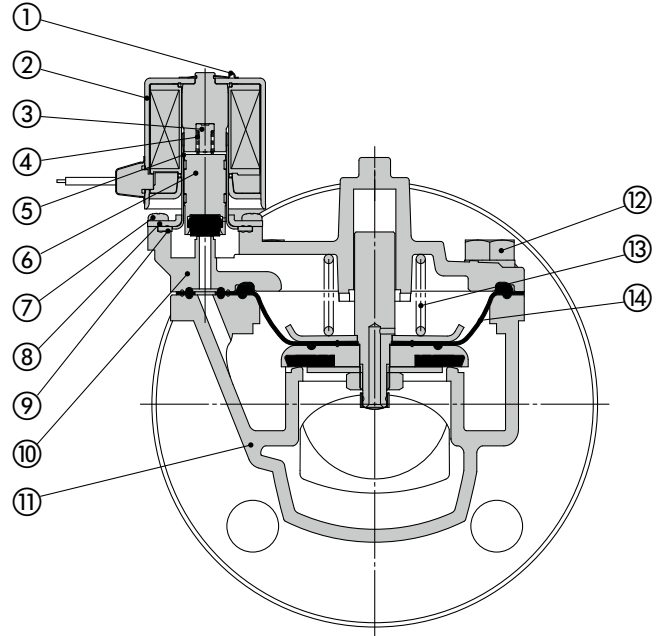
JSXD50, 60, Normally closed (N.C.)
Body material: Brass, Stainless steel



Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Stopper	PPS	
4	Spring	Stainless steel	
5	Tube assembly	Stainless steel	
6	Armature assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	Gasket	NBR, (FKM, EPDM)	
10	Body	Brass	Stainless steel
11	O-ring	NBR, (FKM, EPDM)	
12	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	
13	Valve spring	Stainless steel	
14	Bonnet	Brass	Stainless steel
15	Bolt	Fe	
16	Bracket	Fe	

JSXD70, 80, 90, Normally closed (N.C.)
Body material: Bronze



Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Stopper	PPS	
4	Spring	Stainless steel	
5	Tube assembly	Stainless steel	
6	Armature assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	Gasket	NBR, (FKM, EPDM)	
10	Bonnet	Bronze	
11	Body	Bronze	
12	Bolt	Fe	
13	Valve spring	Stainless steel	
14	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance
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Pilot Operated 2-Port Solenoid Valve *JSXD Series*



Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

Stainless Steel	Brass	Bronze	Aluminum
Normally Closed (N.C.)			
▶ p. 43			

Stainless Steel	Brass	Bronze
Normally Open (N.O.)		
▶ p. 47		

How to Order

JSXD **3** **2** - **C** **N** **02** **R** - **5** **G** - **D** - **B**

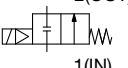
1 2 3 4 5 6 7 8 9 10



1 Size

Symbol	Size
3	30
4	40
5	50
6	60
7	70
8	80
9	90

2 Valve type

Symbol	Valve type
2	N.O.  2(OUT) 1(IN)

3 Body material

Symbol	Body material	Size		
		30	40, 50, 60	70, 80, 90
C	Brass	●	●	—
S	Stainless steel	●	●	—
B	Bronze	—	—	●

4 Seal material

Symbol	Seal material
N	NBR
F	FKM
E	EPDM

5 Port size

Symbol	Connection	Port size	Size						
			30	40	50	60	70	80	90
02	Thread	1/4	●	—	—	—	—	—	—
03		3/8	●	●	—	—	—	—	—
04		1/2	—	●	—	—	—	—	—
06		3/4	—	—	●	—	—	—	—
10		1	—	—	—	●	—	—	—
12		1 1/4	—	—	—	—	●	—	—
14		1 1/2	—	—	—	—	—	●	—
20	Flange	2	—	—	—	—	—	—	●
32		32A	—	—	—	—	●	—	—
40		40A	—	—	—	—	—	●	—
50		50A	—	—	—	—	—	—	●

6 Thread type

Symbol	Thread type	Connection
R	Rc	Thread
N	NPT	
F	G	
Nil	—	Flange

7 Rated voltage

AC				DC			
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC		
2	200 VAC	8	48 VAC	6	12 VDC		
3	120 (110) VAC	B	24 VAC				
4	220 VAC	J	230 VAC				

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Bracket

Symbol	With bracket	Size			
		30	40, 50, 60	70, 80, 90	
Nil	None	●	●	●	
B	With bracket	●	●	—*1	

*1 Sizes 70 to 90 are not available with a bracket.

8 Electrical entry

Symbol	Electrical entry		CE/UKCA-compliant
G	Grommet*1		Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)		
CS	Conduit (With surge voltage suppressor)		
DS	DIN terminal (With surge voltage suppressor)		
DZ	DIN terminal with light (With surge voltage suppressor)		
DN	DIN terminal without connector (With surge voltage suppressor)		
WN	M12 connector without cable (With surge voltage suppressor)*2		

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Size	Body material	Port size	Orifice diameter [mmø]	Flow rate characteristics*1						Min. operating pressure differential [MPa]	Max. operating pressure differential [MPa]	Model	Weight*2 [g]	
				Air				Water, Oil						
				C [dm³/s·bar]	b	Cv	Effective area [mm²]	Kv	Cv					
30	Brass	1/4	10	8.5	0.35	—	1.6	1.9	0.02	0.7	JSXD32-□□02	530		
	Stainless steel	3/8		9.2			2.4	2.0			2.4	JSXD32-□□03	530	
40	Brass	3/8	18	0.35	5.0		3.9	4.5			JSXD42-□□03	750		
	Stainless steel	1/2	20		5.5		4.6	5.5			JSXD42-□□04	750		
50	Brass/ Stainless steel	3/4	20	38	0.30		9.5	8.2			9.5	JSXD52-□□06	910	
60	Brass/ Stainless steel	1	25	—			225	11.0	13.0	JSXD62-□□10	1490			
70	Bronze	1 1/4, 32A	35				415	19.6	23.0	0.03	0.7	JSXD72-□□(12, 32)	5530/3030	
80	Bronze	1 1/2, 40A	40				560	26.4	31.0			JSXD82-□□(14, 40)	6930/4130	
90	Bronze	2, 50A	50				880	42.8	49.0			JSXD92-□□(20, 50)	8530/5530	

*1 The flow rate characteristics of this product vary.

*2 The values were calculated based on the combination of an Rc or NPT thread and a grommet. Add 30 g for the G thread type. Add 20 g for the grommet type with PCB, 70 g for the conduit type, and 50 g for the DIN terminal type.

Applicable Fluid Checklist

Applicable fluid	Seal material		
	NBR	FKM	EPDM
Air	●	●	●
Water	●	●	●
Oil	—	●	—

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use. If something is not clear, please contact SMC.

Common Specifications

Size			30	40	50	60	70	80	90
Valve specifications	Body material		Brass, Stainless steel				Bronze		
	Valve construction		Pilot operated diaphragm						
	Valve type		Normally open (N.O.)						
	Fluid and fluid temperature	Air*1	Air: -10 to 60°C						
		Water, Oil	Water: 1 to 60°C (No freezing), Oil: -5 to 60°C (Kinematic viscosity: 50 mm²/s or less)						
	Withstand pressure		2 MPa						
	Max. system pressure		1 MPa						
	Ambient temperature		-20 to 60°C						
	Valve leakage*2	Air	2 cm³/min (ANR) or less				10 cm³/min (ANR) or less		
		Water, Oil	0.2 cm³/min or less				1 cm³/min or less		
	External leakage*2	Air	1 cm³/min (ANR) or less						
		Water, Oil	0.1 cm³/min or less						
	Mounting orientation		Unrestricted						
	Enclosure*3		IP67 (IP65 for the DIN connector)						
Standards*4		CE/UKCA							
Operating environment		Location without the presence of corrosive gases or explosive gases							
Seal material		NBR, FKM, EPDM							
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V						
		DC	12 V, 24 V						
	Allowable voltage fluctuation		±10% of the rated voltage						
	Allowable leakage voltage	AC	5% or less of the rated voltage						
		DC	2% or less of the rated voltage						
	Apparent power*5, *6	AC	8 VA				9.5 VA		
	Power consumption*5	DC	6 W				8 W		
	Temperature rise*7		AC/DC	70/65°C					

*1 Dew point temperature: -10°C or less

*2 Valve leakage: The value at an ambient temperature of 20°C

*3 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*4 Standards compliance varies depending on the model. For details, refer to page 84.

*5 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*6 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

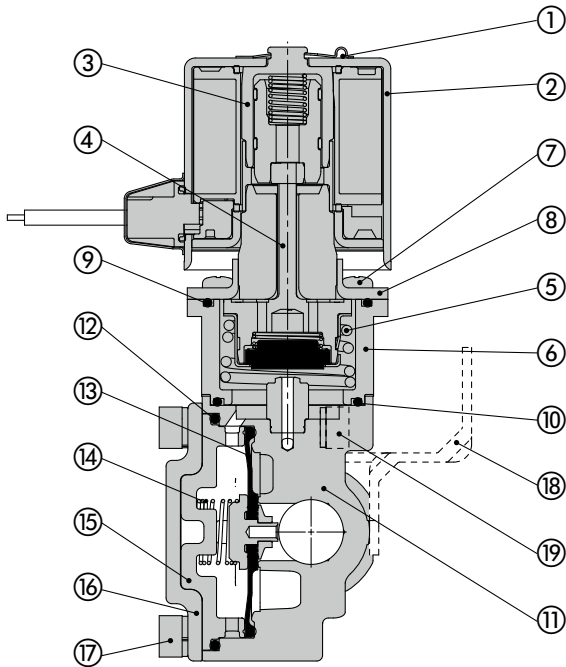
*7 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

Construction

JSXD30, Normally open (N.O.)

Body material: Brass, Stainless steel

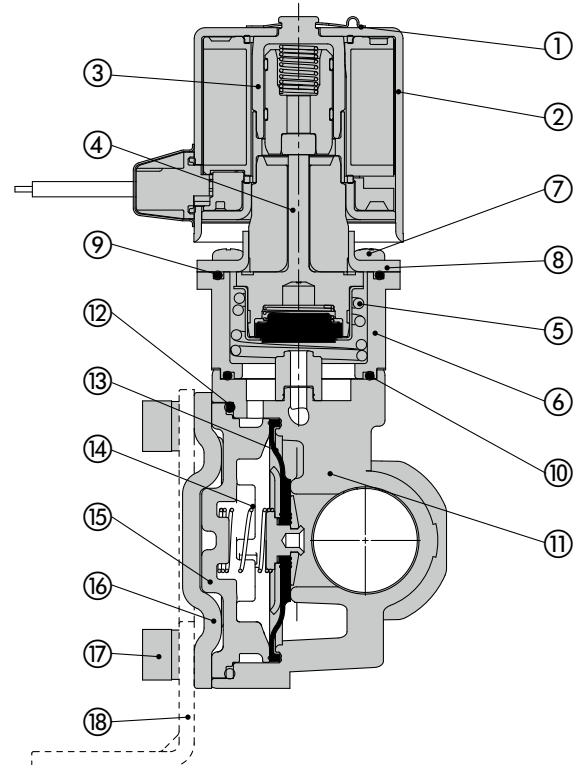


Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Sleeve assembly	Stainless steel, PPS	
4	Push rod assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
5	Spring	Stainless steel	
6	Adapter	PPS	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	O-ring	NBR, (FKM, EPDM)	
10	O-ring	NBR, (FKM, EPDM)	
11	Body	Brass	Stainless steel
12	O-ring	NBR, (FKM, EPDM)	
13	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	
14	Valve spring	Stainless steel	
15	Buffer	PPS	
16	Bonnet	Stainless steel	
17	Bolt	Fe	
18	Bracket	Fe	
19	Bolt for bracket	Fe	

JSXD40, Normally open (N.O.)

Body material: Brass, Stainless steel

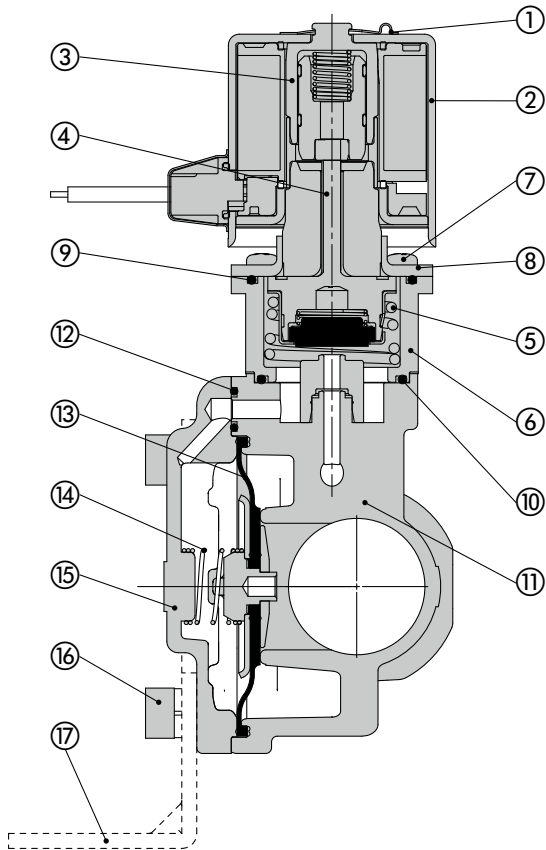


Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Sleeve assembly	Stainless steel, PPS	
4	Push rod assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
5	Spring	Stainless steel	
6	Adapter	PPS	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	O-ring	NBR, (FKM, EPDM)	
10	O-ring	NBR, (FKM, EPDM)	
11	Body	Brass	Stainless steel
12	O-ring	NBR, (FKM, EPDM)	
13	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	
14	Valve spring	Stainless steel	
15	Buffer	PPS	
16	Bonnet	Stainless steel	
17	Bolt	Fe	
18	Bracket	Fe	

Construction

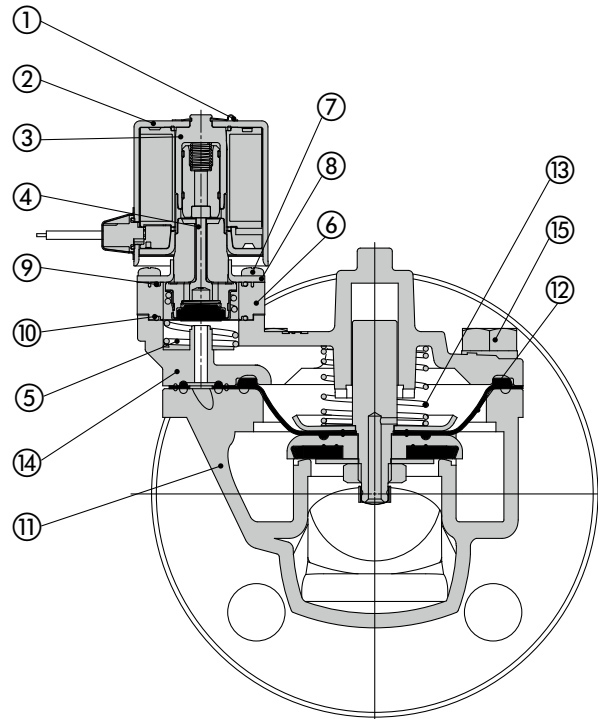
JSXD50, 60, Normally open (N.O.)
Body material: Brass, Stainless steel



Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Sleeve assembly	Stainless steel, PPS	
4	Push rod assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
5	Spring	Stainless steel	
6	Adapter	Resin	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	O-ring	NBR, (FKM, EPDM)	
10	O-ring	NBR, (FKM, EPDM)	
11	Body	Brass	Stainless steel
12	O-ring	NBR, (FKM, EPDM)	
13	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	
14	Valve spring	Stainless steel	
15	Bonnet	Stainless steel	
16	Bolt	Fe	
17	Bracket	Fe	

JSXD70, 80, 90, Normally open (N.O.)
Body material: Bronze



Component Parts

No.	Description	Material	
		Brass	Stainless steel
1	Clip	Stainless steel	
2	Solenoid coil	Stainless steel, Cu, Resin	
3	Sleeve assembly	Stainless steel, PPS	
4	Push rod assembly	Stainless steel, PPS, NBR, (FKM, EPDM)	
5	Spring	Stainless steel	
6	Adapter	Resin	
7	Mounting screw	Fe	
8	Bonnet	Stainless steel	
9	O-ring	NBR, (FKM, EPDM)	
10	O-ring	NBR, (FKM, EPDM)	
11	Body	Brass	Stainless steel
12	Diaphragm assembly	Stainless steel, NBR, (FKM, EPDM)	
13	Valve spring	Stainless steel	
14	Bonnet	Stainless steel	
15	Bolt	Fe	

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance
Table

UL-compliance
Table

Option

Replacement
Parts

Glossary
of Terms

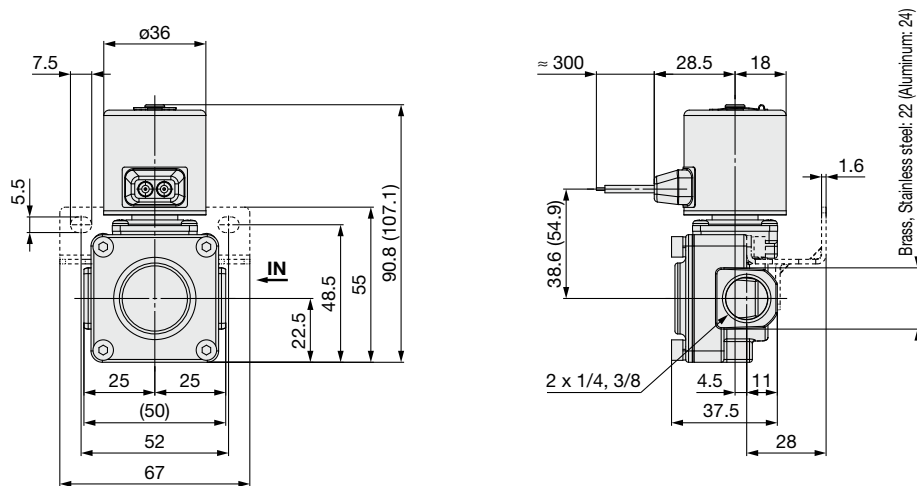
Flow Rate
Characteristics

Specific Product
Precautions

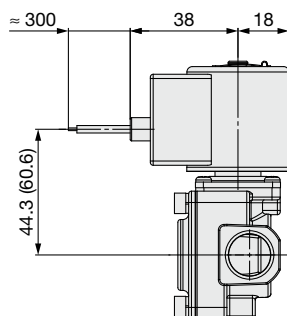
JSXD Series

Normally Closed (N.C.) 1/4, 3/8		Body Material	Aluminum, Brass, Stainless Steel
Dimensions: JSXD 30 Port Size	Normally Open (N.O.) 1/4, 3/8	Body Material	Brass, Stainless Steel

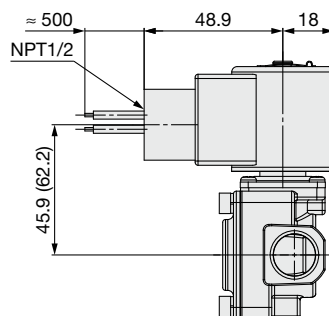
G: Grommet



GS: Grommet with PCB

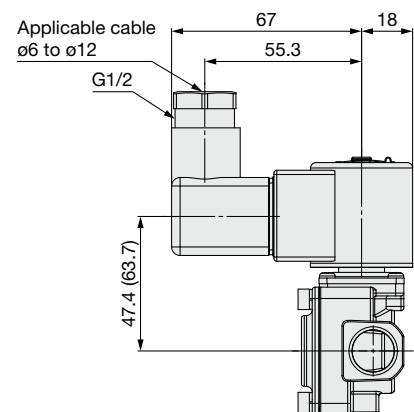


CS: Conduit

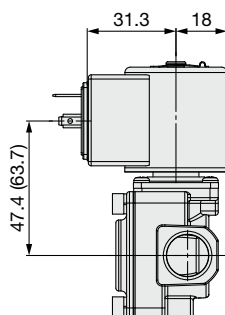


DS: DIN terminal

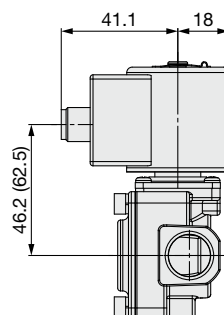
DZ: DIN terminal with light



DN: DIN terminal without connector



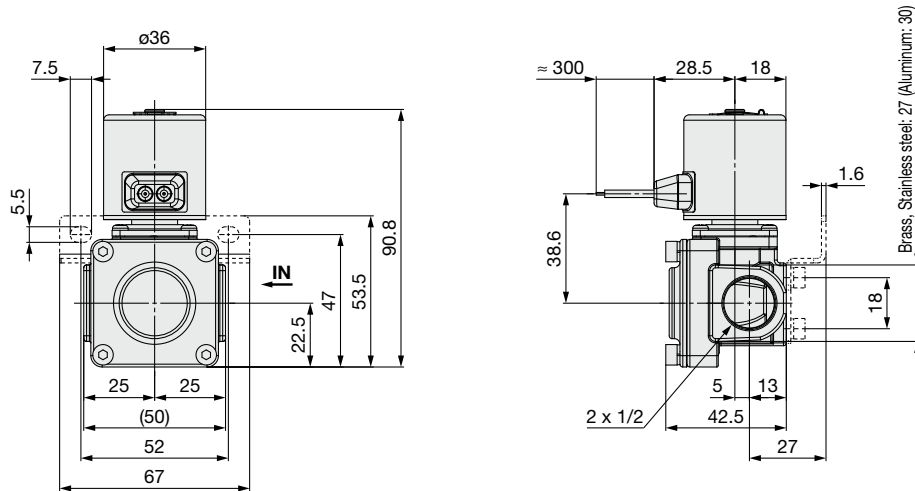
WN: M12 connector



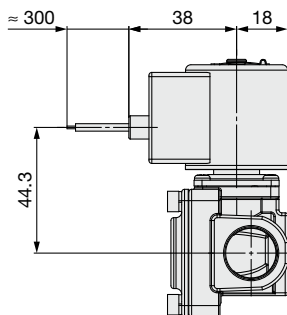
* (): Denotes the Normally Open (N.O.) dimensions

Dimensions: JSXD**30** Port Size **Normally Closed (N.C.) 1/2** Body Material **Aluminum, Brass, Stainless Steel**

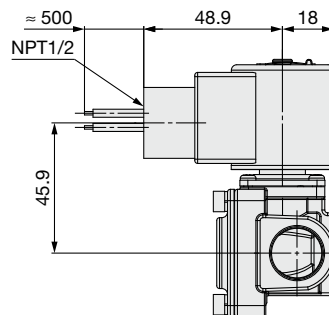
G: Grommet



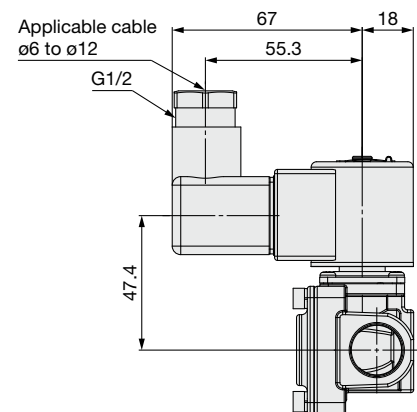
GS: Grommet with PCB



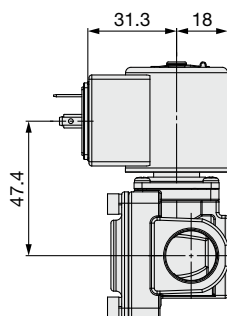
CS: Conduit



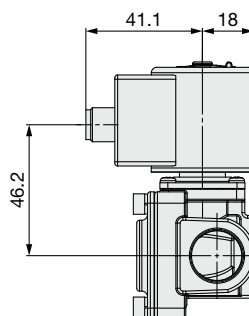
DS: DIN terminal
DZ: DIN terminal with light



DN: DIN terminal without connector

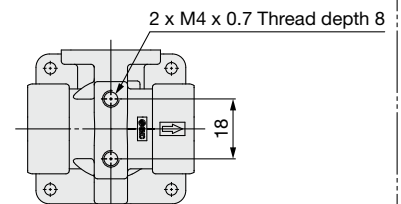


WN: M12 connector



JSXD31-□□04

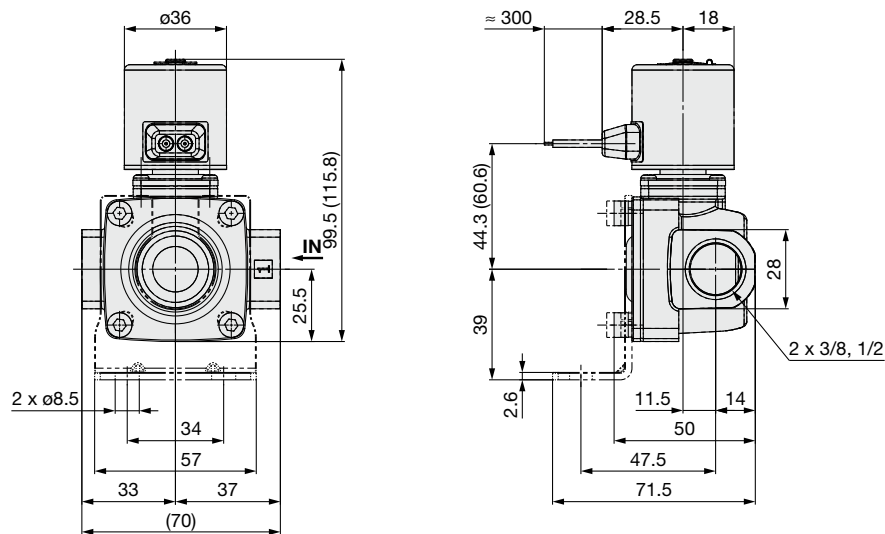
* Only the JSXD31 with port size of 04 (1/2) has threads on the bottom of the body.



JSXD Series

Dimensions: JSXD**40** Port Size **3/8, 1/2** Body Material **Brass, Stainless Steel**

G: Grommet

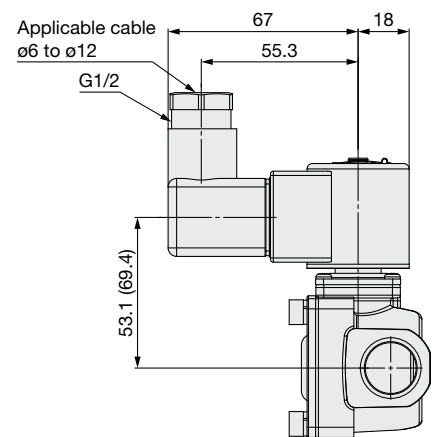
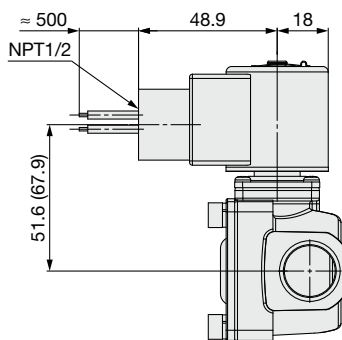
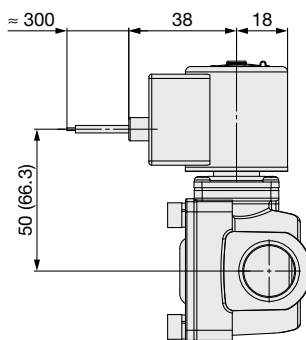


GS: Grommet with PCB

CS: Conduit

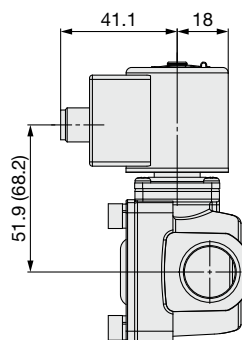
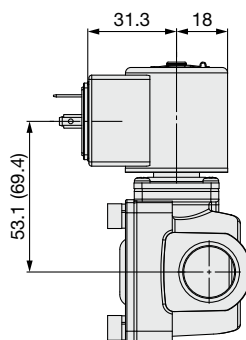
DS: DIN terminal

DZ: DIN terminal with light



DN: DIN terminal without connector

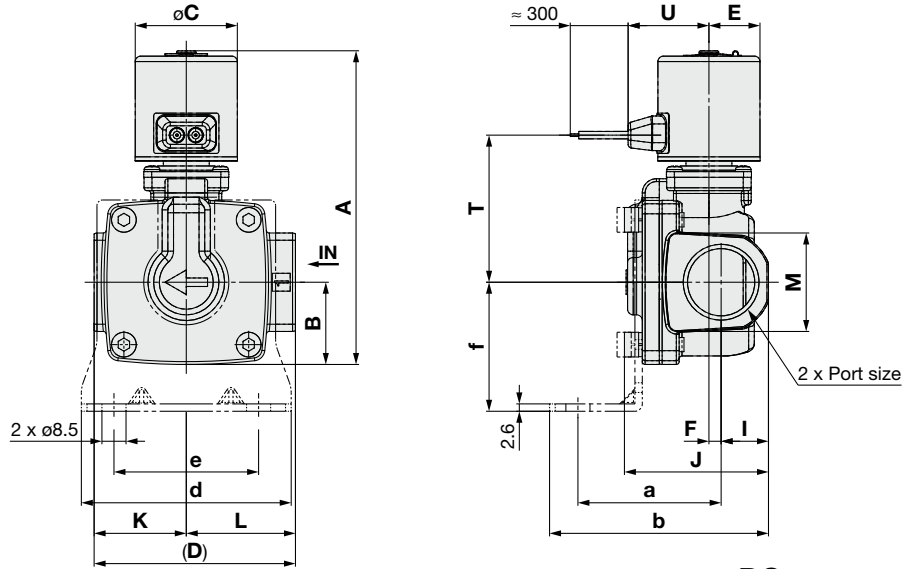
WN: M12 connector



* (): Denotes the Normally Open (N.O.) dimensions

Dimensions: **JSXD50, 60** Port Size **3/4, 1** Body Material **Brass, Stainless Steel**

G: Grommet

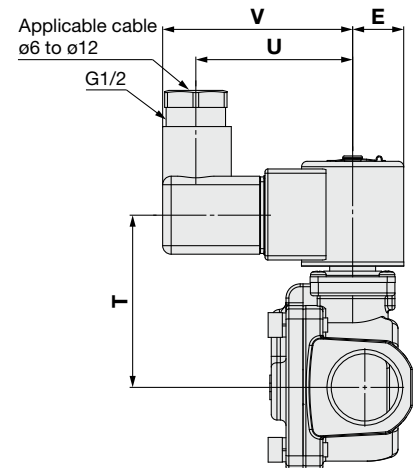
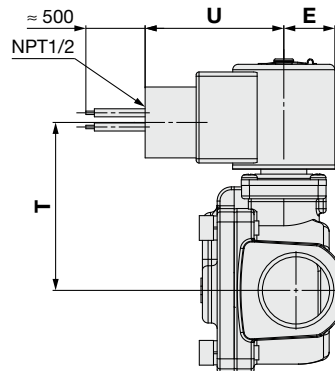
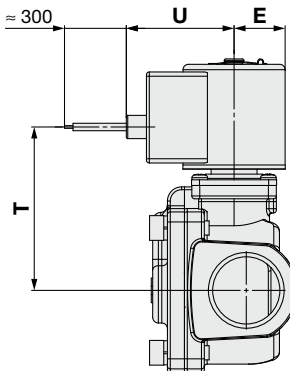


GS: Grommet with PCB

CS: Conduit

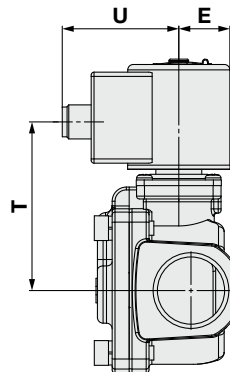
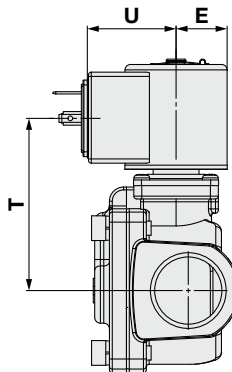
DS: DIN terminal

DZ: DIN terminal with light



DN: DIN terminal without connector

WN: M12 connector



Size	Port size	A	B	C	D	E	F	I	J	K	L	M	Grommet		Grommet with PCB	
													T	U	T	U
50	3/4	50 (126.9)	29	36	71	18	4.5	17	51	32.5	38.5	35	51.9 (68.2)	28.5	57.6 (73.9)	38
60	1	60 (140.6)	33	42	95	21	4.5	20	59.5	45.5	49.5	42	60.4 (70)	31.1	66 (75.6)	41

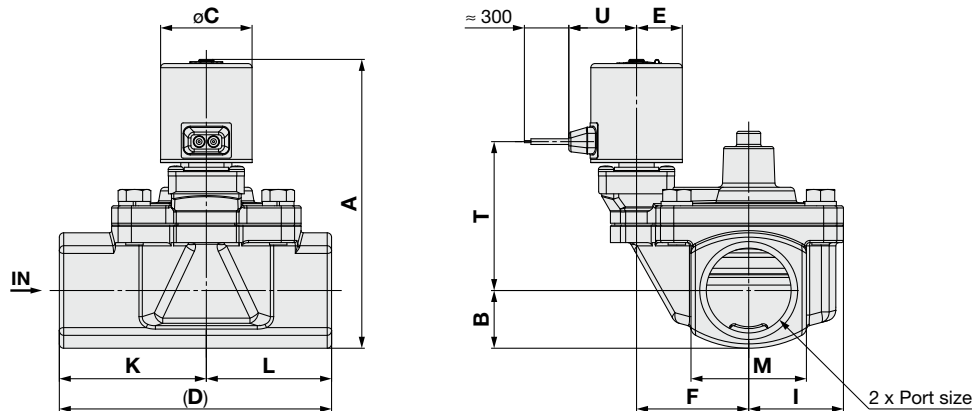
Size	Port size	Conduit		DIN terminal		DIN terminal without connector	M12 connector		Bracket mount dimensions						
		T	U	T	U		T	U	a	b	d	e	f		
50	3/4	59.2 (75.5)	48.9	60.7 (77)	55.3	67	60.7 (77)	31.3	59.5 (75.8)	41.1	50.5	77.5	74	51	45.5
60	1	67.6 (77.2)	51.9	69.1 (78.7)	58.3	70	69.1 (78.7)	34.3	67.9 (77.5)	44.1	55.5	85.5	81	58	49.5

* (): Denotes the Normally Open (N.O.) dimensions

JSXD Series

Dimensions: JSXD **70, 80, 90** Port Size **1 1/4, 1 1/2, 2** Body Material **Bronze**

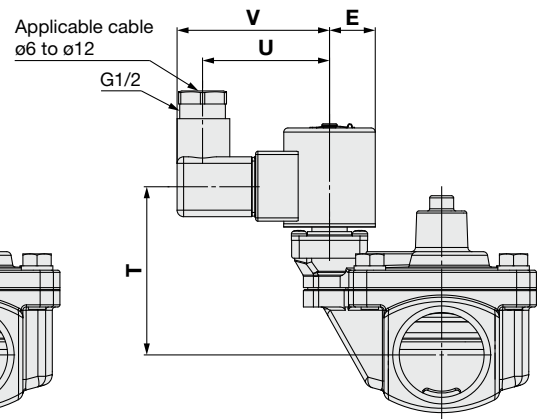
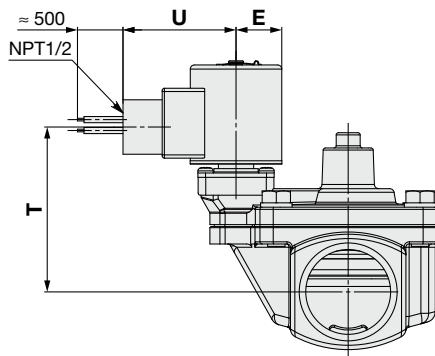
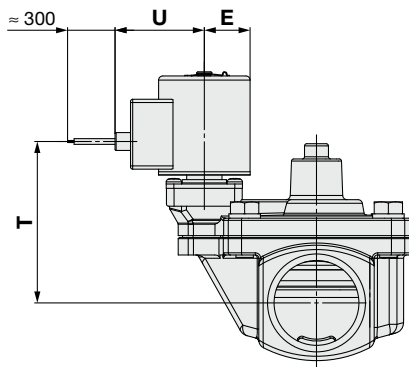
G: Grommet



GS: Grommet with PCB

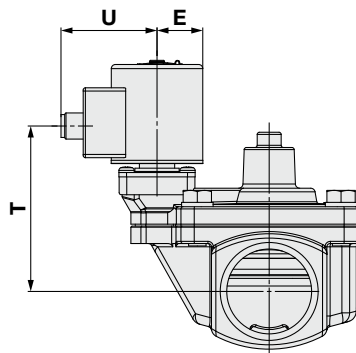
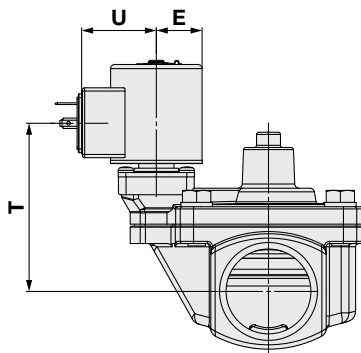
CS: Conduit

DS: DIN terminal
DZ: DIN terminal with light



DN: DIN terminal without connector

WN: M12 connector



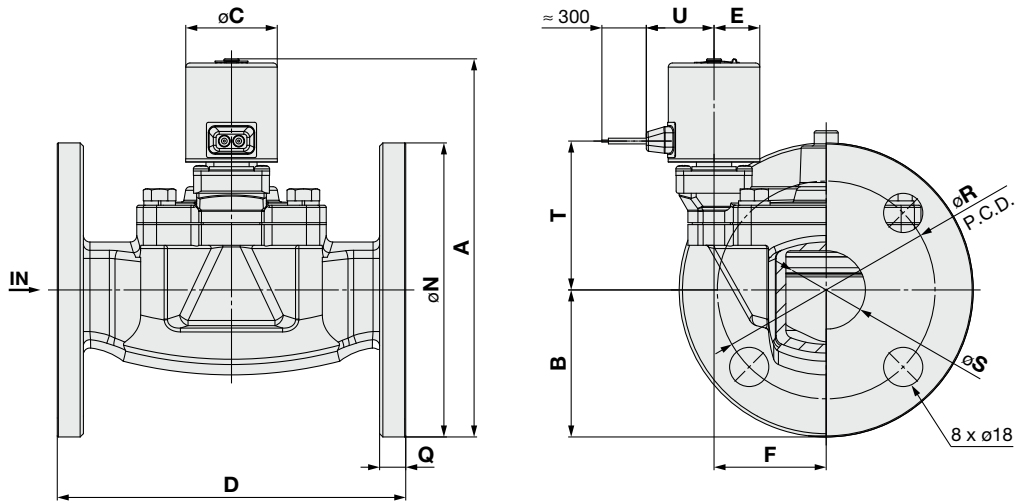
[mm]											
Size	Port size	A	B	C	D	E	F	I	K	L	M
70	1 1/4	70 (142.2)	26.5	42	125	21	51.5	43.5	67.5	57.5	53
80	1 1/2	80 (148.9)	30	42	132	21	54.5	46.5	72	60	60
90	2	90 (159.9)	35.5	42	150	21	59	52	81	69	71

Size	Port size	Grommet		Grommet with PCB		Conduit		DIN terminal			DIN terminal without connector		M12 connector	
		T	U	T	U	T	U	T	U	V	T	U	T	U
70	1 1/4	68.4 (78)	31.1	74.1 (83.7)	41	75.7 (85.3)	51.9	77.2 (86.8)	58.3	70	77.2 (86.8)	34.3	76 (85.6)	44.1
80	1 1/2	71.6 (81.2)	31.1	77.3 (86.9)	41	78.9 (88.5)	51.9	80.4 (90)	58.3	70	80.4 (90)	34.3	79.2 (88.8)	44.1
90	2	77.1 (86.7)	31.1	82.8 (92.4)	41	84.4 (94)	51.9	85.9 (95.5)	58.3	70	85.9 (95.5)	34.3	84.7 (94.3)	44.1

* (): Denotes the Normally Open (N.O.) dimensions

Dimensions: JSXD **70, 80, 90** Applicable Flange **32A, 40A, 50A** Body Material **Bronze**

G: Grommet

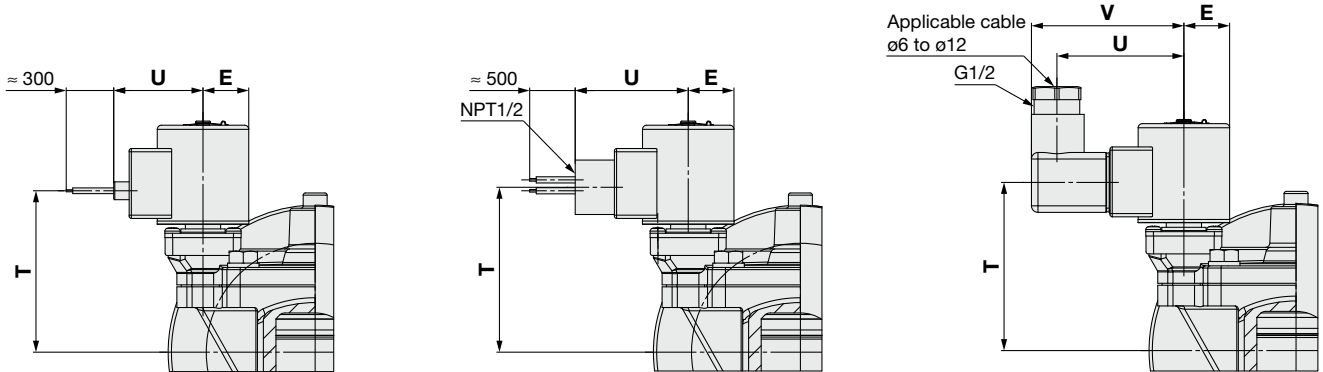


GS: Grommet with PCB

CS: Conduit

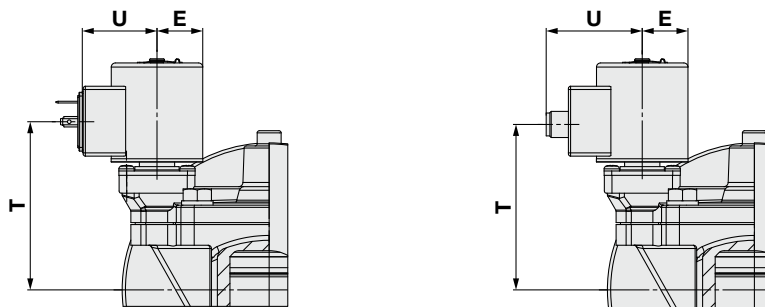
DS: DIN terminal

DZ: DIN terminal with light



DN: DIN terminal without connector

WN: M12 connector



Size	Applicable flange	A	B	C	D	E	F	N	Q	R	S
70	32A	70 (183.2)	67.5	42	160	21	51.5	135	12	100	36
80	40A	80 (188.9)	70	42	170	21	54.5	140	14	105	42
90	50A	90 (201.9)	77.5	42	180	21	59	155	14	120	52

[mm]

Size	Applicable flange	Grommet		Grommet with PCB		Conduit		DIN terminal			DIN terminal without connector		M12 connector	
		T	U	T	U	T	U	T	U	V	T	U	T	U
70	32A	68.4 (78)	31.1	74.1 (83.7)	41	75.7 (85.3)	51.9	77.2 (86.8)	58.3	70	77.2 (86.8)	34.3	76 (85.6)	44.1
80	40A	71.6 (81.2)	31.1	77.3 (86.9)	41	78.9 (88.5)	51.9	80.4 (90)	58.3	70	80.4 (90)	34.3	79.2 (88.8)	44.1
90	50A	77.1 (86.7)	31.1	82.8 (92.4)	41	84.4 (94)	51.9	85.9 (95.5)	58.3	70	85.9 (95.5)	34.3	84.7 (94.3)	44.1

* (): Denotes the Normally Open (N.O.) dimensions

Steam Type

For **Steam**
Heated Water

Pilot Operated 2-Port Solenoid Valve

JSXP Series



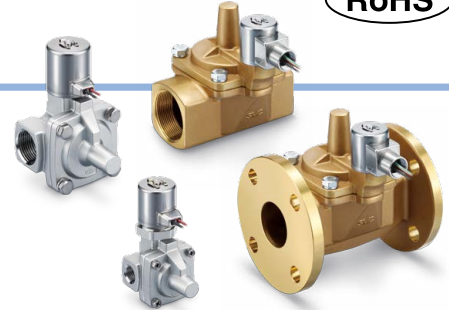
Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

How to Order

JSXP **4** **1** - **C** **F** **03** **R** - **5** **G** - **D** - **B**

1 2 3 4 5 6 7 8 9 10



1 Size

Symbol	Size
4	40
5	50
6	60
7	70
8	80
9	90

2 Valve type

Symbol	Valve type
1	N.C.

3 Body material

Symbol	Body material	Size	
		40, 50, 60	70, 80, 90
C	Brass	●	—
S	Stainless steel	●	—
B	Bronze	—	●

4 Seal material

Symbol	Seal material
F	FKM
T	PTFE

5 Port size

Symbol	Connection	Port size	Size					
			40	50	60	70	80	90
03	Thread	3/8	●	—	—	—	—	—
04		1/2	●	—	—	—	—	—
06		3/4	—	●	—	—	—	—
10		1	—	—	●	—	—	—
12		1 1/4	—	—	—	●	—	—
14		1 1/2	—	—	—	—	●	—
20		2	—	—	—	—	—	●
32	Flange	32A	—	—	—	●	—	—
40		40A	—	—	—	—	●	—
50		50A	—	—	—	—	—	●

8 Electrical entry

Symbol	Electrical entry		CE/UKCA-compliant
G	Grommet*1		Refer to page 84.
GR	Grommet (With surge voltage suppressor)*2		
CR	Conduit/NPT thread (With surge voltage suppressor)*2		
FR	Conduit/G thread (With surge voltage suppressor)*2		

6 Thread type

Symbol	Thread type	Connection
R	Rc	Thread
N	NPT	
F	G	
Nil	—	Flange

7 Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

10 Bracket

Symbol	With bracket	Size	
		40, 50, 60	70, 80, 90
Nil	None	●	●
B	With bracket	●	—*1

*1 Sizes 70 to 90 are not available with a bracket.

*1 DC voltage only

*2 The circuit board is installed between the lead wires.

Flow Rate Characteristics

Size	Body material	Port size	Orifice diameter [mmø]	Flow rate characteristics*1		Min. operating pressure differential [MPa]	Max. operating pressure differential [MPa]	Model	Weight*2 [g]
				Steam/Heated water Kv	Conversion Cv				
40	Brass	3/8	15	3.6	4.2	0.04	1.0	JSXP41- $\frac{C}{S}$ □03	900
	Stainless steel	1/2		4.6	5.3			JSXP41- $\frac{C}{S}$ □04	900
50	Brass/Stainless steel	3/4	20	7.9	9.2			JSXP51- $\frac{C}{S}$ □06	1320
60	Brass/Stainless steel	1	25	10.0	12.0			JSXP61- $\frac{C}{S}$ □10	1930
70	Bronze	1 1/4, 32A	35	20.0	23.0	0.03		JSXP71-B□(12, 32)	6100/3500
80	Bronze	1 1/2, 40A	40	26.0	31.0			JSXP81-B□(14, 40)	7500/4400
90	Bronze	2, 50A	50	43.0	49.0			JSXP91-B□(20, 50)	9400/5600

*1 The flow rate characteristics of this product vary.

*2 Indicates case of grommet type

Add 70 g for the conduit type.

For sizes 70, 80, and 90, the weight on the left is for the flange type, and the weight on the right is for the thread type.

Applicable Fluid Checklist

Applicable fluid	Seal material	
	FKM	PTFE
Steam/Heated water	●	●

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use. If something is not clear, please contact SMC.

Common Specifications

Size			40	50	60	70	80	90
Valve specifications	Body material		Brass, Stainless steel			Bronze		
	Valve construction		Pilot operated diaphragm					
	Valve type		Normally closed (N.C.)					
	Fluid and fluid temperature	Steam	183°C or less					
		Heated water	99°C or less					
	Withstand pressure		2 MPa					
	Max. system pressure		1 MPa					
	Ambient temperature		-20 to 60°C					
	Valve leakage*1	Steam	10 cm ³ /min or less (Seal material: FKM), 500 cm ³ /min or less (Seal material: PTFE)					
		Heated water	1 cm ³ /min or less (Seal material: FKM), 50 cm ³ /min or less (Seal material: PTFE)					
	External leakage*1	Steam	1.0 cm ³ /min or less					
		Heated water	0.1 cm ³ /min or less					
	Mounting orientation		Unrestricted					
	Enclosure*2		IP67					
	Standards*3		CE/UKCA					
Operating environment		Location without the presence of corrosive gases or explosive gases						
Seal material		FKM, PTFE						
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V					
		DC	12 V, 24 V					
	Allowable voltage fluctuation		±10% of the rated voltage					
	Allowable leakage voltage	AC	5% or less of the rated voltage					
		DC	2% or less of the rated voltage					
	Apparent power*4, *5	AC	8 VA	9.5 VA			16 VA	
	Power consumption*4	DC	6 W	8 W			13 W	
Temperature rise*6		AC/DC	70/65°C				80/75°C	

*1 Leakage: The value at a differential pressure the same as or higher than the min. operating pressure differential, and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

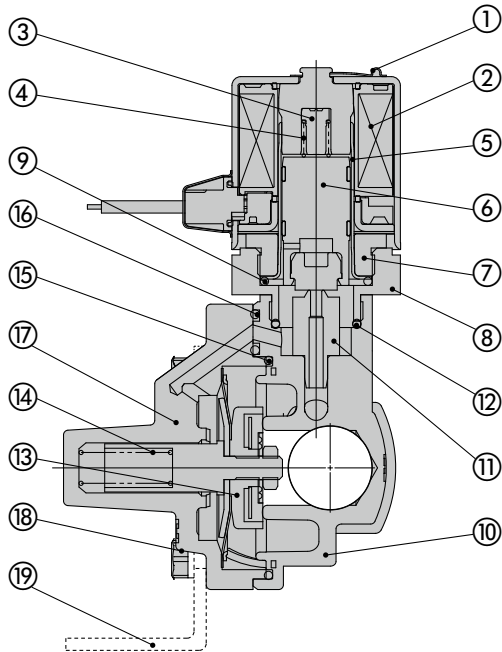
*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

Construction

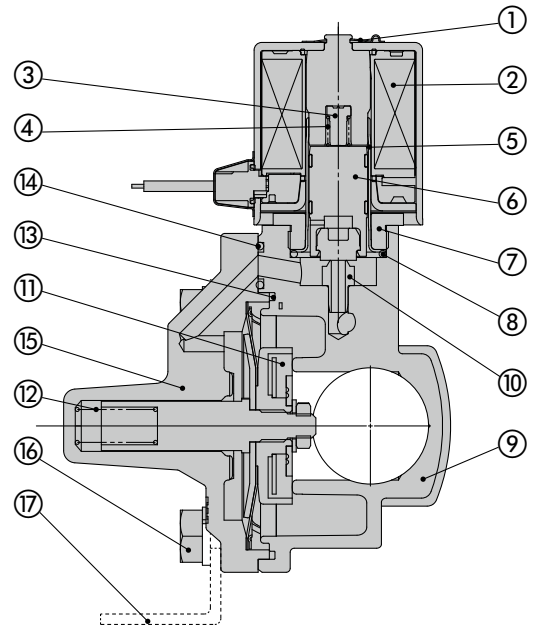
JSXP40, 50, Normally closed (N.C.)
Body material: Brass, Stainless steel



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, FKM (PTFE)
7	Set nut	Stainless steel
8	Adapter	Stainless steel
9	O-ring	FKM (PTFE)
10	Body	Brass Stainless steel
11	Orifice	Stainless steel
12	O-ring	FKM (PTFE)
13	Disk assembly	Brass, FKM (PTFE) Stainless steel, FKM (PTFE)
14	Valve spring	Stainless steel
15	O-ring	FKM (PTFE)
16	O-ring	FKM (PTFE)
17	Bonnet	Brass Stainless steel
18	Bolt	Fe
19	Bracket	Fe

JSXP60, Normally closed (N.C.)
Body material: Brass, Stainless steel



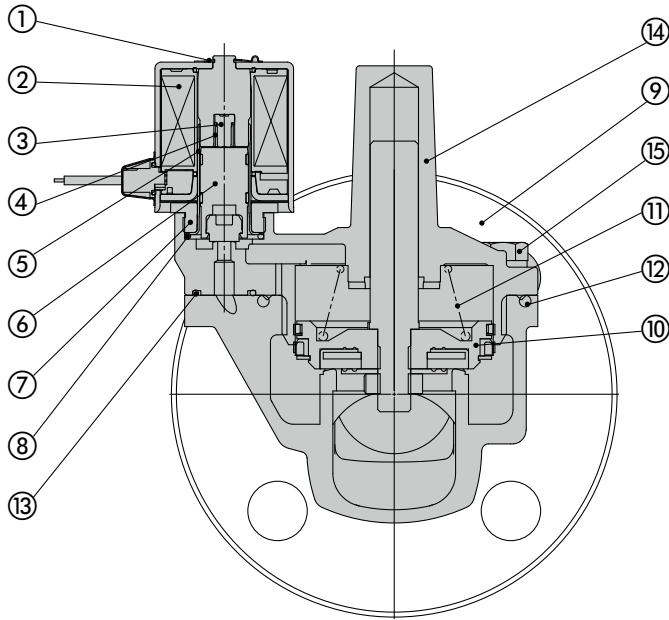
Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, FKM (PTFE)
7	Set nut	Stainless steel
8	O-ring	FKM (PTFE)
9	Body	Brass Stainless steel
10	Orifice	Stainless steel
11	Disk assembly	Brass, FKM (PTFE) Stainless steel, FKM (PTFE)
12	Valve spring	Stainless steel
13	O-ring	FKM (PTFE)
14	O-ring	FKM (PTFE)
15	Bonnet	Brass Stainless steel
16	Bolt	Fe
17	Bracket	Fe

Construction

JSXP70, 80, 90, Normally closed (N.C.)

Body material: Bronze



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, FKM (PTFE)
7	Set nut	Stainless steel
8	O-ring	FKM (PTFE)
9	Body	Bronze
10	Disk assembly	Stainless steel, Brass, FKM (PTFE)
11	Valve spring	Stainless steel
12	O-ring	FKM (PTFE)
13	O-ring	FKM (PTFE)
14	Bonnet	Bronze
15	Bolt	Fe

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance
Table

UL-compliance
Table

Option

Replacement
Parts

Glossary
of Terms

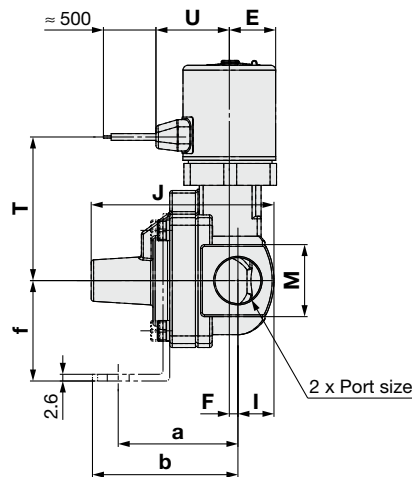
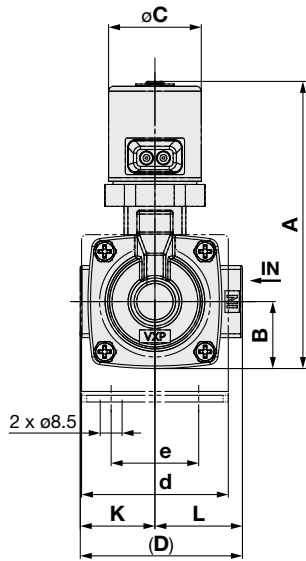
Flow Rate
Characteristics

Specific Product
Precautions

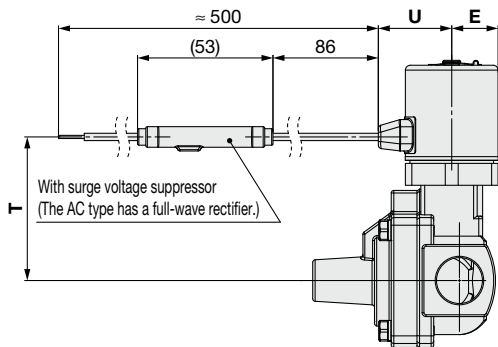
JSXP Series

Dimensions: JSXP**40, 50, 60** Port Size **3/8, 1/2, 3/4, 1** Body Material **Brass, Stainless Steel**

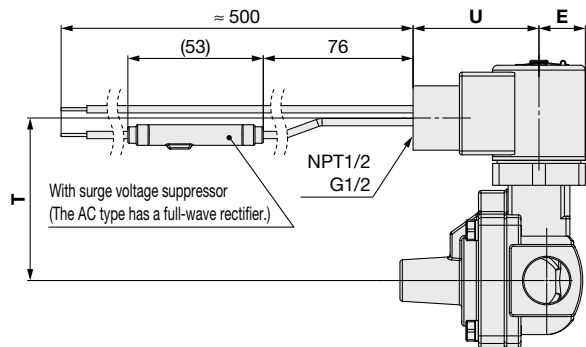
G: Grommet



GR: Grommet with PCB



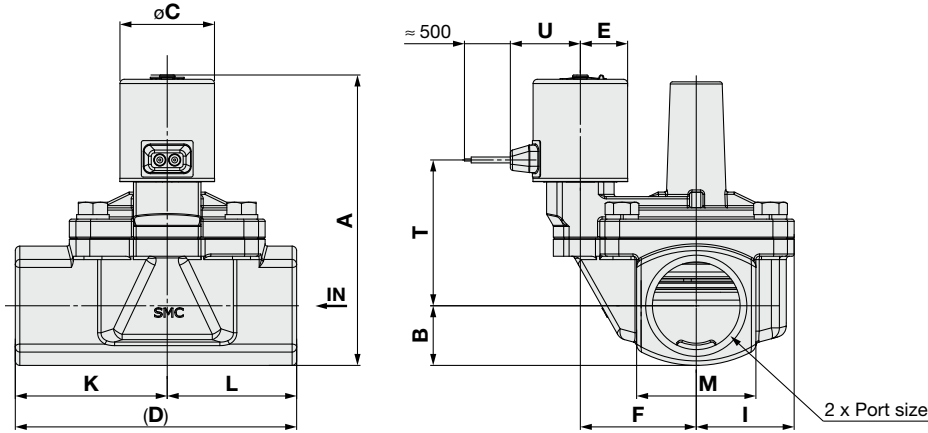
CR: Conduit/NPT thread
FR: Conduit/G thread



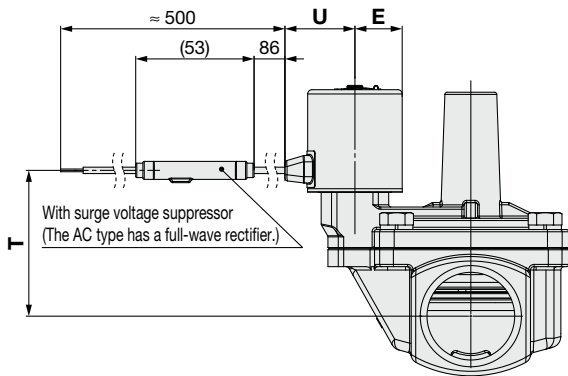
</

Dimensions: JSXP **70, 80, 90** Port Size **1 1/4, 1 1/2, 2** Body Material **Bronze**

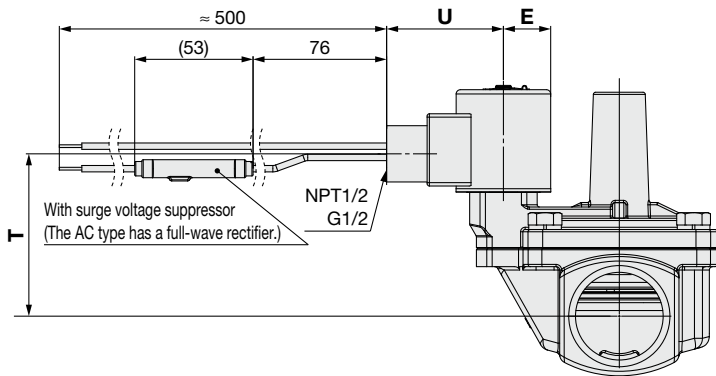
G: Grommet



GR: Grommet with PCB



CR: Conduit/NPT thread
FR: Conduit/G thread



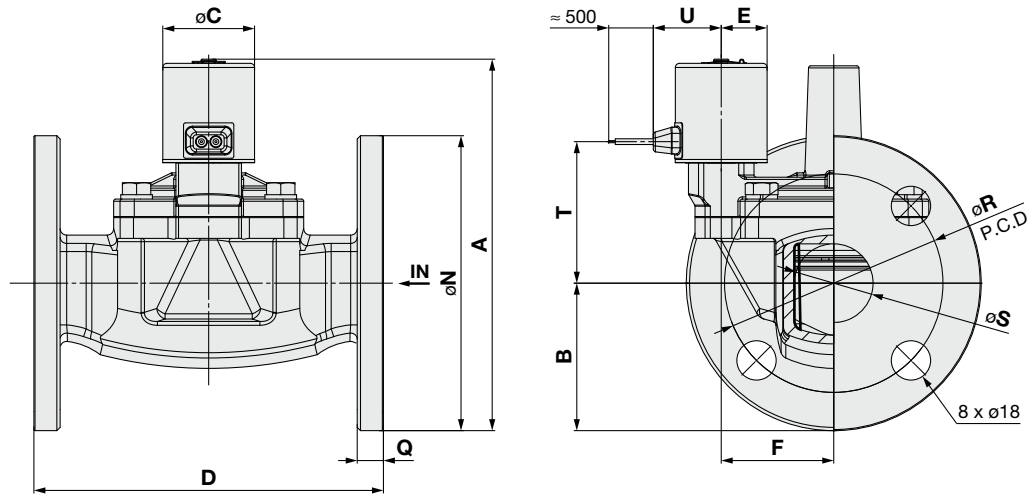
Size	Port size	A	B	C	D	E	F	I	K	L	M	[mm]			
												Grommet		Grommet with PCB	
												T	U	T	U
70	1 1/4	129	26.5	42	125	21	51.5	43.5	67.5	57.5	53	64.8	31.1	64.8	31.1
80	1 1/2	138.5	30	42	132	21	54.5	46.5	72	60	60	70.8	31.1	70.8	31.1
90	2	153.6	35.5	42	150	21	59	52	81	69	71	80.4	31.1	80.4	31.1

Size	Port size	Conduit	
		T	U
70	1 1/4	72.1	51.9
80	1 1/2	78.1	51.9
90	2	87.7	51.9

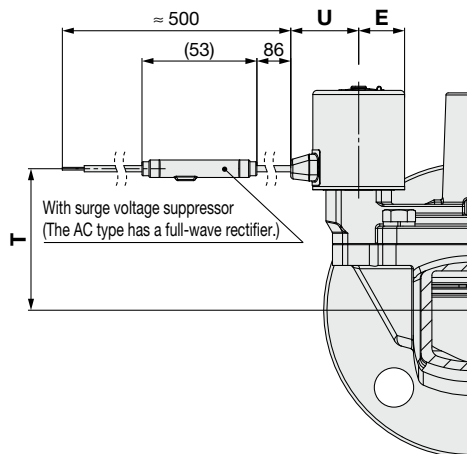
JSXP Series

Dimensions: JSXP **70, 80, 90** Applicable Flange **32A, 40A, 50A** Body Material **Bronze**

G: Grommet

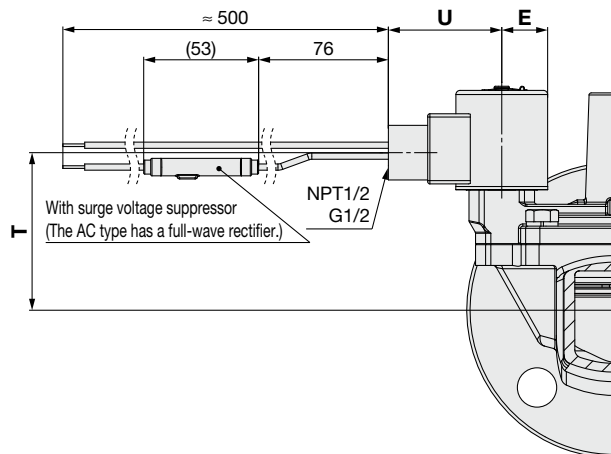


GR: Grommet with PCB



CR: Conduit/NPT thread

FR: Conduit/G thread



[mm]											
Size	Applicable flange	A	B	C	D	E	F	N	Q	R	S
70	32A	170	67.5	42	160	21	51.5	135	12	100	36
80	40A	178.5	70	42	170	21	54.5	140	14	105	42
90	50A	195.6	77.5	42	180	21	59	155	14	120	52

Size	Applicable flange	Grommet		Grommet with PCB		Conduit	
		T	U	T	U	T	U
70	32A	64.8	31.1	64.8	31.1	72.1	51.9
80	40A	70.8	31.1	70.8	31.1	78.1	51.9
90	50A	80.4	31.1	80.4	31.1	87.7	51.9

Specific Product Precautions	Flow Rate Characteristics	Glossary of Terms	Replacement Parts	Option	UL-compliance Table	CE/UKCA-compliance Table	JSXM	JSXZ	JSXR	JSXP	JSXD	JSX
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For **Water**

Water Hammer Relief Pilot Operated 2-Port Solenoid Valve *JSXR Series*



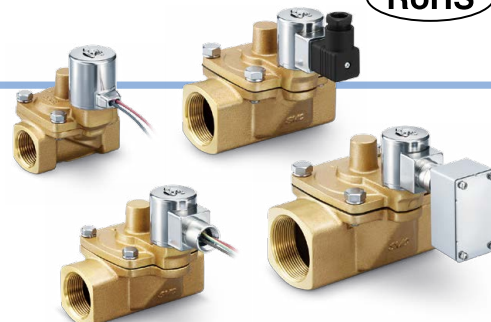
Differs depending on the voltage and electrical entry. Refer to page 84.

RoHS

How to Order

JSXR **5** **1** - **B** **N** **04** **R** - **5** **G** - **D**

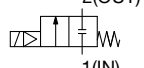
1 2 3 4 5 6 7 8 9



1 Size

Symbol	Size
5	50
6	60
7	70
8	80
9	90

2 Valve type

Symbol	Valve type
1	N.C. 

3 Body material

Symbol	Body material
B	Bronze

4 Seal material

Symbol	Seal material
N	NBR
F	FKM

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

9 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

5 Port size

Symbol	Connection	Port size	Size				
			50	60	70	80	90
04	Thread	1/2	●	—	—	—	—
06		3/4	●	—	—	—	—
10		1	—	●	—	—	—
12		1 1/4	—	—	●	—	—
14		1 1/2	—	—	—	●	—
20		2	—	—	—	—	●

7 Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

8 Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant
G	Grommet*1 	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor) 	
CS	Conduit/NPT thread (With surge voltage suppressor) 	
FS	Conduit/G thread (With surge voltage suppressor) 	
DS	DIN terminal (With surge voltage suppressor) 	
DZ	DIN terminal with light (With surge voltage suppressor) 	
DN	DIN terminal without connector (With surge voltage suppressor) 	
WN	M12 connector/Without connector cable (With surge voltage suppressor)*2 	
TS	Conduit terminal (With surge voltage suppressor) 	

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

Flow Rate Characteristics

Size	Body material	Port size	Orifice diameter [mmø]	Flow rate characteristics*1		Min. operating pressure differential [MPa]	Max. operating pressure differential [MPa]	Model	Weight*2 [g]
				Water					
				Kv	Conversion Cv				
50	Bronze	1/2	20	5.7	6.5	0.04	1.0	JSXR51-B□04	1320
		3/4	20	6.4	7.5			JSXR51-B□06	1320
60	Bronze	1	25	10.3	12.0			JSXR61-B□10	1800
70	Bronze	1 1/4	35	18.9	22.0			JSXR71-B□12	2970
80	Bronze	1 1/2	40	25.7	30.0			JSXR81-B□14	3770
90	Bronze	2	50	42.8	48.0			JSXR91-B□20	4670

*1 The flow rate characteristics of this product vary.

*2 Indicates case of grommet type

Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, 15 g for the M12 connector type, and 310 g for the conduit terminal type.

Applicable Fluid Checklist

Applicable fluid	Seal material	
	NBR	FKM
Water	●	●

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use. If something is not clear, please contact SMC.

Common Specifications

Size			50	60	70	80	90
Valve specifications	Body material		Bronze				
	Valve construction		Pilot operated diaphragm				
	Valve type		Normally closed (N.C.)				
	Fluid and fluid temperature	Water	1 to 60°C				
	Withstand pressure		2 MPa				
	Max. system pressure		1 MPa				
	Ambient temperature		−20 to 60°C				
	Valve leakage*1	Water	1 cm³/min or less				
	External leakage*1	Water	2 cm³/min or less				
	Mounting orientation		Unrestricted				
	Enclosure*2		IP67 (IP65 for the DIN terminal and conduit terminal)				
	Standards*3		CE/UKCA				
	Operating environment		Location without the presence of corrosive gases or explosive gases				
	Seal material		NBR, FKM				
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V				
		DC	12 V, 24 V				
	Allowable voltage fluctuation		±10% of the rated voltage				
	Allowable leakage voltage	AC	5% or less of the rated voltage				
		DC	2% or less of the rated voltage				
	Apparent power*4, *5	AC	8 VA	9.5 VA			16 VA
	Power consumption*4	DC	6 W	8 W			13 W
	Temperature rise*6	AC/DC	70/65°C				80/75°C

*1 Leakage: The value at a differential pressure the same as or higher than the min. operating pressure differential, and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

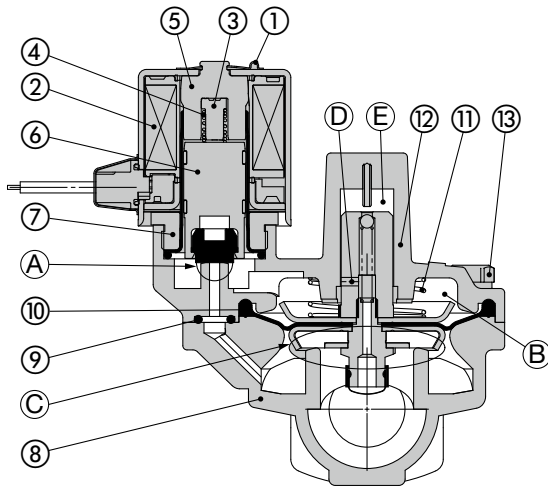
*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.

Construction

JSXR, Normally closed (N.C.)

Body material: Bronze



Operation

< Valve opened > When the solenoid coil ② is energized, the armature assembly ⑥ is attracted into the core of the tube assembly ⑤ and the pilot valve ① opens. Then the pressure in the pressure action chamber ⑧ falls to open the main valve ③.

< Valve closed > When the solenoid coil ② is not energized, the pilot valve ① is closed and the pressure in the pressure action chamber ⑧ rises and the main valve ③ closes.

< Water hammer relieving >

Check valve mechanism is provided in the ⑤ side of the supply orifice ④ and ⑤ and supply into the pressure action chamber ⑧ can be controlled with two stages by moving the diaphragm assembly ⑩. After release of the energy, when the open amount of the main valve ③ becomes small, ⑤ is blocked. A low valve closing speed relieves the water hammer.

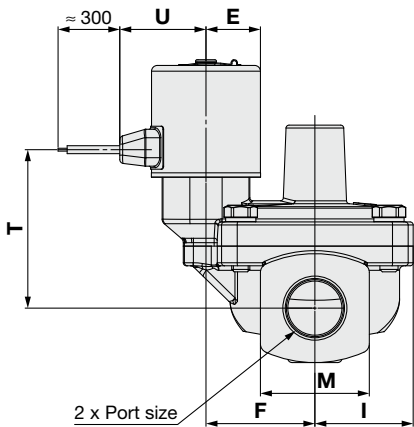
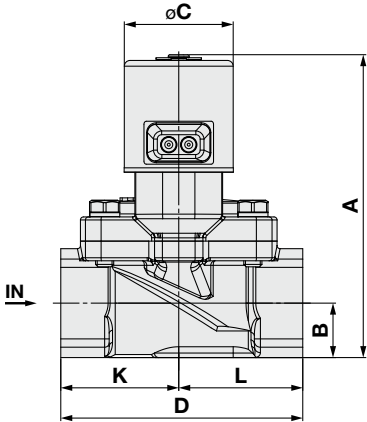
Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR (FKM)
7	Set nut	Brass
8	Body	Bronze
9	O-ring	NBR (FKM)
10	Diaphragm assembly	Stainless steel, NBR (FKM)
11	Valve spring	Stainless steel
12	Bonnet	Bronze
13	Bolt	Fe

Dimensions: JSXR Normally Closed (N.C.)

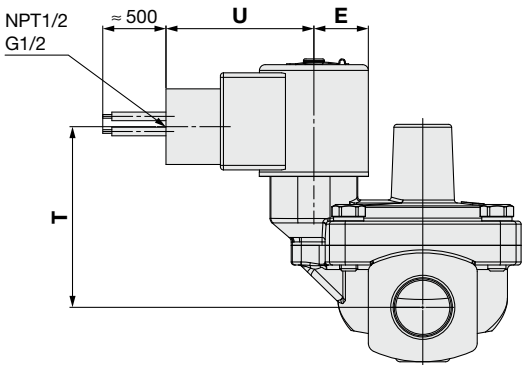
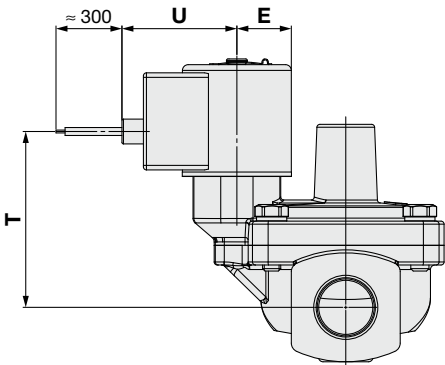
Body MaterialBronze

G: Grommet



GS: Grommet with PCB

CS: Conduit/NPT thread
FS: Conduit/G thread



[mm]											
Size	Port size	A	B	C	D	E	F	I	K	L	M
50	1/2, 3/4	100	18	36	80	18	36	32.5	39	41	36
60	1	119	21	42	90	21	40	36.5	45	45	42
70	1 1/4	127.6	26.5	42	125	21	51.5	43.5	67.5	57.5	53
80	1 1/2	134.5	30	42	132	21	54.5	46.5	72	60	60
90	2	145	35.5	42	150	21	59	52	81	69	70

Size	Grommet		Grommet with PCB		Conduit	
	T	U	T	U	T	U
50	52.4	28.5	58.1	38	59.7	48.9
60	60.4	31.1	66.1	41	67.7	51.9
70	63.4	31.1	69.1	41	70.7	51.9
80	66.8	31.1	72.5	41	74.1	51.9
90	71.8	31.1	77.5	41	79.1	51.9

JSX
JSXD
JSXP
JSXR
JSXZ
JSXM
CE/UKCA-compliance Table
UL-compliance Table
Option
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Glossary of Terms
Flow Rate Characteristics
Specific Product Precautions

JSXR Series

Dimensions:
JSXR
Normally Closed (N.C.)

Body Material

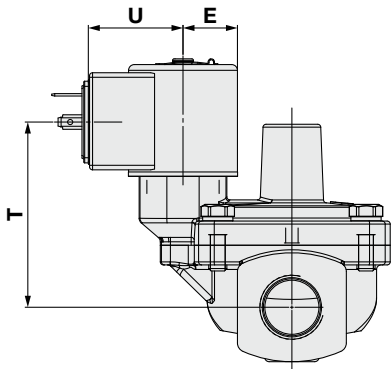
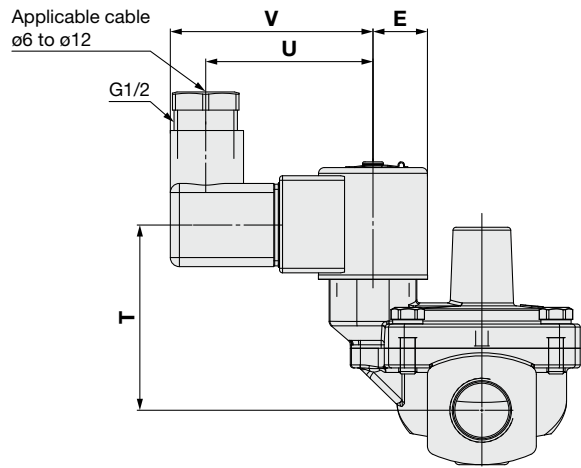
Bronze

DS: DIN terminal

DZ: DIN terminal with light

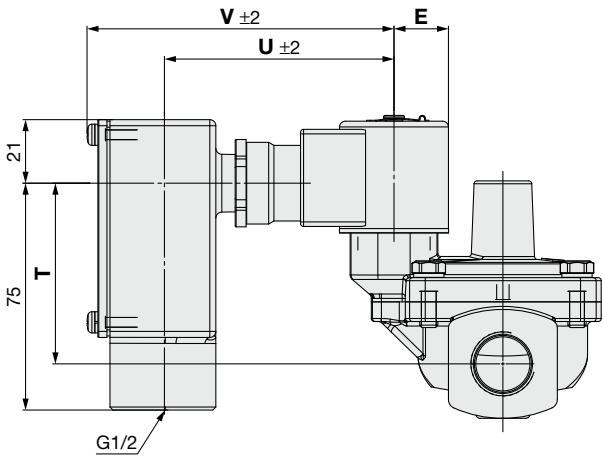
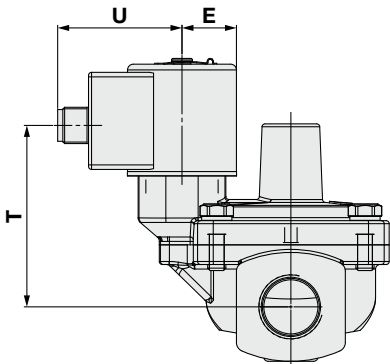
DN: DIN terminal

without connector



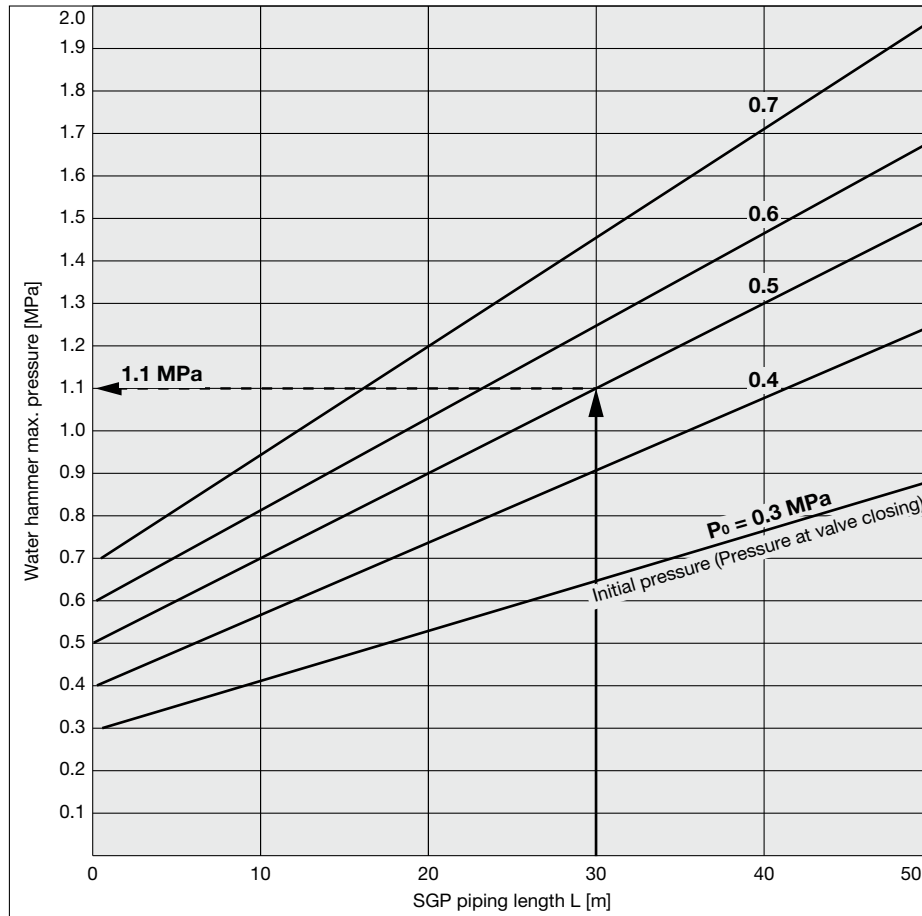
WN: M12 connector

TS: Conduit terminal



Size	E	DIN terminal			DIN terminal without connector		M12 connector		Conduit terminal		
		T	U	V	T	U	T	U	T	U	V
50	18	61.2	55.3	67	61.2	31.3	60	41.1	59.7	75.9	101.5
60	21	69.2	58.3	70	69.2	34.3	68	44.1	67.7	78.9	104.5
70	21	72.2	58.3	70	72.2	34.3	71	44.1	70.7	78.9	104.5
80	21	75.6	58.3	70	75.6	34.3	74.4	44.1	74.1	78.9	104.5
90	21	80.6	58.3	70	80.6	34.3	79.4	44.1	79.1	78.9	104.5

Water Hammer Relieving Characteristics (JSXR51/61)



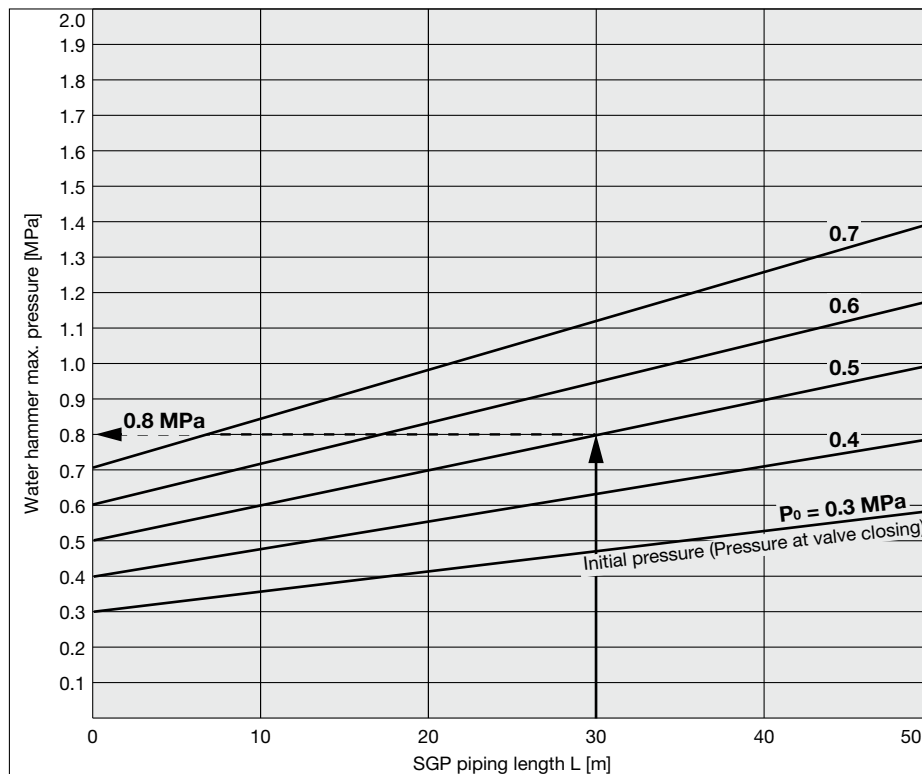
Water hammer

(Example) JSXR series prevents damage of piping, equipment and system and generation of vibration through a great relieving of a water hammer generated using an ordinary solenoid valve.

How to read the graph

When the SGP piping having the same bore as the solenoid valve is 30 m in length, the maximum pressure at the initial pressure of 0.5 MPa results in about 1.1 MPa. (General purpose solenoid valve is 4.0 to 7.0 MPa.)

Water Hammer Relieving Characteristics (JSXR71/81/91)



How to read the graph

When the SGP piping having the same bore as the solenoid valve is 30 m. in the length, the maximum pressure at the initial pressure of 0.5 MPa results in about 0.8 MPa. (General purpose solenoid valve is 2.0 to 4.0 MPa.)

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/JKCA-compliance
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UL-compliance
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Precautions

Zero Differential Pressure Type Pilot Operated 2-Port Solenoid Valve JSXZ Series



Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

Stainless Steel	Brass	Aluminum
Normally Closed (N.C.)		
▶ p. 71		

Stainless Steel	Brass
Normally Open (N.O.)	
▶ p. 73	

How to Order

JSXZ **3** **1** - **C** **N** **02** **R** - **5** **G** - **D** - **B**

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩

① Size

Symbol	Size
3	30
4	40
5	50
6	60

② Valve type

Symbol	Valve type
1	N.C.

③ Body material

Symbol	Body material	Size	
		30	40, 50, 60
C	Brass	●	●
S	Stainless steel	●	●
A	Aluminum	●	—

④ Seal material

Symbol	Seal material
N	NBR
F	FKM
E*1	EPDM

*1 Cannot be used in combination with the aluminum body

⑤ Port size

Symbol	Port size	Size			
		30	40	50	60
02	1/4	●	—	—	—
03	3/8	●	—	—	—
04	1/2	—	●	—	—
06	3/4	—	—	●	—
10	1	—	—	—	●

⑥ Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

⑦ Rated voltage

Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	5	24 VDC	B	24 VAC
2	200 VAC	6	12 VDC	J	230 VAC
3	120 (110) VAC	7	240 VAC		
4	220 VAC	8	48 VAC		

⑧ Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant
G	Grommet*1	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)	
CS	Conduit (With surge voltage suppressor)	
DS	DIN terminal (With surge voltage suppressor)	

Symbol	Electrical entry	CE/UKCA-compliant
DZ	DIN terminal with light (With surge voltage suppressor)	Refer to page 84.
DN	DIN terminal without connector (With surge voltage suppressor)	
WN	M12 connector without connector cable (With surge voltage suppressor)*2	

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

⑨ Oil-free option

Symbol	Option
Nil	None
D	Oil-free

⑩ Bracket option

Symbol	Option
Nil	None
B	With bracket*1

*1 Refer to page 102 for bracket assembly part nos.

Flow Rate Characteristics

Size	Body material	Port size	Orifice diameter [mmø]	Flow rate characteristics*1						Max. operating pressure differential [MPa]	Model	Weight*2 [g]
				Air				Water, Oil				
				C [dm³/(s·bar)]	b	Cv	Effective area [mm²]	Kv	Conversion Cv			
30	Aluminum	1/4	10	8.5	0.44	2.4	—	—		1.0	JSXZ31-A□02	580
		3/8		9.3	0.43	2.6					JSXZ31-A□03	580
	Brass, Stainless steel	1/4		8.5	0.44	2.4		1.6	1.9		JSXZ31- ^C □02	700
		3/8		9.3	0.43	2.6		2.0	2.4		JSXZ31- ^C □03	700
40	Brass, Stainless steel	1/2	15	23	0.34	6.0		4.6	5.3		JSXZ41- ^C □04	820
50	Brass, Stainless steel	3/4	20	36	0.26	9.4		7.8	9.2		JSXZ51- ^C □06	1200
60	Brass, Stainless steel	1	25	—	—	—	185	8.7	10.2		JSXZ61- ^C □10	1400

*1 The flow rate characteristics of this product vary.

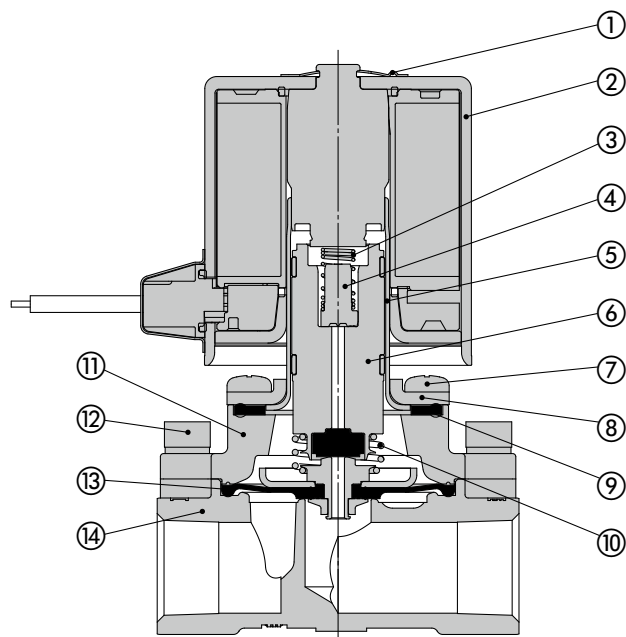
*2 Indicates case of grommet type. Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.

Applicable Fluid Checklist

Applicable fluid	Seal material		
	NBR	FKM	EPDM
Air	●	●	●
Water	●	●	●
Oil	—	●	—

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use.

Construction



Component Parts

No.	Description	Material		
		Aluminum*1	Brass	Stainless steel
1	Clip			Stainless steel
2	Solenoid coil			Stainless steel, Cu, Resin
3	Spring			Stainless steel
4	Stopper			PPS
5	Tube assembly			Stainless steel
6	Armature assembly			Stainless steel, PPS, NBR (FKM, EPDM)
7	Mounting screw			Fe
8	Bonnet			Stainless steel
9	Gasket			NBR (FKM, EPDM)
10	Lift spring			Stainless steel
11	Bonnet	Aluminum	Brass	Stainless steel
12	Bolt			Fe
13	Diaphragm assembly			Stainless steel, NBR (FKM, EPDM)
14	Body	Aluminum	Brass	Stainless steel

*1 Size 30 only

Common Specifications

Series			30		40	50	60
Valve specifications	Body material		Aluminum	Brass, Stainless steel	Brass, Stainless steel		
	Valve construction		Pilot operated diaphragm				
	Valve type		Normally closed (N.C.)				
	Fluid and fluid temperature	Air*1	-10 to 60°C				
		Water, Oil	—	Water: 1 to 60°C (No freezing), Oil: -5 to 60°C (Kinematic viscosity: 50 mm²/s or less)			
	Withstand pressure		2 MPa				
	Max. system pressure		1 MPa				
	Ambient temperature		-20 to 60°C				
	Valve leakage*2/ External leakage*2	Air	15 cm³/min (ANR) or less	1 cm³/min (ANR) or less			
		Water, Oil	—	0.1 cm³/min or less			
	Enclosure*3		IP67 (IP65 for the DIN terminal)				
	Standards*4		CE/UKCA				
Operating environment		Location without the presence of corrosive gases or explosive gases					
Seal material		NBR, FKM, EPDM					
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V				
		DC	12 V, 24 V				
	Allowable voltage fluctuation		±10% of the rated voltage				
	Allowable leakage voltage	AC	5% or less of the rated voltage				
		DC	2% or less of the rated voltage				
	Apparent power*5, *6	AC	9.5 VA			16 VA	
	Power consumption*5	DC	8 W			13 W	
Temperature rise*7		AC/DC	70/65°C			80/75°C	

*1 Dew point temperature: -10°C or less

*2 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*3 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*4 Standards compliance varies depending on the model. For details, refer to page 84.

*5 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*6 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*7 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.



When the differential pressure is less than 0.01 MPa, operation may become unstable. Please contact SMC in case of low-flow operation. (Refer to page 75.)

Zero Differential Pressure Type Pilot Operated 2-Port Solenoid Valve JSXZ Series



Differs depending on the voltage and electrical entry. For details, refer to page 84.

RoHS

Stainless Steel	Brass	Aluminum
Normally Closed (N.C.)		
▶ p. 71		

Stainless Steel	Brass
Normally Open (N.O.)	
▶ p. 73	

How to Order

JSXZ **3** **2** - **C** **N** **02** **R** - **5** **G** - **D** - **B**

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩

① Size

Symbol	Size
3	30
4	40
5	50
6	60

② Valve type

Symbol	Valve type
2	N.O.

③ Body material

Symbol	Body material
S	Stainless steel
C	Brass

④ Seal material

Symbol	Seal material
N	NBR
F	FKM
E	EPDM

⑤ Port size

Symbol	Port size	Size			
		30	40	50	60
02	1/4	●	—	—	—
03	3/8	●	—	—	—
04	1/2	—	●	—	—
06	3/4	—	—	●	—
10	1	—	—	—	●

⑥ Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

⑦ Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

⑧ Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant
G	Grommet*1	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)	
CS	Conduit/NPT thread (With surge voltage suppressor)	
FS	Conduit/G thread (With surge voltage suppressor)	
DS	DIN terminal (With surge voltage suppressor)	

Symbol	Electrical entry	CE/UKCA-compliant
DZ	DIN terminal with light (With surge voltage suppressor)	Refer to page 84.
DN	DIN terminal without connector (With surge voltage suppressor)	
WN	M12 connector without connector cable (With surge voltage suppressor)*2	

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

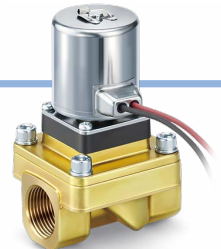
⑨ Oil-free option

Symbol	Option
Nil	None
D	Oil-free

⑩ Bracket option

Symbol	Option
Nil	None
B	With bracket*1

*1 Refer to page 102 for bracket assembly part nos.



Flow Rate Characteristics

Size	Port size	Orifice diameter [mmø]	Flow rate characteristics*1						Max. operating pressure differential [MPa]	Model	Weight*2 [g]
			Air				Water, Oil				
			C [dm³/(s·bar)]	b	Cv	Effective area [mm²]	Kv	Conversion Cv			
30	1/4	10	8.5	0.44	2.4	—	1.6	1.9	0.7	JSXZ32- $\frac{3}{8}$ □02	650
	3/8		9.3	0.43	2.6		2.0	2.4		JSXZ32- $\frac{1}{2}$ □03	
40	1/2	15	23	0.34	6.0		4.6	5.3		JSXZ42- $\frac{1}{2}$ □04	750
50	3/4	20	36	0.26	9.4		7.8	9.2		JSXZ52- $\frac{3}{4}$ □06	1000
60	1	25	—	—	—		185	8.7		10.2	JSXZ62- $\frac{1}{2}$ □10

*1 The flow rate characteristics of this product vary.

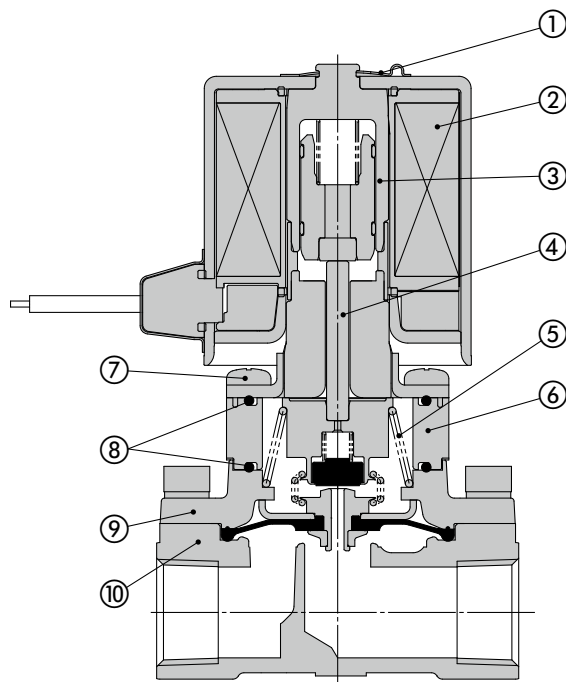
*2 Indicates case of grommet type. Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.

Applicable Fluid Checklist

Applicable fluid	Seal material		
	NBR	FKM	EPDM
Air	●	●	●
Water	●	●	●
Oil	—	●	—

* The list shows the compatibility between general fluids and the seal materials. Consider the operating environment and application sufficiently before selecting the seal material. Fluid and component compatibility should be checked before use.

Construction



Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Sleeve assembly	Stainless steel, PPS
4	Push rod/ Diaphragm assembly	Stainless steel, NBR (FKM, EPDM)
5	Spring	Stainless steel
6	Adapter	PPS
7	Mounting screw	Stainless steel
8	O-ring	NBR (FKM, EPDM)
9	Bonnet	Stainless steel, Brass
10	Body	Stainless steel, Brass

Common Specifications

Size			30	40	50	60
Valve specifications	Valve construction		Pilot operated diaphragm			
	Valve type		Normally open (N.O.)			
	Fluid and fluid temperature		Air: -10 to 60°C (Dew point temperature: -10°C or less) Water: 1 to 60°C (No freezing) Oil: -5 to 60°C (Kinematic viscosity: 50 mm ² /s or less)			
	Withstand pressure		2 MPa			
	Max. system pressure		1 MPa			
	Ambient temperature		-20 to 60°C			
	Valve leakage/ External leakage*1	Air	1 cm ³ /min (ANR) or less			
		Water, Oil	0.1 cm ³ /min or less			
	Mounting orientation		Unrestricted			
	Enclosure*2		IP67 (IP65 for the DIN terminal)			
Coil specifications	Standards*3		CE/UKCA			
	Operating environment		Location without the presence of corrosive gases or explosive gases			
	Seal material		NBR, FKM, EPDM			
	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V			
		DC	12 V, 24 V			
	Allowable voltage fluctuation		±10% of the rated voltage			
	Allowable leakage voltage	AC	5% or less of the rated voltage			
		DC	2% or less of the rated voltage			
	Apparent power*4, *5	AC	9.5 VA		16 VA	
	Power consumption*4	DC	8 W		13 W	
	Temperature rise*6	AC/DC	70/65°C		80/75°C	

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read the "Specific Product Precautions" before handling the product.



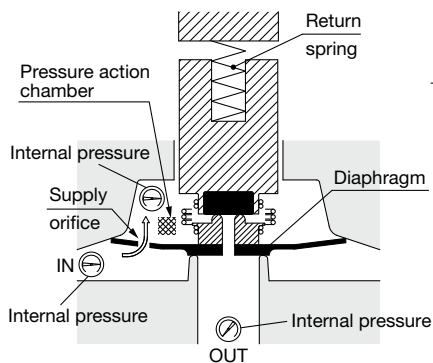
When the differential pressure is less than 0.01 MPa, operation may become unstable. Please contact SMC in case of low-flow operation. (Refer to page 75.)

Working Principle

N.C. type

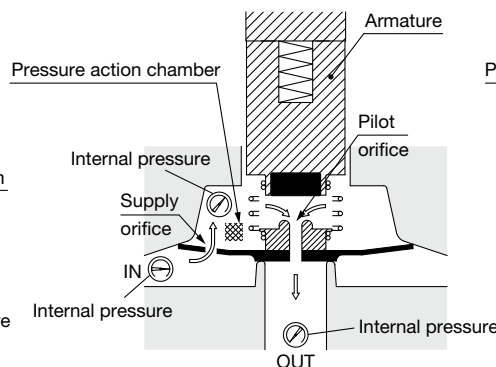
De-energized

The fluid enters from the IN goes through the supply orifice to fill the pressure action chamber. Main valve is closed by the pressure in the pressure action chamber and the reaction force of the return spring.



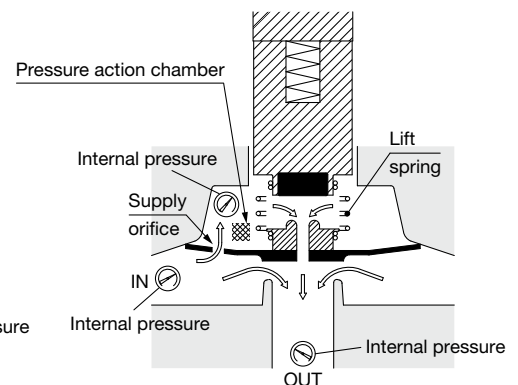
Energized (Pilot valve open)

When the coil is energized, the armature is attracted causing the pilot orifice to opening. The fluid filling the pressure action chamber flows to the OUT side through the pilot orifice.



Energized (Main valve open)

The pressure in the pressure action chamber decreases by discharging fluid through the pilot orifice. Because the force which pushes down the valve is reduced by the discharge of the fluid, the force that pushes up the main valve overcomes the push down force and opens the main valve. The main valve opens by the lift spring reaction force even if pressure on the IN side is 0 MPa or very low pressure.



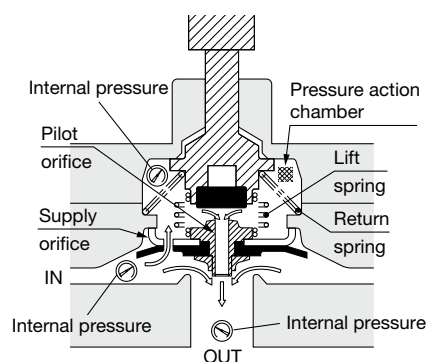
N.O. type

De-energized

The main valve is opened by the return spring reaction force.

For the JSXZ, the main valve is opened by the lift spring reaction force even if the pressure on the IN side is 0 MPa or very low pressure.

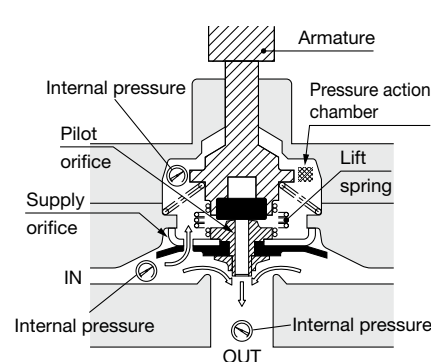
The fluid that flowed into the pressure action chamber flows to the OUT side through the pilot orifice.



Energized (Pilot valve closed)

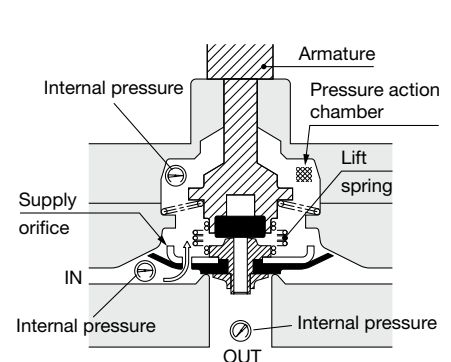
When the coil is energized, the armature is attracted causing the pilot orifice to close.

The fluid enters from the IN goes through the supply orifice to fill the pressure action chamber.



Energized (Main valve closed)

The pressure in the pressure action chamber increases by closing the pilot orifice. When the force that pushes down the valve increases, the main valve closes.



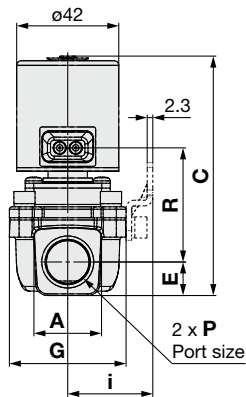
Warning

Unstable flow may occur with the product under the following conditions: • low flow from the pump or compressor, etc. • use of several elbows or tees in the circuit, or • thin nozzles installed at the end of the piping, etc. This can cause valve opening/closing failure, or oscillation, and cause a valve malfunction. If products are used with vacuum, then the vacuum level can be unstable due to these conditions. Please contact SMC to check if the valve can be used in the application by providing the relevant fluid circuit.

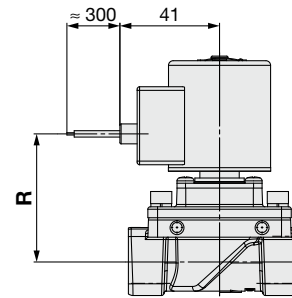
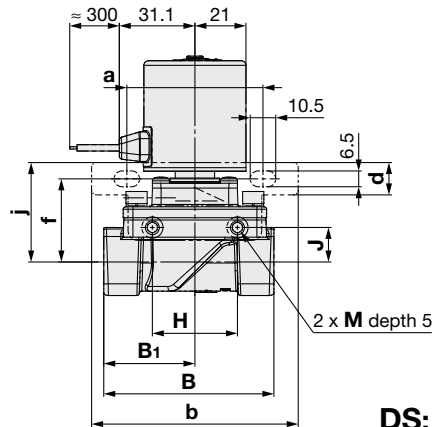
JSXZ30 Port Size **1/4, 3/8** Body Material **Stainless Steel, Brass, Aluminum**

Dimensions: **JSXZ40, 50, 60** Port Size **1/2, 3/4, 1** Body Material **Stainless Steel, Brass**

G: Grommet

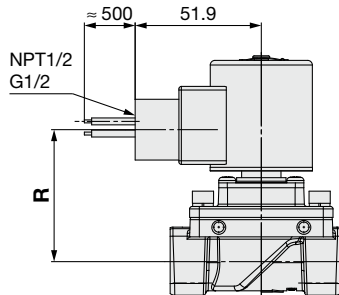


GS: Grommet with PCB



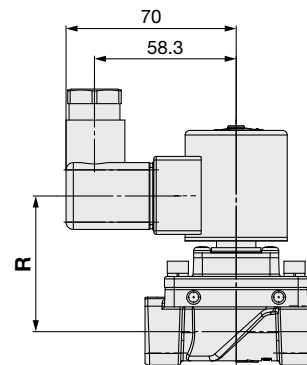
CS: Conduit/NPT thread

FS: Conduit/G thread

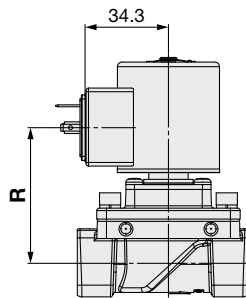


DS: DIN terminal

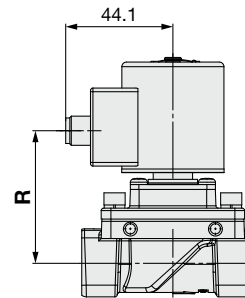
DZ: DIN terminal with light



DN: DIN terminal without connector



WN: M12 connector without cable



Size	Port size P	A	B	B ₁	C	E	G	H	J	M
30	1/4, 3/8	21 <22>	57	28.5	89.8 (96.8)	10.5	40	35	10	M5
40	1/2	28	70	37.5	98.5 (104.8)	13.8	48	35	14.2	M5
50	3/4	33.5	71	38.5	104.6 (111.4)	16.7	62	33	15.2	M6
60	1	42	95	49.5	110.6 (117.3)	19.8	66	37	17.2	M6

Size	Grommet	Grommet with PCB	Conduit	DIN terminal	DIN terminal without connector	M12 connector without cable
	R	R	R	R	R	R
30	41.6 (48.6)	47.3 (54.3)	48.9 (55.9)	50.4 (57.4)	50.4 (57.4)	49.2 (56.2)
40	47 (53.3)	52.7 (59)	54.3 (60.6)	55.8 (62.1)	55.8 (62.1)	54.6 (60.9)
50	50.2 (57)	55.9 (62.7)	57.5 (64.3)	59 (65.8)	59 (65.8)	57.8 (64.6)
60	53.1 (59.8)	58.8 (65.5)	60.4 (67.1)	61.9 (68.6)	61.9 (68.6)	60.7 (67.4)

Size	a	b	d	f	i	j
30	56	75	13.3	30	31	36.7
40	56	75	13.3	34.2	35	40.9
50	70.5	92	18	39	43	45.7
60	70.5	92	18	41	45	47.7

(): Denotes the Normally Open (N.O.) dimensions.
The value in < > is for the aluminum body.
There is no aluminum body option for the N.O. type.

Modular Mounting Type 2-Port Solenoid Valve *JSXM Series*

**CE UK
CA**
Differs depending on the voltage and electrical entry. For details, refer to page 84.

How to Order

JSXM **2** **1** - **A** **N** **301** **R** - **5** **G** - **U** - **F** - **D**

1 2 3 4 5 6 7 8 9 10 11

1 Size

Symbol	Size
2	20
3	30
4	40

2 Valve type

Symbol	Valve type
1	N.C.

3 Body material

Symbol	Body material
A	Aluminum

4 Seal material

Symbol	Seal material
N	NBR
F	FKM

5 Orifice diameter and port size

Symbol	Orifice diameter [mmφ]	Port size	Size		
			20	30	40
301	3.2	1/8	●	—	—
302		1/4	●	—	—
402	4.0	1/4	—	●	●
403		3/8	—	●	●
404		1/2	—	—	●

6 Thread type

Symbol	Thread type
R	Rc
N	NPT
F	G

7 Rated voltage

AC				DC	
Symbol	Rated voltage	Symbol	Rated voltage	Symbol	Rated voltage
1	100 VAC	7	240 VAC	5	24 VDC
2	200 VAC	8	48 VAC	6	12 VDC
3	120 (110) VAC	B	24 VAC		
4	220 VAC	J	230 VAC		

8 Electrical entry

Symbol	Electrical entry	CE/UKCA-compliant
G	Grommet*1	Refer to page 84.
GS	Grommet with PCB (With surge voltage suppressor)	
CS	Conduit (With surge voltage suppressor)	
DS	DIN terminal (With surge voltage suppressor)	
DZ	DIN terminal with light (With surge voltage suppressor)	
DN	DIN terminal without connector (With surge voltage suppressor)	
WN	M12 connector without cable (With surge voltage suppressor)*2	

*1 DC voltage only

*2 A cable for the M12 connector is not included with the product. Refer to the "Option" on page 90 to order it separately.

9 Coil orientation

Symbol	Orientation
Nil	Upward
U	Downward

10 Blow port position

Coil orientation: Upward
(When "Nil" is selected for 9)

Coil orientation: Downward
(When "U" is selected for 9)

Symbol	Position	Symbol	Position
Nil	Bottom	Nil	Top
F	Front	F	Front

11 Oil-free option

Symbol	Option
Nil	None
D	Oil-free

Simple Specials System

A system designed to respond quickly and easily to your special ordering needs

For modular connection units (shipped assembled), the simple specials system can be used.

Short lead times

This system enables us to respond to your special needs (additional machining, accessory assembly, or the designing of a modular unit) and deliver your personalized products as quickly as standard products.

Please contact your local sales representative for more details.

Repeat orders

Once we receive a simple special part number from one of your previous orders, we will process the order, manufacture the product, and deliver it to you as quickly as possible.

Simple
Specials
System

Connection Example



Ordering Example

① Air combination AC20B-02E-D	1 pc.
② Modular mounting type 2-port solenoid valve JSXM21-AN302R-5G-U-F	3 pcs.
③ Spacer with bracket Y200T-D	1 pc.
④ Spacer Y200-D	2 pcs.

Flow Rate Characteristics

Size	Port size	Orifice diameter [mmø]	Flow rate characteristics*1			Max. operating pressure differential [MPa]	Model	Weight*2 [g]
			Air					
			C [dm³/(s·bar)]	b	Cv			
20	1/8	3.2	1.36	0.47	0.40	0.7	JSXM21-A□01	300
	1/4						JSXM21-A□02	300
30	1/4	4.0	1.55	0.59	0.50	1.0	JSXM31-A□02	500
	3/8						JSXM31-A□03	500
40	1/4	4.0	1.55	0.59	0.50	1.0	JSXM41-A□02	630
	3/8						JSXM41-A□03	630
	1/2						JSXM41-A□04	630

*1 The flow rate characteristics of this product vary.

*2 Indicates case of grommet type

Add 20 g for the grommet type with PCB, 70 g for the conduit type, 50 g for the DIN terminal type, and 15 g for the M12 connector type.

Common Specifications

Size			20	30	40
Valve specifications	Valve construction		Direct operated poppet		
	Valve type		Normally closed (N.C.)		
	Fluid and fluid temperature		Air: −10 to 60°C (Dew point temperature: −10°C or less)		
	Withstand pressure		2 MPa		
	Max. system pressure		1 MPa		
	Ambient temperature		−20 to 60°C		
	Valve leakage*1/External leakage*1	Air	1 cm³/min (ANR) or less		
	Mounting orientation		Unrestricted		
	Enclosure*2		IP67 (IP65 for the DIN terminal)		
	Standards*3		CE/UKCA		
	Operating environment		Location without the presence of corrosive gases or explosive gases		
Body material		Aluminum			
Seal material		NBR, FKM			
Coil specifications	Rated voltage	AC	24 V, 48 V, 100 V, 110 V, 120 V, 200 V, 220 V, 230 V, 240 V		
		DC	12 V, 24 V		
	Allowable voltage fluctuation		±10% of the rated voltage		
	Allowable leakage voltage	AC	5% or less of the rated voltage		
		DC	2% or less of the rated voltage		
	Apparent power*4, *5	AC	8 VA	9.5 VA	
	Power consumption*4	DC	6 W	8 W	
	Temperature rise*6	AC/DC	70/65°C		

*1 Leakage: The value at a differential pressure of 0.01 MPa or higher and an ambient temperature of 20°C

*2 This product has an IP67 enclosure, but if water enters the product, it may result in malfunction or breakage.

Therefore, take appropriate measures to prevent water from entering the product when using outdoors or in an environment where it is constantly exposed to water.

*3 Standards compliance varies depending on the model. For details, refer to page 84.

*4 Power consumption/Apparent power: The value at an ambient temperature of 20°C and when the rated voltage is applied (Variation: ±10%)

*5 There is no difference in the frequency and the inrush and energized apparent power, since a rectifying circuit is used in the AC.

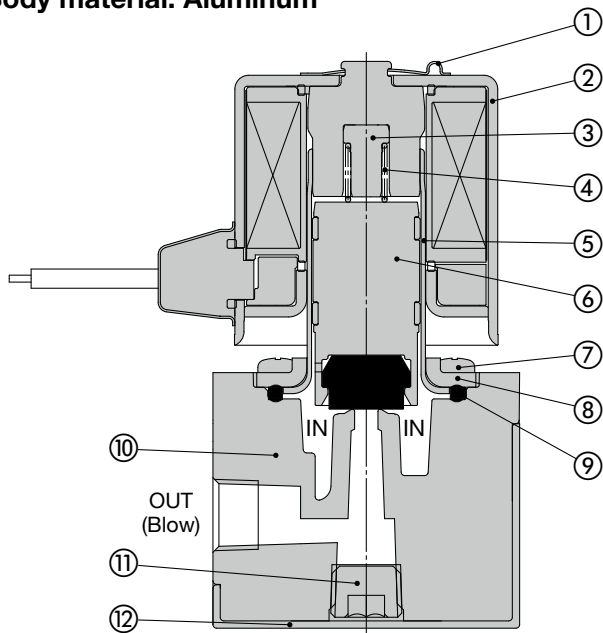
*6 Temperature rise: The value at an ambient temperature of 20°C and when the rated voltage is applied. Use this value as a reference as the actual value varies depending on the ambient environment.

Be sure to read "Specific Product Precautions" before handling the product.

JSXM Series

Construction

JSXM20, 30, 40, Normally closed (N.C.)
Body material: Aluminum

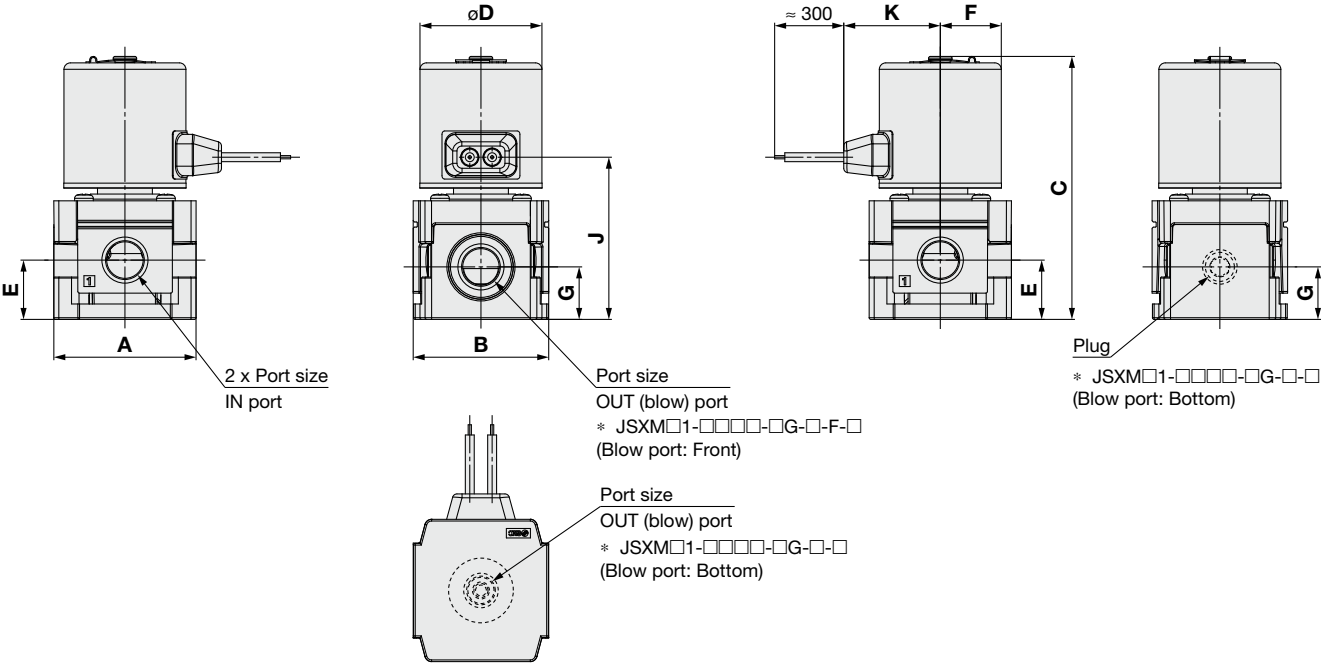


Component Parts

No.	Description	Material
1	Clip	Stainless steel
2	Solenoid coil	Stainless steel, Cu, Resin
3	Stopper	PPS
4	Spring	Stainless steel
5	Tube assembly	Stainless steel
6	Armature assembly	Stainless steel, PPS, NBR, (FKM)
7	Screw	Fe
8	Bonnet	Stainless steel
9	Gasket	NBR, (FKM)
10	Body	Aluminum
11	Plug	Fe
12	Cover	POM

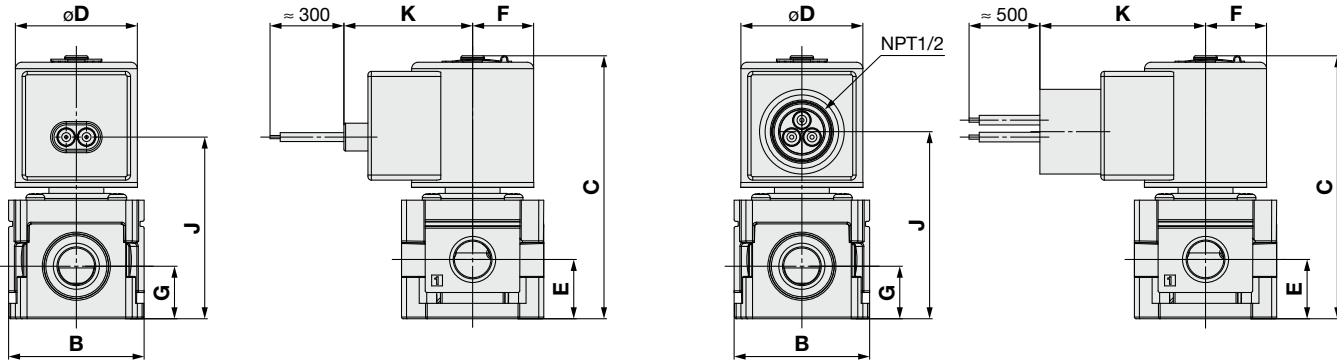
Dimensions

G: Grommet



GS: Grommet with PCB

CS: Conduit



[mm]								
Size	Port size	A	B	C	D	E	F	G
20	1/8, 1/4	42	40	77.6	36	17.5	18	15.5
30	1/4, 3/8	53	53	94.5	42	21.5	21	18
40	1/4, 3/8, 1/2	71	70	102.5	42	25.5	21	22.5

Size	Port size	Grommet		Grommet with PCB		Conduit	
		J	K	J	K	J	K
20	1/8, 1/4	47.9	28.5	53.6	38	55.2	48.9
30	1/4, 3/8	56.8	31.1	62.5	41	64.1	51.9
40	1/4, 3/8, 1/2	64.8	31.1	70.5	41	72.1	51.9

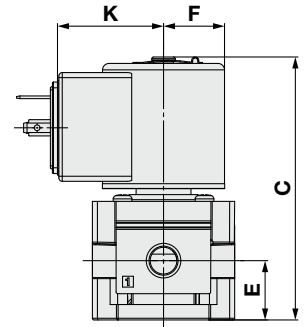
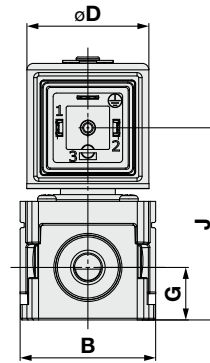
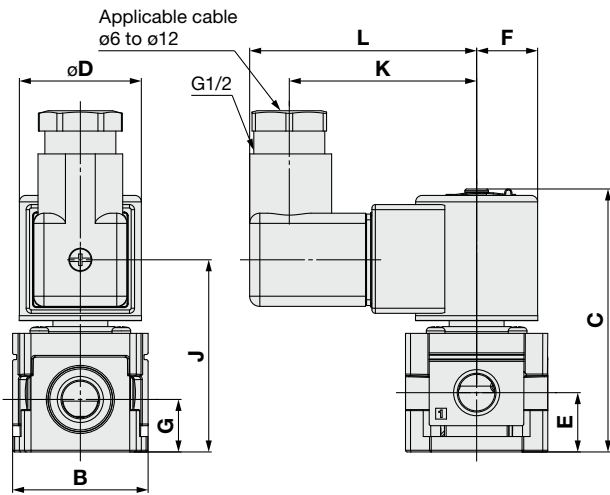
JSXM Series

Dimensions

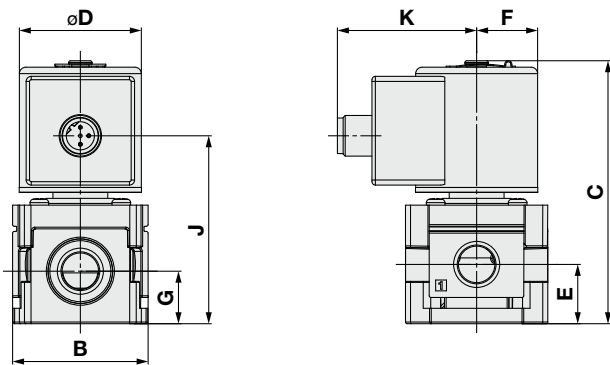
DS: DIN terminal

DS: DIN terminal with light

DN: DIN terminal without connector



WN: M12 connector



[mm]

Size	Port size	A	B	C	D	E	F	G
20	1/8, 1/4	42	40	77.6	36	17.5	18	15.5
30	1/4, 3/8	53	53	94.5	42	21.5	21	18
40	1/4, 3/8, 1/2	71	70	102.5	42	25.5	21	22.5

Size	Port size	DIN terminal			DIN terminal without connector		M12 connector	
		J	K	L	J	K	J	K
20	1/8, 1/4	56.7	55.3	67	56.7	31.3	55.5	41.1
30	1/4, 3/8	65.6	58.3	70	65.6	34.3	64.4	44.1
40	1/4, 3/8, 1/2	73.6	58.3	70	73.6	34.3	72.4	44.1

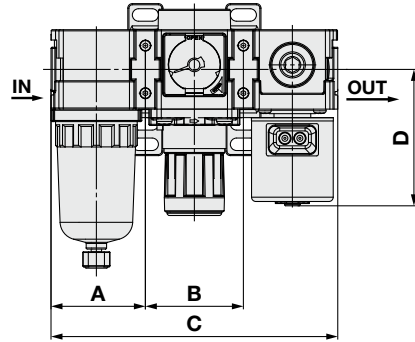
Modular Connection Examples (Dimensions)

Please note that products do not come assembled. They should be ordered separately and assembled by the customer.

For modular connection units (shipped assembled), the simple specials system can be used. For details, refer to page 9.

Combination example ①

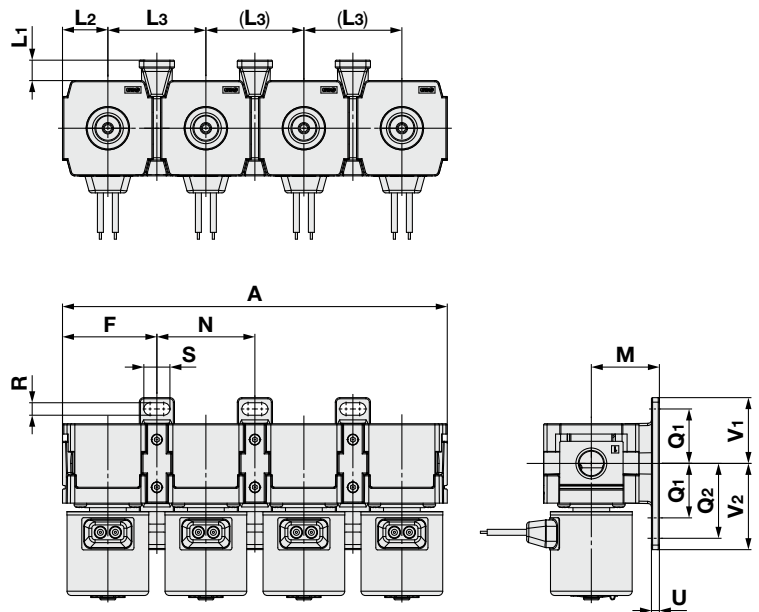
Air combination AC20B-02E-D ————— 1 pc.
 Spacer with bracket Y200T-D ————— 1 pc.
 Modular mounting type 2-port solenoid valve
 JSXM21-AN301R-5G-U-F ————— 1 pc.



Applicable air combination model	A	B	C	D
AC20-D	41.6	43.2	126.4	60.12
AC30-D	55.1	57.2	167.4	73.01
AC40-D	72.6	75.2	220.3	77.01

Combination example ②

Modular mounting type 2-port solenoid valve
 JSXM21-AN301R-5G-U ————— 4 pcs.
 Spacer with bracket Y200T-D ————— 3 pcs.



Series	Bracket mount dimensions													
	A	F	L1	L2	L3	M	N	Q1	Q2	R	S	U	V1	V2
JSXM20	169.6	41.6	9	20	43.2	30	43.2	24	33	5.5	11.5	3.5	29	38
JSXM30	224.6	55.1	14.5	26.4	57.2	41	57.2	35	—	7	14	6	42.5	42.5
JSXM40	295.3	72.55	14.5	34.9	75.1	50	75.1	40	55	9	18	7	50	65

Spacer / Spacer with Bracket

Y **300** **□** - D

① ②

	Symbol	Description	① Body size [Applicable size]		
			200 [JSXM20]	300 [JSXM30]	400 [JSXM40]
② Bracket	Nil	Spacer	●	●	●
	T	Spacer with bracket	●	●	●

Standard Specifications

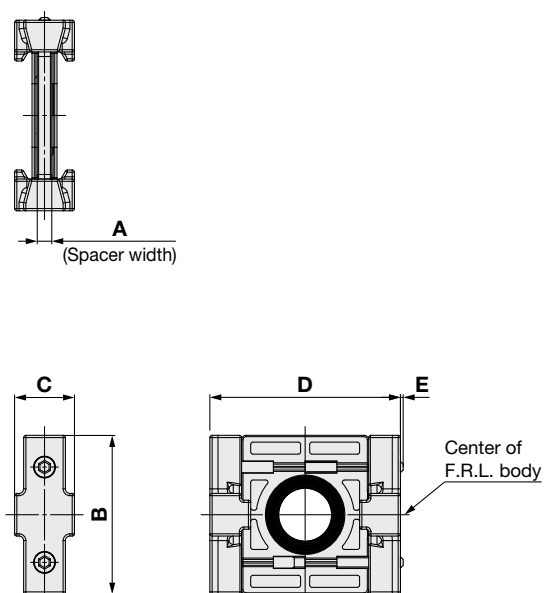
Fluid	Air
Ambient and fluid temperatures	-5 to 60°C (No freezing)
Proof pressure	1.5 MPa
Max. operating pressure	1.0 MPa

Replacement Parts

Description	Material	Part number		
		Y200-D Y200T-D	Y300-D Y300T-D	Y400-D Y400T-D
Seal	HNBR	Y220P-050S	Y320P-050S	Y420P-050S

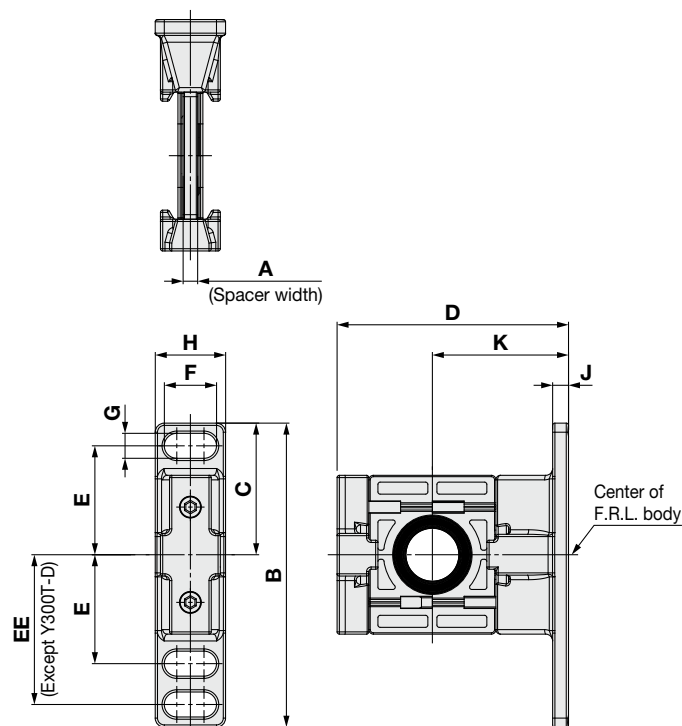
Dimensions

Spacer



Part no.	A	B	C	D	E	Applicable size
Y200-D	3.2	35	13.2	42	0.6	JSXM20
Y300-D	4.2	43	16.2	53	—	JSXM30
Y400-D	5.2	51	19.2	71	—	JSXM40

Spacer with bracket



Part no.	A	B	C	D	E	EE	F	G	H	J	K	Applicable size
Y200T-D	3.2	67	29	51	24	33	11.5	5.5	15.5	3.5	30	JSXM20
Y300T-D	4.2	85	42.5	67.5	35	—	14	7	20	6	41	JSXM30
Y400T-D	5.2	115	50	85.5	40	55	18	9	26	7	50	JSXM40

CE/UKCA-compliance Table

* For CE and UKCA-compliant products, refer to the table below.



G
Grommet



DN
Without DIN connector



GS/GR
Grommet
with PCB



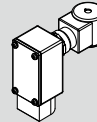
WN
M12 connector/
Without connector cable



**CS/FS/
CR/FR**
Conduit



TS
Conduit terminal



○: Compliant X: Not compliant (Part number available) —: No part number available

Electrical entry	Rated voltage									
	AC specification								DC specification	
	1	2	3	4	7	8	B	J	5	6
	100 VAC	200 VAC	120 VAC (110 VAC)	220 VAC	240 VAC	48 VAC	24 VAC	230 VAC	24 VDC	12 VDC
G Grommet	—	—	—	—	—	—	—	—	○	○
GS/GR Grommet with PCB (With surge voltage suppressor)	○	×	×	×	×	○	○	×	○	○
CS/FS/CR/FR Conduit (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○
DS DIN terminal (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○
DZ DIN terminal with light (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○
DN DIN terminal without connector (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○
WN M12 connector/ Without connector cable (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○
TS Conduit terminal (With surge voltage suppressor)	○	○	○	○	○	○	○	○	○	○

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance
Table

UL-compliance
Table

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JSX10, 20, 30 Series

UL-compliance Table

* Refer to the table below for UL-compliant products.



Recognized

G
Grommet



GS
Grommet
with PCB



DN
DIN terminal
without connector



WN
M12 connector/
Without connector cable



JSX11	Series/Valve type	Body material	Seal material	Orifice diameter/Port size	Thread type	Rated voltage	Electrical entry	Option
	JSX11	S	N F E	101 201	R N F	1 2 3 4 5 6 7 8 B J	G*1 GS DN WN	*

JSX21	Series/Valve type	Body material	Seal material	Orifice diameter/Port size	Thread type	Rated voltage	Electrical entry	Option
	JSX21	S	N F E	301 302 303 402 403 502 503 702 703	R N F	1 2 3 4 5 6 7 8 B J	G*1 GS DN WN	*

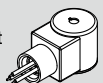
JSX31	Series/Valve type	Body material	Seal material	Orifice diameter/Port size	Thread type	Rated voltage	Electrical entry	Option
	JSX31	S	N F E	402 403 502 503 702 703	R N F	1 2 3 4 5 6 7 8 B J	G*1 GS DN WN	*

*1 Only applicable to rated voltage symbols "5" and "6"



Listed

CS
Conduit



JSX21	Series/Valve type	Body material	Seal material	Orifice diameter/Port size	Thread type	Rated voltage	Electrical entry	Option
	JSX21	S	N F E	301 302 303 402 403 502 503 702 703	R N F	1 2 3 4 5 6 7 8 B J	CS	*

JSX31	Series/Valve type	Body material	Seal material	Orifice diameter/Port size	Thread type	Rated voltage	Electrical entry	Option
	JSX31	S	N F E	402 403 502 503 702 703	R N F	1 2 3 4 5 6 7 8 B J	CS	*

JSXD30, 40, 50, 60, 70, 80, 90 Series

UL-compliance Table

* Refer to the table below for UL-compliant products.



Recognized

G*1
Grommet



*1 Only applicable to rated voltage symbols "5" and "6"

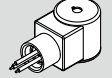
GS
Grommet with PCB



DN
Without DIN connector



CS*2
Conduit



*2 Only applicable to the flange type in sizes 70, 80, and 90

WN
M12 connector/
Without connector cable



JSXD31

Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
JSXD31	C	N	02	R	1	G	None	None
	S	F	03	N	2	GS	D	B
	A	E*3	04	F	3	DN		
					4	WN		
					5			
					6			
					7			
					8			
					B			
					J			

*3 Cannot be used in combination with body material symbol "A"

JSXD41

Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
JSXD41	C	N	03	R	1	G	None	None
	S	F	04	N	2	GS	D	B
		E		F	3	DN		
					4	WN		
					5			
					6			
					7			
					8			
					B			
					J			

JSXD51

Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
JSXD51	C	N	06	R	1	G	None	None
	S	F		N	2	GS	D	B
		E		F	3	DN		
					4	WN		
					5			
					6			
					7			
					8			
					B			
					J			

JSXD61

Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
JSXD61	C	N	10	R	1	G	None	None
	S	F		N	2	GS	D	B
		E		F	3	DN		
					4	WN		
					5			
					6			
					7			
					8			
					B			
					J			

JSXD71

Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option
JSXD71	B	N	12	R	1	G	None
		F		N	2	GS	D
		E		F	3	DN	
					4	WN	
					5		
					6		
					7		
					8		
					B		
					J		

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance Table

UL-compliance Table

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JSXD30, 40, 50, 60, 70, 80, 90 Series



Recognized

G*1
Grommet



*1 Only applicable to rated voltage symbols "5" and "6"

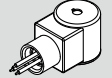
GS
Grommet with PCB



DN
Without DIN connector



CS*2
Conduit



*2 Only applicable to the flange type in sizes 70, 80, and 90

WN
M12 connector/
Without connector cable



JSXD71

Series/ Valve type	Body material	Seal material	Port size
JSXD71	B	N	32
		F	
		E	

Rated voltage	Electrical entry	Oil-free option
1	G	None
2	GS	D
3	DN	
4	CS	
5	WN	
6		
7		
8		
B		
J		

JSXD81

Series/ Valve type	Body material	Seal material	Port size	Thread type
JSXD81	B	N	14	R
		F		N
		E		F

Rated voltage	Electrical entry	Oil-free option
1	G	None
2	GS	D
3	DN	
4	WN	
5		
6		
7		
8		
B		
J		

JSXD81

Series/ Valve type	Body material	Seal material	Port size
JSXD81	B	N	40
		F	
		E	

Rated voltage	Electrical entry	Oil-free option
1	G	None
2	GS	D
3	DN	
4	CS	
5	WN	
6		
7		
8		
B		
J		

JSXD91

Series/ Valve type	Body material	Seal material	Port size	Thread type
JSXD91	B	N	20	R
		F		N
		E		F

Rated voltage	Electrical entry	Oil-free option
1	G	None
2	GS	D
3	DN	
4	WN	
5		
6		
7		
8		
B		
J		

JSXD91

Series/ Valve type	Body material	Seal material	Port size
JSXD91	B	N	50
		F	
		E	

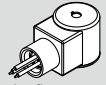
Rated voltage	Electrical entry	Oil-free option
1	G	None
2	GS	D
3	DN	
4	CS	
5	WN	
6		
7		
8		
B		
J		

UL-compliance Table **JSXD30, 40, 50, 60, 70, 80, 90 Series**



Listed

CS*1
Conduit



*1 Excludes the flange type
in sizes 70, 80, and 90

JSXD31	Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
	JSXD31	C	N	02	R	1	CS	None	None
		S	F	03	N	2		D	B
		A	E*2	04	F	3			
*2 Cannot be used in combination with body material symbol "A"									
						4			
						5			
						6			
						7			
						8			
						B			
						J			
JSXD41	Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
	JSXD41	C	N	03	R	1	CS	None	None
		S	F	04	N	2		D	B
			E		F	3			
						4			
						5			
						6			
						7			
						8			
						B			
						J			
JSXD51	Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
	JSXD51	C	N	06	R	1	CS	None	None
		S	F		N	2		D	B
			E		F	3			
						4			
						5			
						6			
						7			
						8			
						B			
						J			
JSXD61	Series/ Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	Bracket option
	JSXD61	C	N	10	R	1	CS	None	None
		S	F		N	2		D	B
			E		F	3			
						4			
						5			
						6			
						7			
						8			
						B			
						J			
JSXD71	Series/Valve type	Body material	Seal material	Port size	Thread type	Rated voltage	Electrical entry	Oil-free option	
	JSXD71	B	N	12	R	1	CS	None	
			F		N	2		D	
			E		F	3			
						4			
						5			
						6			
						7			
						8			
						B			
						J			

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

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JSXD30, 40, 50, 60, 70, 80, 90 Series



*1 Excludes the flange type
in sizes 70, 80, and 90

JSXD81		Series/ Valve type		Body material		Seal material		Port size		Thread type		Rated voltage		Electrical entry		Oil-free option	
		JSXD81		B		N		14		R		1		CS		None	
						F				N		2				D	
						E				F		3					
												4					
												5					
												6					
												7					
												8					
												B					
												J					

JSX/JSX□ Series Option

Cable for M12 Connector (Female Connector with Cable)

The solenoid valve does not come with a cable for the M12 connector.
Please order it separately if necessary.

JSX022-30-1-1

● Specification

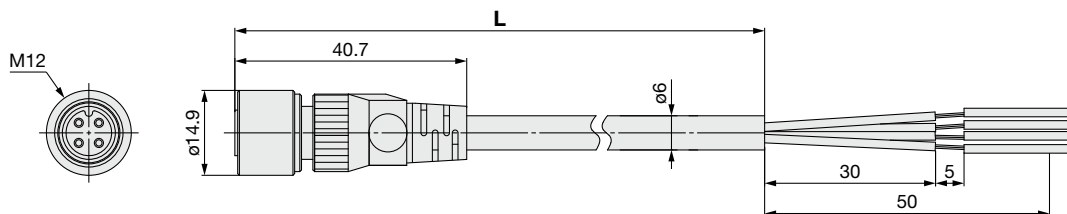
1	For DC voltages
2	For AC voltages

● Cable length L [mm]

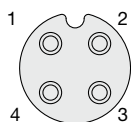
1	1000
2	2000
5	5000

Specifications

Part number		JSX022-30-1-□	JSX022-30-2-□
Key type		A-coded	B-coded
Rating/Performance	Rated current	4 A	
	Rated voltage	250 V	
	Contact resistance	40 mΩ or less	
	Insulation resistance	1000 MΩ or more	
	Withstand voltage	1500 VAC	
	Operating temperature range	-25 to 70°C	
	Min. bending radius (Fixed)	50 mm	
	Protection class	IP67 (Only with screw tightened)	
Material	Allowable repeated insertion/withdrawal	200	
	Material of knurl	Brass (Ni plating)	
	Contact (Surface treatment)	Copper alloy (Au plating)	
	Connector material	PBT	
	Cover	Soft PBT	



**For DC voltages
(A-coded)**

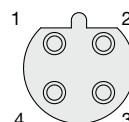


**Socket connector
pin arrangement**

Terminal no.	Core wire color
1	Brown: Unused
2	White: Unused
3	Blue: Valve power supply
4	Black: Valve power supply

Connections

**For AC voltages
(B-coded)**



**Socket connector
pin arrangement**

Terminal no.	Core wire color
1	Brown: Grounding
2	White: Unused
3	Blue: Valve power supply
4	Black: Valve power supply

Connections

* The solenoid valve has no polarity for DC voltages.
However, the high flow/ power saving type has
polarity. Refer to the "Electrical Circuits" on page 106.

JSX/JSX□ Series Replacement Parts

Solenoid Coil Assembly (Applicable to the JSX, JSX□□U, JSX□□V, JSX□□H, JSX□□S, JSXD, JSXP, JSXR, JSXZ, and JSXM series)

When ordering, be sure to add the “-KT1” suffix to the end of the part number of the valve currently in use.

JSX 2 1 - S N 101 R - 5 G - B - KT1

● Enter the standard product number.

JSX (Stainless steel/Brass).....	p. 13	JSX□□S (Steam).....	p. 39
JSX (Aluminum)	p. 15	JSXD.....	p. 43
JSX (N.O.).....	p. 17	JSXP.....	p. 57
JSX□□U (High flow/ Power saving)....	p. 19	JSXR.....	p. 65
JSX□□V (Vacuum)	p. 23	JSXZ	p. 71
JSX□□H (High pressure)	p. 25	JSXM	p. 77

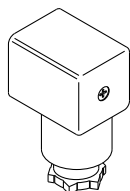
The solenoid coil assembly is shipped with a name plate with the valve part number printed on it. In addition, the name plate has the marks of all applicable standards printed on it.

For the solenoid coil assembly, eligibility for CE/UKCA marking and UL/CSA standard certification varies depending on the electrical entry type and the rated voltage.

When ordering a solenoid coil assembly with different specifications than the valve currently in use, refer to the “How to Order” in the catalog to confirm the status of standard compliance.

For solenoid coil replacement instructions, refer to “Specific Product Precautions 9” on page 108.

DIN Connector Part No.



<For JSX20/30, JSXD, JSXZ, JSXR, JSXM>

Electrical option	Rated voltage	Connector part no.
None	24 VDC	3G-GDM2A
	12 VDC	
	100 VAC	
	120 (110) VAC	
	200 VAC	
	220 VAC	
	230 VAC	
	240 VAC	
	24 VAC	
	48 VAC	
With light	24 VDC	GDM2A-L5
	12 VDC	GDM2A-L6
	100 VAC	GDM2A-L1
	120 (110) VAC	GDM2A-L1
	200 VAC	GDM2A-L2
	220 VAC	GDM2A-L2
	230 VAC	GDM2A-L2
	240 VAC	GDM2A-L2
	24 VAC	GDM2A-L5
	48 VAC	GDM2A-L15

* Contact SMC for details on the type for the JSXZ series.

<For JSX10>

Electrical option	Rated voltage	Connector part no.
None	24 VDC	JSX021-1-18
	12 VDC	
	100 VAC	
	120 (110) VAC	
	200 VAC	
	220 VAC	
	230 VAC	
	240 VAC	
	24 VAC	
	48 VAC	
With light	24 VDC	SY100-82-3-05
	12 VDC	SY100-82-3-06
	100 VAC	SY100-82-2-01
	120 (110) VAC	SY100-82-2-03
	200 VAC	SY100-82-2-02
	220 VAC	SY100-82-2-04
	230 VAC	SY100-82-2-04
	240 VAC	SY100-82-2-04

* Contact SMC for details on the 24 and 48 VAC types with a light for the JSX10.

Gasket Part No. for DIN Connector

VCW20-1-29-1 (For JSX20/30, JSXD, JSXZ, JSXR, JSXM)

* Contact SMC for details on the type for the JSXZ or JSX10.

Clip (Applicable to the JSX, JSXD, JSXZ, JSXP, JSXR, and JSXM series)

For JSX10 VDW20-10

For JSX20/30, JSXD, JSXZ, JSXP, JSXR, JSXM VX021N-10S

JSX/JSX□ Series

Glossary of Terms

Pressure Terminology

1. Max. operating pressure differential

The max. pressure differential (the difference between the inlet and outlet pressure) which is allowed for operation. When the outlet pressure is 0 MPa, this becomes the max. operating pressure.

2. Min. operating pressure differential

The min. pressure differential (the difference between the inlet pressure and outlet pressure) required to keep the main valve fully open.

3. Max. system pressure

The max. pressure that can be applied inside the pipelines (line pressure).

[The pressure differential of the solenoid valve portion must not exceed the max. operating pressure differential.]

4. Withstand pressure

The pressure in which the valve must be withstood without a drop in performance after holding for one minute under prescribed pressure and returning to the operating pressure range. (value under the prescribed conditions)

Electrical Terminology

1. Apparent power (VA)

Volt-ampere is the product of voltage (V) and current (A).

Power consumption (W): For AC, $W = V \cdot A \cdot \cos \theta$.

For DC, $W = V \cdot A$.

* $\cos \theta$ shows power factor. $\cos \theta \approx 0.9$

2. Surge voltage

A high-voltage which is momentarily generated by shutting off the power in the shut-off area.

3. Degrees of protection

A degree defined in the "JIS C 0920: Waterproof test of electric machinery/appliance and the degree of protection against the intrusion of solid foreign objects."

IP –

First digit • Second digit

● First Digit:

Degree of protection against solid foreign objects

0	Not protected
1	Protected against solid foreign objects of 50 mmø and larger
2	Protected against solid foreign objects of 12 mmø and larger
3	Protected against solid foreign objects of 2.5 mmø and larger
4	Protected against solid foreign objects of 1.0 mmø and larger
5	Dust protected
6	Dust-tight

● Second Digit:

Degree of protection against water

0	Not protected	—
1	Protected against vertically falling water droplets	Drip-proof type 1
2	Protected against vertically falling water droplets when enclosure is tilted up to 15°	Drip-proof type 2
3	Protected against rainfall when enclosure is tilted up to 60°	Rainproof type
4	Protected against splashing water	Splashproof type
5	Protected against water jets	Water-jet-proof type
6	Protected against powerful water jets	Powerful water-jet-proof type
7	Protected against the effects of temporary immersion in water	Immersible type
8	Protected against the effects of continuous immersion in water	Submersible type

Others

1. Material

NBR: Nitrile rubber

FKM: Fluororubber

EPDM: Ethylene propylene rubber

2. Symbol

In the symbol () , when the valve is closed, flow is blocked from port 1 to port 2. However, if the pressure in port 2 is higher than port 1, the valve will not be able to block the fluid and it will flow from port 2 to port 1.

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Solenoid Valve Flow Rate Characteristics

(How to indicate flow rate characteristics)

Scan the QR code to access software
for easy flow rate calculation.
For details ►



1. Indication of flow rate characteristics

The flow rate characteristics of equipment, such as a solenoid valve, etc., are indicated in their specifications as shown in Table (1).

Table (1) Indication of Flow Rate Characteristics

Corresponding equipment	Indication by international standard	Other indications	Compliant standards
Pneumatic equipment	C, b	—	ISO 6358:1989 JIS B 8390:2000
	—	S	JIS B 8390:2000 Equipment: JIS B 8379, 8381-1, 8381-2
		Cv	ANSI/(NFPA)T3.21.3 R1-2008
Process fluid control equipment	Kv	—	IEC 60534-1:2005 IEC 60534-2-3:1997 JIS B 2005-1:2012
	—	Cv	JIS B 2005-2-3:2004 Equipment: JIS B 8471, 8472, 8473

2. Pneumatic equipment

2.1 Indication according to the international standards

(1) Compliant standards

**ISO 6358:1989 : Pneumatic fluid power—Components using compressible fluids—
Determination of flow rate characteristics**

**JIS B 8390:2000 : Pneumatic fluid power—Components using compressible fluids—
How to test flow rate characteristics**

(2) Definition of flow rate characteristics

The flow rate characteristics are indicated as a result of a comparison between the sonic conductance **C** and the critical pressure ratio **b**.

Sonic conductance C : Value which divides the passing mass flow rate of a piece of equipment in a choked flow condition by the product of the upstream absolute pressure and the density in a standard condition.

Critical pressure ratio b : Pressure ratio (downstream pressure/upstream pressure) which will turn to a choked flow when the value is smaller than this ratio.

Choked flow : Flow in which the upstream pressure is higher than the downstream pressure and where sonic speed in a certain part of a piece of equipment is reached.

Gaseous mass flow rate is in proportion to the upstream pressure and not dependent on the downstream pressure.

Subsonic flow : Flow greater than the critical pressure ratio.

Standard condition : Air in a temperature state of 20°C, absolute pressure 0.1 MPa (= 100 kPa = 1 bar), relative humidity 65%.

It is stipulated by adding the “(ANR)” after the unit depicting air volume.

(Standard reference atmosphere)

Compliant standards: ISO 8778:1990 Pneumatic fluid power—Standard reference

atmosphere, JIS B 8393:2000: Pneumatic fluid power—Standard reference atmosphere

(3) Formula for flow rate

It is described by the practical units as following.

When

$$\frac{P_2 + 0.1}{P_1 + 0.1} \leq b, \text{ choked flow}$$

$$Q = 600 \times C (P_1 + 0.1) \sqrt{\frac{293}{273 + T}} \quad \text{.....(1)}$$

When

$$\frac{P_2 + 0.1}{P_1 + 0.1} > b, \text{ subsonic flow}$$

$$Q = 600 \times C (P_1 + 0.1) \sqrt{1 - \left[\frac{\frac{P_2 + 0.1}{P_1 + 0.1} - b}{1 - b} \right]^2} \sqrt{\frac{293}{273 + T}} \quad \text{.....(2)}$$

Q : Air flow rate [L/min (ANR)]

C : Sonic conductance [dm³/(s·bar)], dm³ (Cubic decimeter) of SI units = L (liter)

b : Critical pressure ratio [—]

P₁ : Upstream pressure [MPa]

P₂ : Downstream pressure [MPa]

T : Temperature [°C]

* Formula of subsonic flow is the elliptic analogous curve.

Flow rate characteristics are shown in Graph (1). For details, please use the calculation software available from the SMC website.

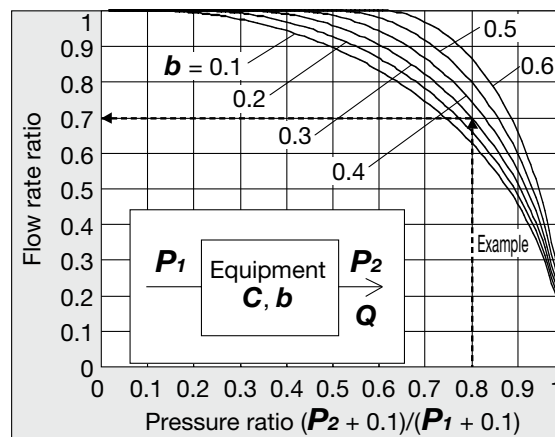
Example)

Obtain the air flow rate for **P₁** = 0.4 [MPa], **P₂** = 0.3 [MPa], **T** = 20 [°C] when a solenoid valve is performed in **C** = 2 [dm³/(s·bar)] and **b** = 0.3.

According to formula 1, the max. flow rate = $600 \times 2 \times (0.4 + 0.1) \times \sqrt{\frac{293}{273 + 20}} = 600$ [L/min (ANR)]

Pressure ratio = $\frac{0.3 + 0.1}{0.4 + 0.1} = 0.8$

Based on Graph (1), it will be 0.7 if the pressure ratio is 0.8 and the flow rate ratio is **b** = 0.3.
Hence, the flow rate = Max. flow x flow ratio = 600 x 0.7 = 420 [L/min (ANR)]



Graph (1) Flow rate characteristics

(4) Test method

Connect the piece of test equipment to the test circuit as shown in Fig. (1). While maintaining the upstream pressure at a fixed value above 0.3 MPa, measure the max. flow to be saturated initially. Next, measure this flow rate at 80%, 60%, 40%, and 20%, as well as the upstream and downstream pressure. The sonic conductance **C** can be calculated based on this max. flow rate. Use the data of the others and the subsonic flow formula to find **b**, and calculate the critical pressure ratio **b** from that average.

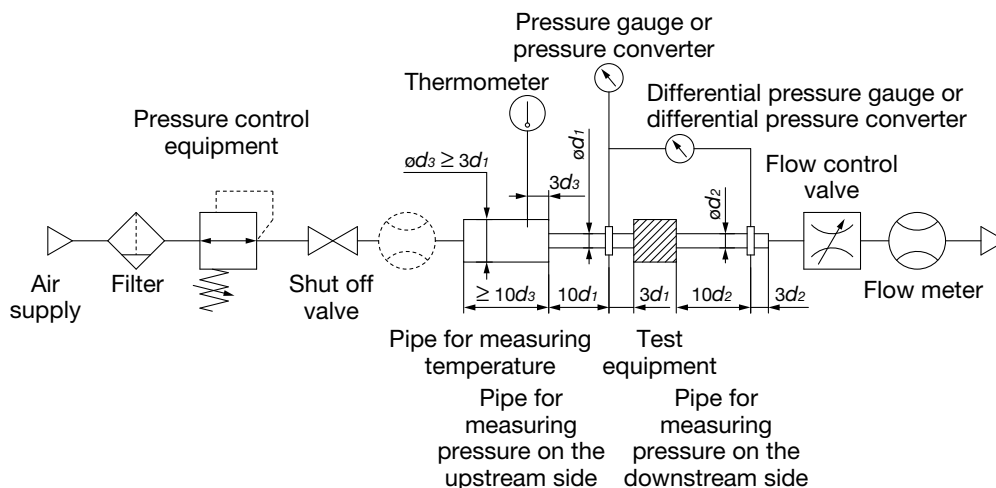


Fig. (1) Test circuit based on ISO 6358:1989, JIS B 8390:2000

2.2 Effective area **S**

(1) Compliant standards

JIS B 8390:2000: Pneumatic fluid power—Components using compressible fluids—

How to test flow rate characteristics

Equipment standards: JIS B 8373: Solenoid valve for pneumatics

JIS B 8379: Silencer for pneumatics

JIS B 8381-1: Fittings for pneumatics—Part 1: Push-in fittings for thermoplastic resin tubing

JIS B 8381-2: Fittings for pneumatics—Part 2: Compression fittings for thermoplastic resin tubing

(2) Definition of flow rate characteristics

Effective area **S**: Cross-sectional area that has an ideal throttle without friction or reduced flow. The value is derived by calculating pressure changes inside of an air tank when the compressed air is discharged from a piece of equipment mounted on the tank in a choked flow. The value of the effective area **S**, like that of sonic conductance **C**, expresses the “ease of flow.”

(3) Formula for flow rate

When

$$\frac{P_2 + 0.1}{P_1 + 0.1} \leq 0.5, \text{ choked flow}$$

$$Q = 120 \times S (P_1 + 0.1) \sqrt{\frac{293}{273 + T}} \quad \text{.....(3)}$$

When

$$\frac{P_2 + 0.1}{P_1 + 0.1} > 0.5, \text{ subsonic flow}$$

$$Q = 240 \times S \sqrt{(P_2 + 0.1) (P_1 - P_2)} \sqrt{\frac{293}{273 + T}} \quad \text{.....(4)}$$

Conversion with sonic conductance **C**:

$$S = 5.0 \times C \quad \text{.....(5)}$$

Q : Air flow rate [L/min (ANR)]

S : Effective area [mm²]

P₁ : Upstream pressure [MPa]

P₂ : Downstream pressure [MPa]

T : Temperature [°C]

* The formula for subsonic flow (4) is only applicable when the critical pressure ratio **b** is the unknown piece of equipment. In the sonic conductance **C** formula (2), it is the same formula as when **b** = 0.5.

(4) Test method

Connect the piece of test equipment to the test circuit as shown in Fig. (2). Discharge the air from the air tank filled with compressed air at a fixed value above 0.6 MPa (0.5 MPa) into the atmosphere until the pressure inside the tank falls to 0.25 MPa (0.2 MPa). Measure the discharge time and the residual pressure inside the tank after discharging until it has returned to the normal value. Then, calculate the effective area **S** using the following formula. Select an air tank with a volume within the specified range of the test equipment's effective area. For JIS B 8379, the pressure values are in parentheses and the coefficient of the formula is 12.9.

$$S = 12.1 \frac{V}{t} \log_{10} \left(\frac{P_s + 0.1}{P + 0.1} \right) \sqrt{\frac{293}{T}} \quad \text{.....(6)}$$

S : Effective area [mm²]

V : Air tank capacity [L]

t : Discharging time [s]

P_s : Pressure inside air tank before discharging [MPa]

P : Residual pressure inside air tank after discharging [MPa]

T : Temperature inside air tank before discharging [K]

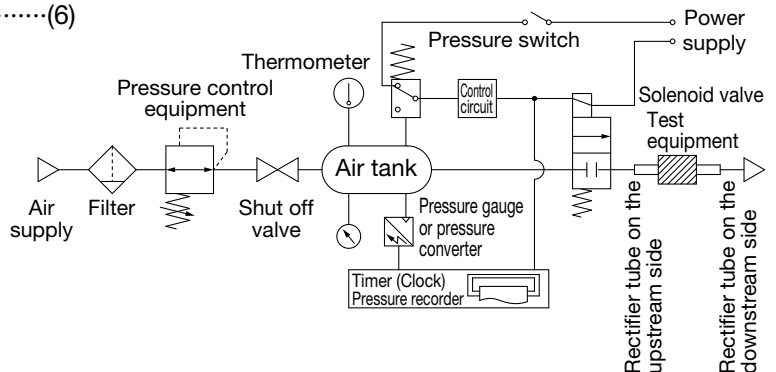


Fig. (2) Test circuit based on JIS B 8390:2000

2.3 Flow coefficient **Cv** factor

The United States Standard ANSI/(NFPA)T3.21.3:R1-2008R: Pneumatic fluid power—Flow rating test procedure and reporting method for fixed orifice components

This standard defines the **Cv** factor of the flow coefficient by the following formula that is based on the test conducted by the test circuit analogous to ISO 6358.

$$Cv = \frac{Q}{114.5 \sqrt{\frac{\Delta P (P_2 + P_a)}{T_1}}} \quad (7)$$

ΔP : Pressure drop between the static pressure tapping ports [bar]

P_1 : Pressure of the upstream tapping port [bar gauge]

P_2 : Pressure of the downstream tapping port [bar gauge]: $P_2 = P_1 - \Delta P$

Q : Flow rate [L/s standard condition]

P_a : Atmospheric pressure [bar absolute]

T_1 : Upstream absolute temperature [K]

The test conditions are $P_1 + P_a = 6.5 \pm 0.2$ bar absolute, $T_1 = 297 \pm 5$ K, $0.07 \text{ bar} \leq \Delta P \leq 0.14 \text{ bar}$.

This is the same concept as the effective area **A** which ISO 6358 stipulates as being applicable only when the pressure drop is smaller than the upstream pressure and the compression of air does not become a problem.

3. Process fluid control equipment

(1) Compliant standards

IEC 60534-1:2005: Industrial-process control valves. Part 1: Control valve terminology and general considerations

IEC 60534-2-3:1997: Industrial-process control valves. Part 2: Flow capacity, Section Three-Test procedures

JIS B 2005-1:2012: Industrial-process control valves – Part 1: Control valve terminology and general considerations

JIS B 2005-2-3:2004: Industrial-process control valves – Part 2: Flow capacity – Section 3: Test procedures

Equipment standards: JIS B 8471: Solenoid valve for water

JIS B 8472: Solenoid valve for steam

JIS B 8473: Solenoid valve for fuel oil

(2) Definition of flow rate characteristics

Kv factor: Value of the clean water flow rate (represented by m³/h) which runs through a valve (test equipment) at 5 to 40°C when the pressure difference is 1×10^5 Pa (1 bar). It is calculated using the following formula.

$$Kv = Q \sqrt{\frac{1 \times 10^5}{\Delta P} \cdot \frac{\rho}{1000}} \quad (8)$$

Kv : Flow coefficient [m³/h]

Q : Flow rate [m³/h]

ΔP : Pressure difference [Pa]

ρ : Density of fluid [kg/m³]

(3) Formula of flow rate

It is described by practical units. Also, the flow rate characteristics are shown in Graph (2).

In the case of liquids:

$$Q = 53 Kv \sqrt{\frac{\Delta P}{G}} \quad (9)$$

Q : Flow rate [L/min]

Kv : Flow coefficient [m³/h]

ΔP : Pressure difference [MPa]

G : Relative density [water = 1]

In the case of saturated aqueous vapor:

$$Q = 232 Kv \sqrt{\Delta P (P_2 + 0.1)} \quad (10)$$

Q : Flow rate [kg/h]

Kv : Flow coefficient [m³/h]

ΔP : Pressure difference [MPa]

P_1 : Upstream pressure [MPa]: $\Delta P = P_1 - P_2$

P_2 : Downstream pressure [MPa]

Conversion of flow coefficient:

$$K_v = 0.865 C_v \dots\dots\dots(11)$$

Here,

C_v factor: Value of the clean water flow rate (represented by US gal/min) which runs through a valve at 40 to 100°F when the pressure difference is 1 lbf/in² (psi)

The values of **K_v** and **C_v** factors for pneumatic purposes are different due to different test methods.

(4) Test method

Connect the piece of test equipment to the test circuit as shown in Fig. (3), and run water at 5 to 40°C. Then, measure the flow rate with a pressure difference where vaporization does not occur in a turbulent flow (pressure difference of 0.035 MPa to 0.075 MPa when the inlet pressure is within 0.15 MPa to 0.6 MPa). However, as the turbulent flow is definitely caused, the pressure difference needs to be set with a large enough difference so that the Reynolds number does not fall below 1×10^5 , and the inlet pressure needs to be set slightly higher to prevent vaporization of the liquid. Substitute the measurement results in formula (8) to calculate **K_v**.

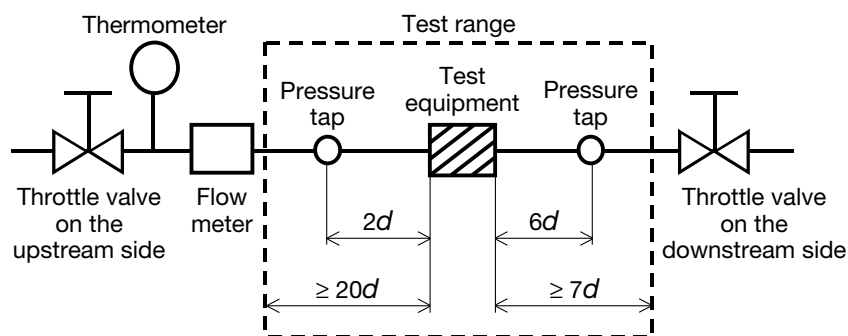
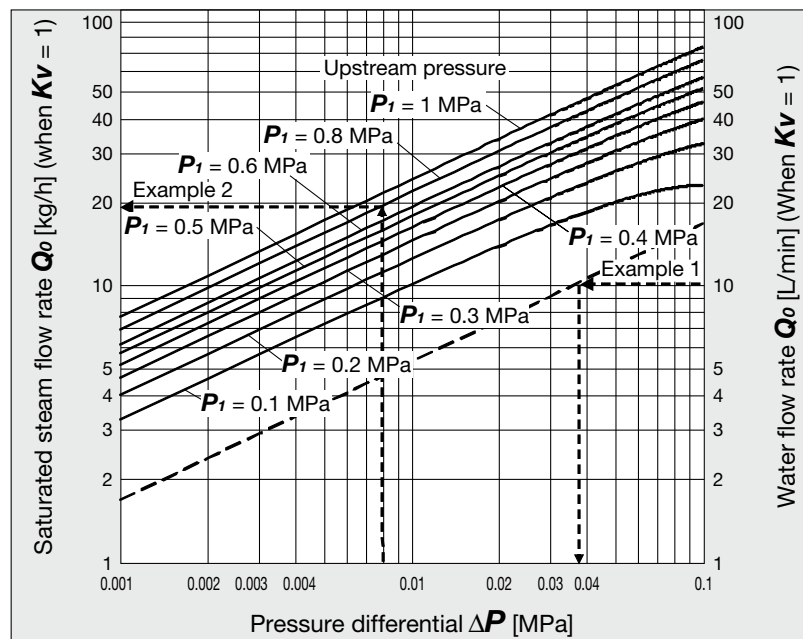


Fig. (3) Test circuit based on IEC 60534-2-3, JIS B 2005-2-3



Graph (2) Flow rate characteristics

Example 1)

Obtain the pressure difference when 15 [L/min] of water runs through a solenoid valve with a **K_v** = 1.5 [m³/h]. As the flow rate when **K_v** = 1 is calculated as the formula: $Q_0 = 15 \times 1/1.5 = 10$ [L/min], read off ΔP when Q_0 is 10 [L/min] in Graph (2). The reading is 0.036 [MPa].

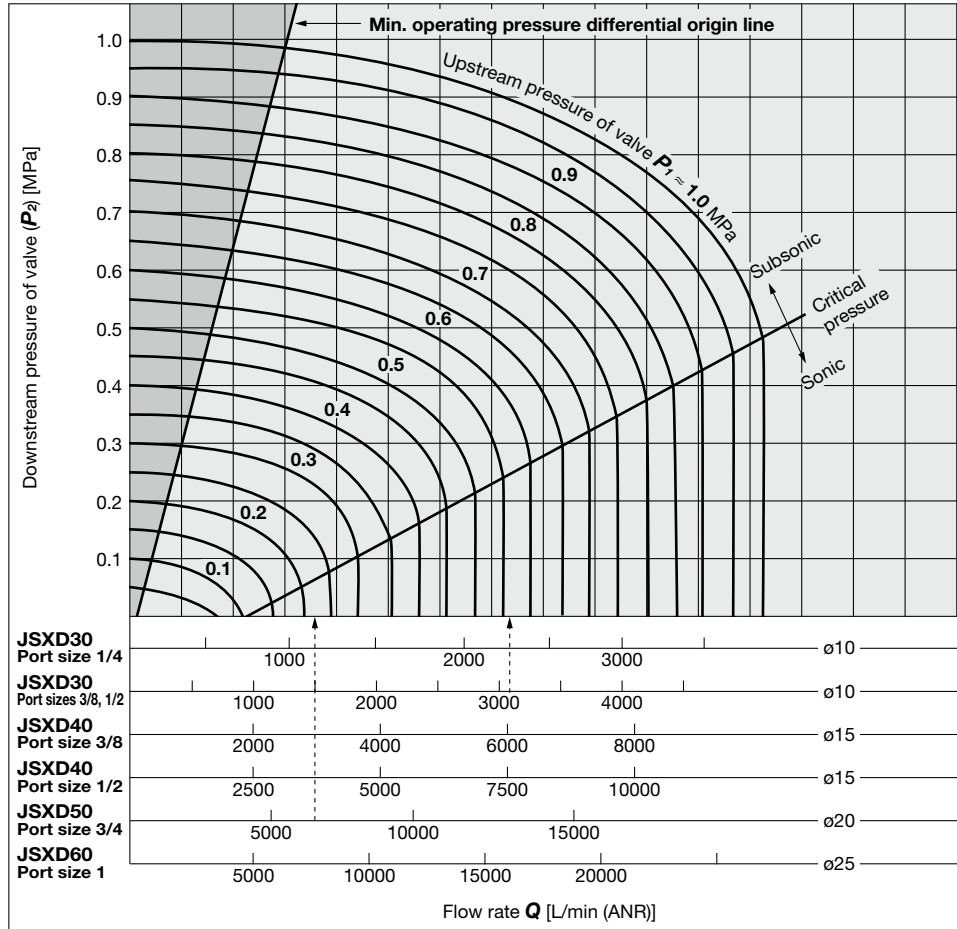
Example 2)

Obtain the saturated steam flow rate when $P_1 = 0.8$ [MPa] and $\Delta P = 0.008$ [MPa] with a solenoid valve with a **K_v** = 0.05 [m³/h]. Read off Q_0 when P_1 is 0.8 and ΔP is 0.008 in Graph (2), the reading is 20 [kg/h]. Therefore, the flow rate is calculated as the formula: $Q = 0.05/1 \times 20 = 1$ [kg/h].

Flow Rate Characteristics

* Use this graph as a guide. In the case of obtaining an accurate flow rate, refer to pages 93 to 97.

For Air (Orifice diameter: $\phi 10$ mm, $\phi 15$ mm, $\phi 20$ mm, $\phi 25$ mm)



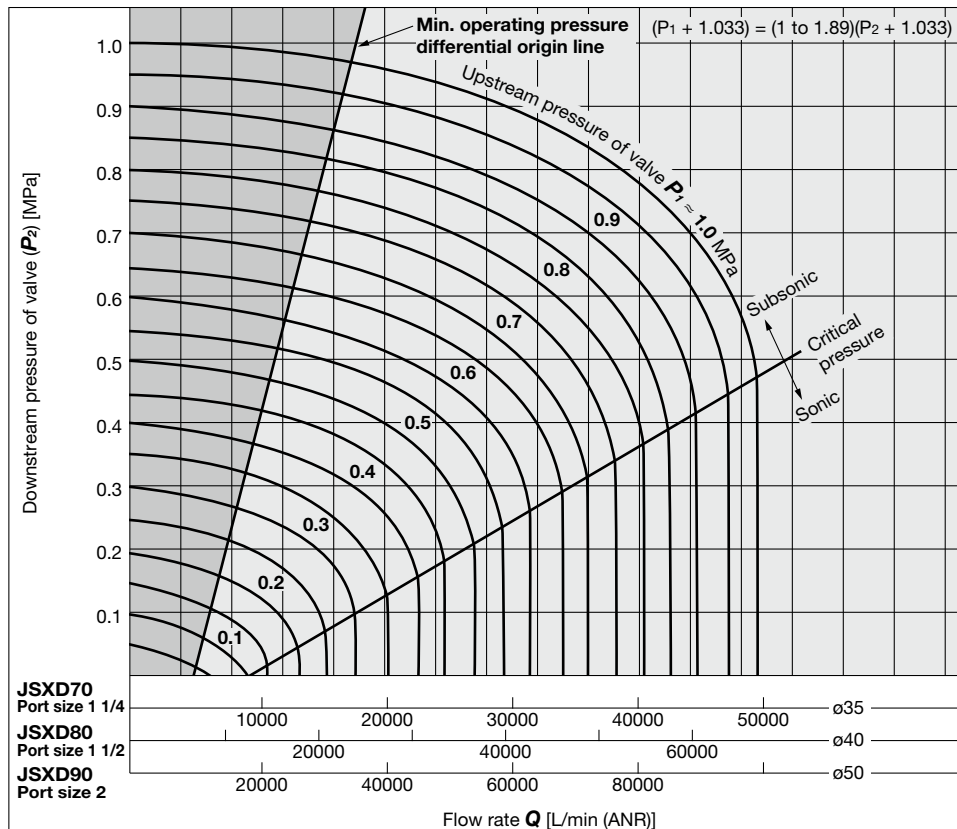
How to read the graph

The sonic range pressure to generate a flow rate of 6000 L/min (ANR) is as follows.
For a $\phi 15$ orifice (JSXD40/Port size 3/8), $P_1 \approx 0.57$ MPa,
for a $\phi 20$ orifice (JSXD50/Port size 3/4), $P_1 \approx 0.22$ MPa

Warning

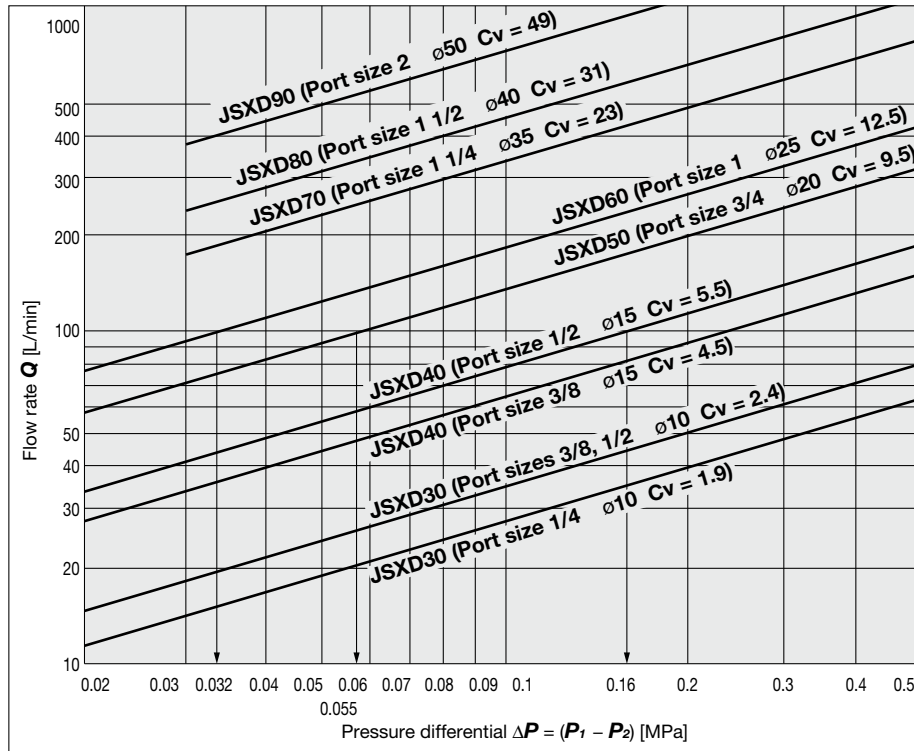
In the area located left to the min. operating pressure differential origin line in the flow rate characteristics table, the min. operating pressure is not generated. Do not use the product in this area as this may cause operation failure (valve opening failure, valve closing failure) or damage of the valve. Select valves with suitable size.

For Air (Orifice diameter: $\phi 35$ mm, $\phi 40$ mm, $\phi 50$ mm)



JSXD Series

For Water



How to read the graph

The pressure differential to generate a flow rate of 100 L/min water is as follows.

For a $\phi 15$ orifice (JSXD40/Port size 1/2), $\Delta P \approx 0.16$ MPa,
 for a $\phi 20$ orifice (JSXD50), $\Delta P \approx 0.055$ MPa,
 for a $\phi 25$ orifice (JSXD60), $\Delta P \approx 0.032$ MPa



JSX/JSX□ Series

Specific Product Precautions 1

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the "Handling Precautions for SMC Products" and the "Operation Manual" on the SMC website: <https://www.smcworld.com>

Design

⚠ Warning

1. Confirm the specifications.

Give careful consideration to the operating conditions, such as the application, fluid, and environment, and use within the specified operating ranges. If the product is used beyond the specification range, this may cause the product to break or malfunction. We do not guarantee against any damage if the product is used outside of the specification range.

2. Cannot be used as an emergency shutoff valve, etc.

This product is not designed for use as an emergency shutoff valve. If the valve is used in this type of system, other reliable safety assurance measures should also be adopted.

3. Cannot be used for pressure (including vacuum) holding

This product cannot be used to hold the pressure (including vacuum) inside of a pressure vessel because valve air leakage is unavoidable.

4. Closed liquid circuit

In a closed circuit, when liquid is static, the pressure could rise due to temperature fluctuations. This pressure rise could cause either a malfunction or damage to components such as valves. To prevent this, install a relief valve in the system.

5. Actuator driving

When an actuator, such as a cylinder, is to be driven using a valve, take appropriate measures to prevent potential danger caused by actuator operation.

6. Extended periods of continuous energization

The solenoid coil will generate heat when continuously energized. Avoid using in a tightly shut container. Install the valve in a well-ventilated area. Furthermore, do not touch it while it is being energized or right after it has been energized.

7. Water hammer

When an impact, such as water hammer, etc., caused by rapid pressure fluctuation is applied, the valve may be damaged. Install water hammer relief equipment (an accumulator, etc.) or use an SMC water hammer relief valve (VXR series). Please contact SMC for details.

8. Back pressure

If there is a possibility that back pressure will be applied, take countermeasures by installing a check valve, etc., on the downstream side.

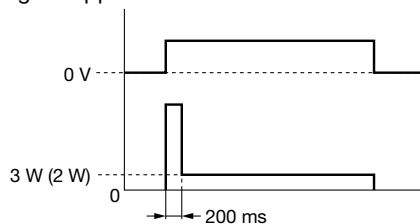
9. Do not disassemble the product or replacement parts or make any modifications to either of them, including additional machining.

Doing so may lead to human injury and/or an accident.

10. High flow/ Power saving type

Power consumption is reduced compared with the standard model by reducing the wattage required to hold the valve in an energized state.

Effective after being energized for more than 200 ms when the voltage is applied



* The value in () is for the JSX10U.

The OFF time should be at least 2 s.

If the OFF time is less than 2 s, the coil may generate an abnormal amount of heat, resulting in damage, depending on the length of ON time.

Do not use in an environment subject to constant vibration and/or impact.

The valve may close when held in an energized state.

Design

⚠ Caution

1. Power saving circuit

The power saving circuit (PWM control) built into the product reduces power consumption via high-speed switching operation with the PWM control circuit after the rated voltage has been applied for approx. 200 ms when energized. Please note that the effect of this PWM control can cause the following problems depending on the type of switch and drive circuit used.

1. When a mechanical relay, etc., is used in the drive circuit, the product may not turn ON normally if chattering occurs within approx. 200 ms of the start of energization.
2. When a filter or another device is installed between the power supply and the product to achieve noise reduction, the current may be reduced due to filtering, which may prevent the product from turning ON normally.
3. When an SSR (solid state relay) with a built-in photo coupler is used in the drive circuit, the photo coupler may not turn OFF, preventing the product from switching OFF (it will remain ON).

Operating Environment

⚠ Warning

Do not use the product in locations such as those described below.

1. Locations with atmospheres in which water vapor is present or locations in which corrosive fluids (chemicals), sea water, or water may come into contact with the product

Implement appropriate protective measures if water will be applied to the product for long periods of time, even for products which have IP65 or IP67 enclosures. Such water may enter through microscopic gaps in the product's external surfaces, resulting in fire damage or short-circuiting of the solenoid valve coils. If installing the product in close proximity to equipment such as machine tools, processing machines, etc., which use large amounts of liquids or oils, be sure to confirm that liquid dispersal or spatter from the peripheral equipment does not come into contact with the product.

2. Locations with explosive atmospheres

3. Locations subject to vibration or impact

4. Locations where radiated heat will be received from nearby heat sources

5. Locations that are outdoors (Excludes outdoor specification valves)

Although using an indoor specification product outdoors voids its product warranty, if outdoor use proves unavoidable, be sure to implement the protective measures mentioned below.

- 1) Install a protective cover, etc., to protect the product from direct sunlight.
- 2) Encase the product in an enclosure to protect it from rain and wind.
 - * If only a roof-type cover is provided for the product, it will not be sufficiently protected from side winds or rain splashing up from the ground, which will result in water adhering to and entering the product. In addition, when the product is encased in an enclosure, be sure to implement proper ventilation measures to prevent overheating due to long-term energizing of the product.
- 3) Be sure to confirm that the location is not one in which condensation is easily generated.
 - * If the product is used in an environment with large temperature fluctuations, etc., condensation may be generated, and water may adhere to the external surface of the product. Be sure to implement protective measures against condensation, such as ambient temperature control, in such locations where condensation is easily generated.

6. Locations where freezing may occur within piping lines [When the fluid is liquid]

If the product is to be used in cold regions or during winter, be sure to implement measures to prevent the freezing of fluids. If the fluid is likely to freeze, implement measures such as draining the water in the piping when the equipment is OFF or installing a heater or insulation in the piping.

If warming the solenoid valve, be sure to avoid the coil portion as warming it will result in poor heat dissipation.

[When the fluid is air]

With high flow rates, drain may be generated due to adiabatic expansion, resulting in freezing.

Be sure to periodically drain the product or conduct drain removal using an air dryer.

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Specific Product Precautions 2

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the "Handling Precautions for SMC Products" and the "Operation Manual" on the SMC website: <https://www.smcworld.com>

Fluid

⚠ Warning

1. Fluid selection

- 1) Compatibility between the components and fluids should be checked in the application before use.
- 2) Since the compatibility of the fluid used may vary depending on its type, additives, concentration, temperature, etc., give sufficient consideration when selecting the material. Please contact SMC if anything is unclear.
- 3) Use a fluid with a kinematic viscosity of 50 mm²/s or less.

2. Do not use the product with the fluids shown below.

- 1) Fluids that are harmful to humans
- 2) Combustion-supporting or flammable fluids
- 3) Corrosive gas
- 4) Sea water, Saline solution

3. Take measures to prevent static electricity, since some fluids can cause static electricity.

4. Fluid temperature

Operate within the specified operating fluid temperature range.

5. Install a filter (strainer) to ensure clean fluids.

- 1) The use of a fluid that contains foreign matter can cause problems, such as malfunction and seal failure by promoting the wear of the valve seat and armature, by sticking to the sliding parts of the armature, etc. Install a filter (strainer) on the upstream side of the valve to remove foreign matter.
Air: 5 μm or less Water: 100 mesh or more
- 2) Replace or clean the filter (strainer) when the pressure drop reaches 0.1 MPa to prevent them from getting clogged.

Fluid Quality

⚠ Warning

1. Air

- 1) Do not use compressed air that contains chemicals, synthetic oils that include organic solvents, salt, corrosive gases, etc., as it can cause malfunction or damage.
- 2) Compressed air that contains excessive drainage may cause the malfunction of valves and other pneumatic equipment. Install an aftercooler or an air dryer on the inlet side of the valve as a countermeasure against drainage.
- 3) If excessive carbon powder is generated by the compressor, it may adhere to the inside of the valves and cause malfunction. Install a mist separator on the inlet side of the valve as a countermeasure to remove any carbon powder.
- 4) For compressed air quality, refer to the **Web Catalog**.
- 5) When operating fluid air with a dew point of -70°C or lower, the inside of the valve may wear and the product life will be shortened.

2. Water

- 1) Be aware that rust stains, chloride separation, etc., from the piping may cause malfunction, leakage, or, in worse case scenarios, damage due to corrosion. Also, such damage may result in the spraying of fluids or scattering of parts. Please be sure to have protective measures in place in case such incidents should occur.
- 2) In the case that water contains substances such as calcium and magnesium, which generate hard scale and sludge, install water softening equipment and a filter (strainer) directly upstream from the valve to remove these substances, as this scale and sludge can cause the valve to malfunction.
- 3) The water pressure of tap water is usually 0.4 MPa or less, but the pressure can sometimes increase to 1.0 MPa in tall buildings. Therefore, pay attention to the max. operating pressure differential.

Fluid Quality

⚠ Warning

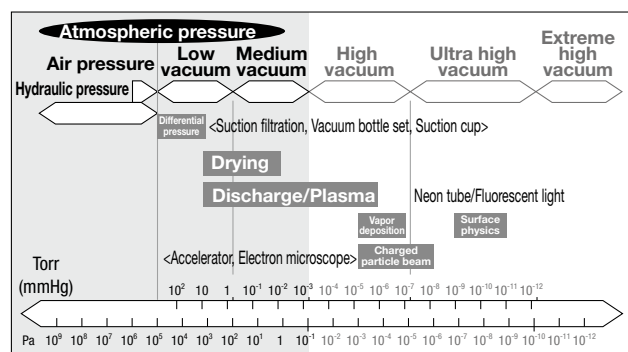
3. Oil

Generally, FKM is used as seal material, as it is resistant to oil. The resistance of the seal material may deteriorate depending on the type of oil, manufacturer, or additives. Check the resistance before use.

The kinematic viscosity must not exceed 50 mm²/s.

4. Vacuum

Please be aware that there is a range of pressure that can be used.



Vacuum piping direction: if the system uses a vacuum pump, we ask that you install the vacuum pump on the secondary side.

Also, install a filter on the primary side, and be careful that no foreign object is picked up.

Please replace the valve after operating the device approximately 300,000 times.

5. Steam

The use of a steam that contains foreign matter can cause problems, such as malfunction and seal failure, by promoting the wear of the valve seat and armature, and by sticking to the sliding parts of the armature, etc. Install a suitable filter (strainer) immediately upstream from the valve.

As per standard, the mesh count for the strainer should be 100 mesh. However, the size and shape of the foreign matter that occur depends on the operating environment. Check the fluid status and choose an appropriate mesh count.

The supply water to a boiler includes materials that create a hard sediment or sludge, such as calcium and magnesium. Sediment and sludge from steam can cause the valve to not operate properly. Install a water softening device which removes these materials.

Do not use operation steam which contains chemicals, synthetic oils that contain organic solvents, salts, corrosive gases, etc., as these can cause damage or deterioration.

The seal material (special FKM) used for wetted parts of the product can withstand steam in standard conditions.

However, the resistance of the sealing material can deteriorate depending on the types of additives such as boiler compounds and water conditioners within the boiler steam. Please only utilize the product after determining the sealing material resistance within the actual usage conditions.



JSX/JSX□ Series Specific Product Precautions 3

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Mounting

⚠ Warning

1. Ensure sufficient space for maintenance and inspection.
2. When mounting the product, avoid sources of vibration, or adjust the arm from the body to the min. length so that resonance will not occur.
3. Do not install the product near a heat source and install it in locations where the product is not affected by radiant heat.

4. Do not apply external force to the coil section.

When the product is installed, apply a wrench to the outside of the piping connection while paying attention that it will not come into contact with the coil.

5. Do not warm the coil section with a heat insulator, etc.

When insulation is used as a countermeasure against freezing, the insulation should be limited to the piping and body only. Do not insulate the coil. This can cause the coil to burn out.

6. If air leakage increases or equipment does not operate properly, stop operation.

After installation or during maintenance, check that the product is correctly mounted with appropriate functional and leakage inspections by supplying compressed air and power supplies. Do not use the product when the equipment does not operate correctly.

7. Do not touch the valve while it is being energized or right after it has been energized.

Valves will reach high temperatures after operation. Use caution, as there is a danger of being burnt if a valve is touched directly.

⚠ Caution

1. Painting and coating

Warnings or specifications printed or labeled on the product should not be erased, removed, or covered up.

How to Assemble Brackets

⚠ Caution

1. JSX series

Body material: Stainless steel, Brass, Aluminum

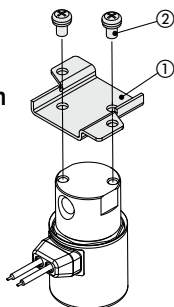
How to assemble

- 1) Mount the bracket ① to the bottom of the valve using the mounting screws ②.

Tightening torque

JSX10: 0.6 N·m ±5%

JSX20/30: 1.5 N·m ±5%



Bracket Assembly Part Nos. (With mounting screws)

Size	Body material	Port size	Thread type	Bracket assembly part no.	Weight [g]	Bracket material
10	Brass, Stainless steel	1/8	Rc NPT G	JSX021-12A-3	10	Stainless steel
20	Stainless steel	1/8		JSX022-12A-3	30	
30	Brass	1/8, 1/4, 3/8		JSX20-12A-4	35	
30	Stainless steel*1	1/8, 1/4, 3/8		VX021N-12A	20	
20	Aluminum	1/8, 1/4, 3/8		VX022N-12A	30	

*1 Only N.O. specification is available.

How to Assemble Brackets

⚠ Caution

2. JSX series

Body material: Stainless steel

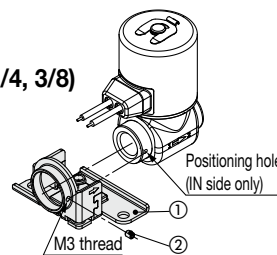
(N.C. specification, Port size: 1/4, 3/8)

How to assemble

- 1) Insert the bracket ① into the IN port side of the valve.

- 2) Secure it with the hexagon socket head set screw ②.

Tightening torque: 0.4 N·m ±5%



Caution regarding assembly

- 1) Pay attention to the bracket insertion direction.

There is only a positioning hole on the IN port side. Therefore, the bracket cannot be mounted to the OUT port side.

- 2) The bracket should be mounted after connecting the fitting. (Refer to the “Piping” section in the “Specific Product Precautions.”)

* The bracket is shipped together with the product.

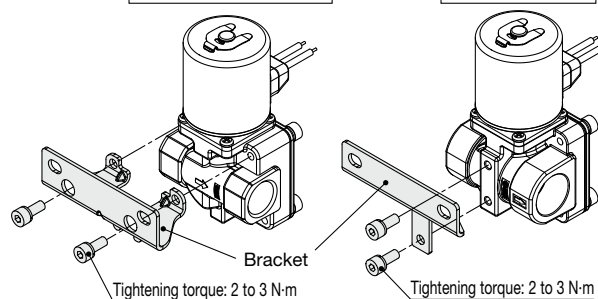
Bracket Assembly Part Nos. (With set screw)

Size	Port size	Thread type	Bracket assembly part no. (With set screw)	Weight [g]	Material
20, 30	1/4	Rc, NPT, G	JSX022-12A-2-1	30	Stainless steel
	3/8	Rc, NPT	JSX022-12A-2-1		
		G	JSX022-12A-2-2		

3. JSXD30 series: How to assemble brackets

For port sizes 1/4, 3/8

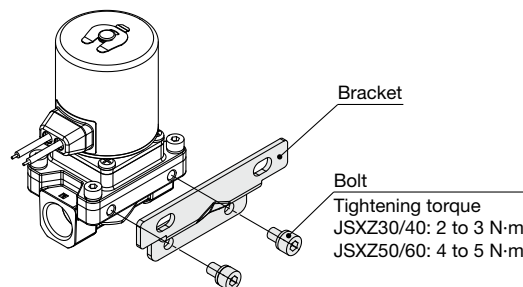
For port size 1/2



Size	Port size	Bracket assembly part no. (With screws)	Weight [g]
30	1/4, 3/8	VXD30S-14A-1	40
	1/2	VXD30S-14A-3	30

* For the JSXD30 series, the bracket is shipped together with the product.

4. JSXZ series: How to assemble brackets



* The bracket is shipped together with the product.

* For the JSXZ50/60, the mounting bolts and washers are separable, so be careful not to lose the washers.

Size	Port size	Bracket assembly part no. (With screws)	Weight [g]
30, 40	1/4, 3/8, 1/2	VXZ30S-14A-1	45
50, 60	3/4, 1	VXZ50S-14A-1	60

JSX

JSXD

JSXP

JSXR

JSXZ

JSXM

CE/UKCA-compliance Table

UL-compliance Table

Option

Replacement Parts

Glossary of Terms

Flow Rate Characteristics

Specific Product Precautions



JSX/JSX□ Series

Specific Product Precautions 4

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Piping

Warning

1. There may be cases in which the tubing detaches from the fitting and thrashes around uncontrollably due to tubing degradation or fitting breakage. To prevent this, fit the tubing with a protective cover or secure it in place.
2. If using tube piping, secure the product to a permanent fixture. Do not suspend it from the tubing.

Caution

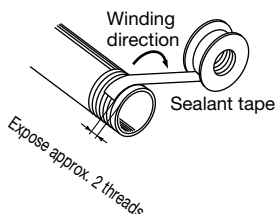
1. For handling One-touch fittings, refer to the “Fittings and Tubing Precautions” in the “Handling Precautions for SMC Products.”

2. Preparation before piping

Before piping is connected, it should be thoroughly blown out with air (flushing) or washed to remove chips, cutting oil, and other debris from inside the pipe. Install piping so that it does not apply pulling, pressing, bending, or other forces on the valve body.

3. Winding of sealant tape

When connecting pipes, fittings, etc., be sure that chips from the pipe threads and sealing material do not enter the valve. Furthermore, when sealant tape is used, leave 1.5 to 2 thread ridges exposed at the end of the threads.



4. Screw tightening torque for piping

When connecting piping to the valve, tighten with the proper tightening torque shown below.

Tightening Torque for Piping

Connection thread	Proper tightening torque [N·m]	Connection thread	Proper tightening torque [N·m]
1/8	3 to 5	1	36 to 38
1/4	8 to 12	1 1/4	40 to 42
3/8	15 to 20	1 1/2	48 to 50
1/2	20 to 25	2	48 to 50
3/4	28 to 30		

5. When using a fitting other than an SMC fitting

Follow the instructions given by the fitting manufacturer.

6. Avoid connecting ground lines to piping, as this may cause the electric corrosion of the system.

7. When connecting piping to a product, avoid mistakes regarding the supply port, etc.



If the tightening torque is applied to the fitting while the valve is secured to the bracket, the bracket might break.

Caution

8. Recommended piping conditions

When connecting piping to the One-touch fitting, use a pipe length with sufficient margin, in accordance with the piping conditions shown in Fig. 1. Also, when using a tying band, etc., to bind the piping together, make sure that external force does not come to bear on the fitting. (See Fig. 2.)

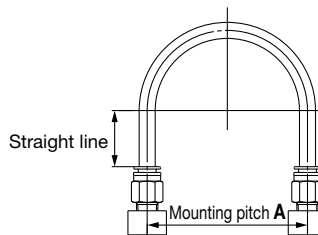


Fig. 1 Recommended piping

Unit: mm

Tubing size	Mounting pitch A			Straight line length
	Nylon tubing	Soft nylon tubing	Polyurethane tubing	
ø1/8"	44 or more	29 or more	25 or more	16 or more
ø6	84 or more	39 or more	39 or more	30 or more
ø1/4"	89 or more	56 or more	57 or more	32 or more
ø8	112 or more	58 or more	52 or more	40 or more
ø10	140 or more	70 or more	69 or more	50 or more
ø12	168 or more	82 or more	88 or more	60 or more

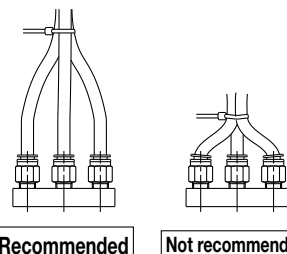
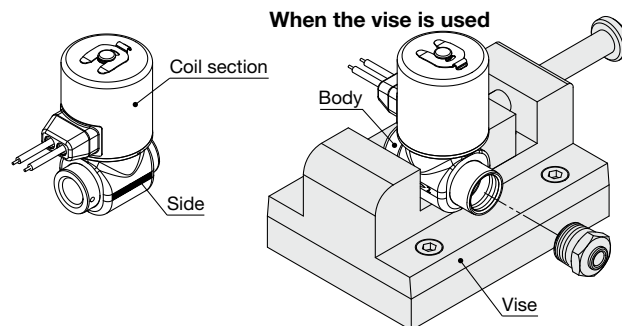


Fig. 2 When using a tying band to bind the piping together

9. When connecting a fitting to the valve, clamp the side of the body with a vise.



10. When using a stainless steel bracket (N.C. specification, Port size: 1/4, 3/8), connect the fitting in accordance with the following procedure.

- Step 1) Connect the fittings to both the IN and OUT sides of the valve.
- Step 2) Insert the IN side port of the valve into the bracket hole.
- Step 3) Secure the valve to the bracket with the hexagon socket set screw.



JSX/JSX□ Series

Specific Product Precautions 5

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Wiring

⚠ Warning

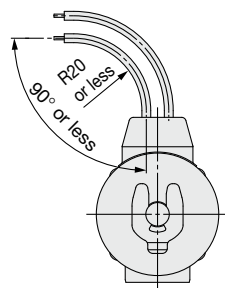
The solenoid valve is an electrical product. For safety, install an appropriate fuse and circuit breaker before use. When using multiple solenoid valves, it is not sufficient to merely install one fuse. For protecting the equipment more safely, select an appropriate fuse to each circuit of the solenoid valve.

⚠ Caution

1. As a rule, use electrical wire with a cross sectional area of 0.5 to 1.25 mm² for wiring.

2. External force applied to the lead wire

If an excessive force is applied to the lead wire, this may cause faulty wiring. Take appropriate measures so that a force of 10 N or more is not applied to the lead wire. Do not bend the lead wires beyond 90° with a radius of less than 20 mm or damage may occur.



3. Use electrical circuits which do not generate chattering in their contacts.

4. Use voltage which is within $\pm 10\%$ of the rated voltage. In cases with a DC power supply where importance is placed on responsiveness, stay within $\pm 5\%$ of the rated value. The voltage drop is the value in the lead wire section connecting the coil.

5. When a surge from the solenoid affects the electrical circuitry, install a surge voltage suppressor, etc., in parallel with the solenoid. Or, use the product with a surge voltage suppressor.

Residual voltage of the surge voltage suppressor

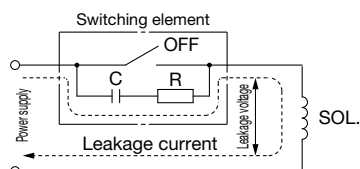
DC specification: Approx. 60 V

AC specification: Approx. 1 V

High flow/ Power saving type: Approx. 1 V

6. Leakage voltage

When the solenoid valve is operated using the controller, etc., the leakage voltage should be the product allowable leakage voltage or less. Particularly when using a resistor in parallel with a switching element and using a C-R element to protect the switching element, take note that leakage current will flow through the resistor, C-R element, etc., creating a possible danger that the valve may not turn off.



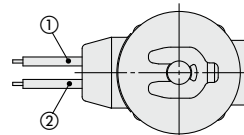
AC coil: 5% or less of the rated voltage
DC coil: 2% or less of the rated voltage

Electrical Connections

⚠ Caution

1. Grommet

Lead wire: AWG20 Insulator O.D.: 2.6 mm

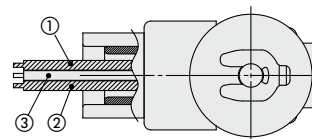


Rated voltage	Lead wire color	
	①	②
DC	Black	Red
DC (High flow/ Power saving type)*1	Black (–)	Red (+)
100 VAC	Blue	Blue
200 VAC	Red	Red
Other AC	Gray	Gray

*1 Only the high flow/power saving type has polarity.

2. Conduit

Lead wire: AWG18 Insulator O.D.: 2.8 mm



Rated voltage	Lead wire color		
	①	②	③
DC	Black	Red	Green/Yellow
DC (High flow/ Power saving type)*1	Black (–)	Red (+)	Green/Yellow
DC	Black	Red	Green/Yellow
100 VAC	Blue	Blue	Green/Yellow
200 VAC	Red	Red	Green/Yellow
Other AC	Gray	Gray	Green/Yellow

*1 Only the high flow/power saving type has polarity.

* ③: Ground wire

3. DIN terminal

Disassembly

- After loosening the binding head screw with flange, then if the housing is pulled in the direction of the arrow, the connector will be removed from the solenoid valve.
- Pull out the binding head screw with flange from the housing.
- There is a cutout on the bottom of the terminal block. Insert a small flat head screwdriver, etc., into this cutout, and remove the terminal block from the housing. (Refer to the figure on the next page.)
- Remove the gland nut, and pull out the washer and the rubber seal.

Wiring

- Pass the cable through the gland nut, washer, and rubber seal in this order, and insert these parts into the housing.
- Loosen the binding head screw of the terminal block, then insert the core wire or the crimped terminal of the lead wire into the terminal, and securely fix it with the binding head screw. The binding head screw of the terminal block is M3.

*1 Tighten the screw to a torque of between 0.5 and 0.6 N·m.

*2 Cable O.D.: $\phi 6$ to $\phi 12$ mm

*3 For an outside cable diameter of $\phi 9$ to $\phi 12$ mm, remove the internal parts of the rubber seal before use.



JSX/JSX□ Series

Specific Product Precautions 6

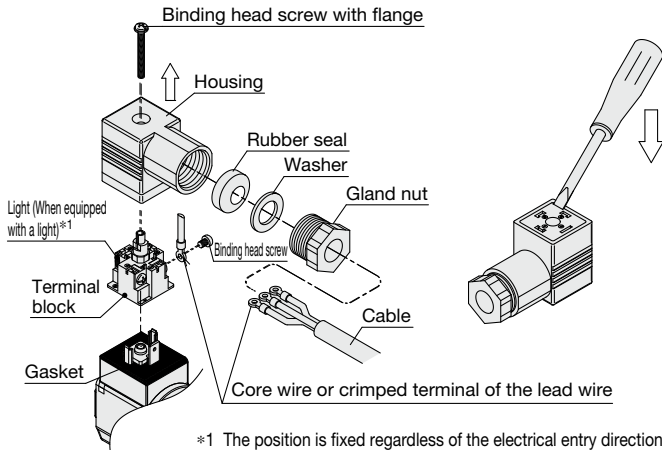
Be sure to read this before handling the products. Refer to the back cover for safety instructions.
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Electrical Connections

⚠ Caution

Assembly

1. Pass the cable through the gland nut, washer, rubber seal, and the housing in this order, and connect to the terminal block. Then, set the terminal block inside the housing. (Push in the terminal block until it snaps into position.)
2. Insert the rubber seal and the washer in this order into the cable entry of the housing, and then tighten the gland nut securely.
3. Insert the gasket between the bottom part of the terminal block and the plug attached to the equipment, and then insert the binding head screw with flange from the top of the housing, and tighten it.
 - *1 Tighten the screw to a torque of between 0.5 and 0.6 N·m.
 - *2 The orientation of the connector can be changed in steps of 90° by changing the method of assembling the housing and the terminal block.



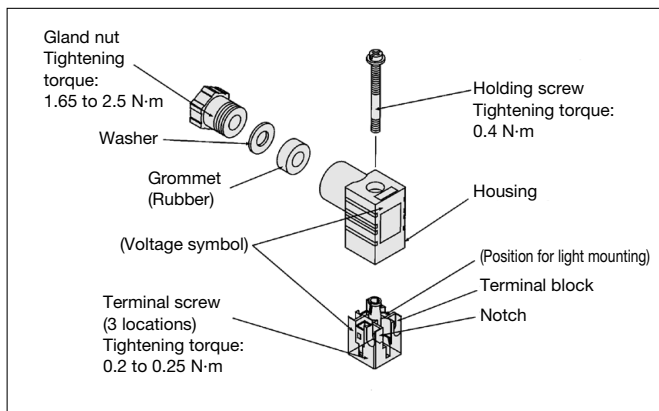
*1 The position is fixed regardless of the electrical entry direction.

For the JSX10

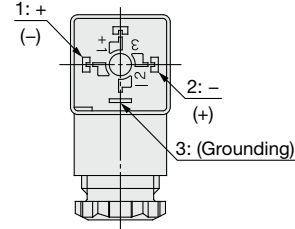
Compatible cable

Cord O.D.: $\phi 3.5$ to $\phi 7$

(Reference) 0.5 mm², 2-core or 3-core, equivalent to JIS C 3306



Internal connections are as shown below. Make connections to the power supply accordingly.



Terminal no.	1	2
DIN terminal*1	- (+)	+ (-)
DIN terminal (High flow/ Power saving type)*2	-	+

*1 There is no polarity.

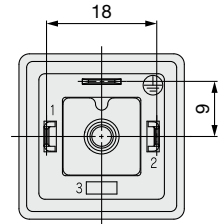
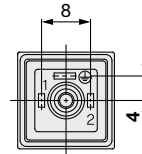
*2 The high flow/ power saving type has polarity.

* No.3: Ground wire

DIN (EN 175301-803) Terminal

This DIN terminal corresponds to the Form C DIN connector with an 8 mm terminal pitch.

This DIN terminal corresponds to the Form A DIN connector with an 18 mm terminal pitch.



Size: 10

Applicable cable O.D.: $\phi 3.5$ to $\phi 7$

Size: 20, 30

Applicable cable O.D.: $\phi 6$ to $\phi 12$

4. M12 connector

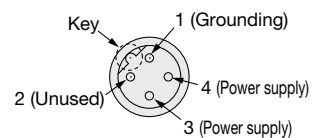
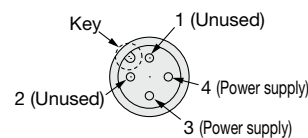
1. The IP67 (enclosure) rating of the valve can be obtained by using a cable with a female connector of IP67 specification. Please note that this product cannot be used in water.
2. Do not use a tool to mount the connector as this may cause damage. Only tighten it by hand. (0.39 to 0.49 N·m)
3. Avoid repeatedly bending or stretching the cable and applying heavy objects or force to it.
4. Do not pull the connector or cable unnecessarily.
5. Do not bend the cable at the root of the connector when installed.

■ Coding and pin arrangement of the M12 connector on the valve side

The shape (coding) and pin arrangement of the M12 connector are as follows.

DC specification: A-coded, 4-pin

AC specification: B-coded, 4-pin



Terminal no.	3	4
Pin terminal*1	+ (-)	- (+)
Pin terminal (High flow/ Power saving type)*2	-	+

*1 There is no polarity.

*2 The high flow/ power saving type has polarity.

* AC specification: No. 1 is the ground wire.

DC specification (including the high flow/power saving type): There is no ground wire.



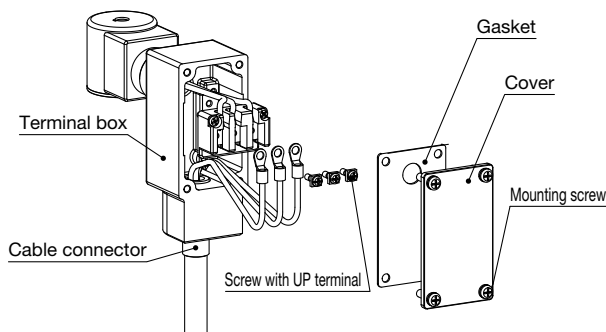
JSX/JSX□ Series Specific Product Precautions 7

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Electrical Connections

When using the cable with a female connector, make sure that the coding is correct. When installing the cable, be sure to align the key on the cable side connector (female side) with the key on the valve side connector (male side). Be careful not to squeeze it in the wrong direction as pin damage, etc., may result.

5. Conduit terminal

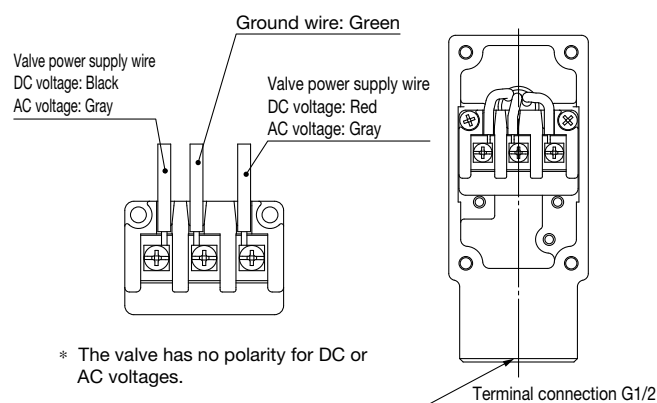


Disassembly

1. Loosen the mounting screws, and remove the cover and gasket from the terminal box.

Wiring

1. Thread the cable through the piping hole on the terminal box.
2. Loosen the screws with an UP terminal on the terminal block inside the box. Then, insert the core wires or crimped terminals of the lead wire into the terminal block, and secure them with the screws with an UP terminal.
 - * Tighten the screws with an UP terminal with a tightening torque of 0.5 to 0.6 N·m.
 - * Applicable crimped terminal: JIS C 2805 R1.25-3
3. Make connections according to the figure below.
 - Properly seal the terminal connection (G1/2) with the cable connector, etc.



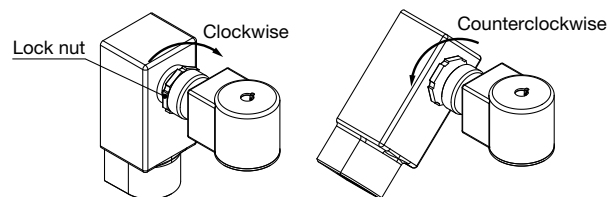
Assembly

1. Place the gasket and cover on the terminal box, and tighten the mounting screws.
 - * Tighten the mounting screws with a tightening torque of 0.7 to 0.8 N·m.

Changing the Conduit Terminal Direction

When changing the orientation of the conduit terminal, carry out the following procedure.

1. Loosen the lock nut by rotating it clockwise.
2. Rotate the conduit terminal counterclockwise to the desired position. (Max. 360°/1 full rotation)
 - * At this time, be sure not to rotate the terminal clockwise, or it may lose its waterproof qualities.
3. Rotate the lock nut counterclockwise to secure the terminal in place.

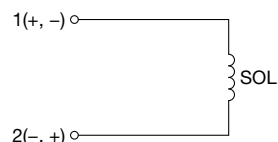


Electrical Circuits

⚠ Caution

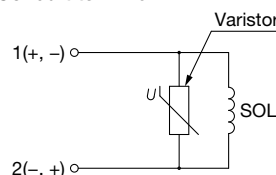
1. DC circuit

● Grommet



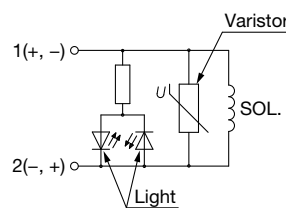
Without electrical option

● Grommet, Conduit, DIN terminal, Conduit terminal



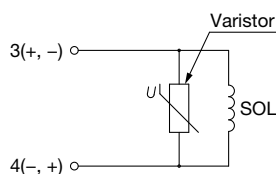
With surge voltage suppressor

● DIN terminal



With light/surge voltage suppressor

● M12 connector

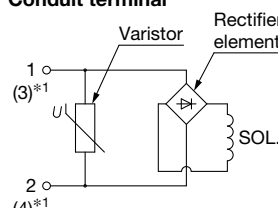


With surge voltage suppressor

2. AC circuit

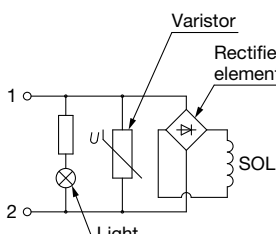
The standard product is equipped with a surge voltage suppressor.

● Grommet, Conduit, DIN terminal, M12 connector, Conduit terminal



*1 For M12 connector
Without electrical option

● DIN terminal



With light



JSX/JSX□ Series

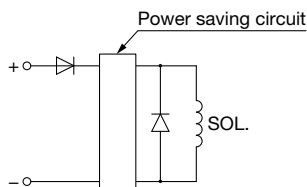
Specific Product Precautions 8

Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Electrical Circuits

⚠ Caution

3. High flow/ Power saving type



· Lead Wire and Terminal Nos.

Polarity	+	-
Grommet	2 (Red)	1 (Black)
Conduit	2 (Red)	1 (Black)
DIN terminal	2	1
M12 connector	4	3

* Be sure to confirm the polarity when connecting.

Maintenance

⚠ Warning

1. Removal of product

- 1) Shut off the fluid supply and release the fluid pressure in the system.
- 2) Shut off the power supply.
- 3) Confirm that the valve temperature has dropped sufficiently before removing the product.

2. Replace or clean filters (strainers) periodically.

- 1) Replace filters after one year of use, or earlier if the pressure drop reaches 0.1 MPa.
- 2) Clean strainers when the pressure drop reaches 0.1 MPa.

3. Exhaust the drainage from air filters periodically.

If condensation in the drain bowl is not emptied on a regular basis, the bowl will overflow and allow the condensation to enter the compressed air lines. This causes the malfunction of pneumatic equipment. If the drain bowl is difficult to check and remove, the installation of a drain bowl with an auto drain option is recommended.

4. Low frequency operation

Switch valves at least once every 30 days to prevent malfunction. Also, in order to use them under the optimum state, conduct a regular inspection biannually.

5. Storage

In the case of long-term storage after use, thoroughly remove all moisture and store it in a location where the product is not exposed to sunlight and higher humidity to prevent rust and deterioration of rubber materials, etc.

6. Perform a maintenance and inspection periodically.

Confirm that the product is mounted correctly by conducting suitable function and leakage tests periodically. If air leakage increases or equipment does not operate properly, stop operation.

Return of Product

⚠ Warning

If the product to be returned is contaminated or is possibly contaminated with substances that are harmful to humans, for safety reasons, please contact SMC beforehand and then employ a specialist cleaning company to decontaminate the product. After the decontamination prescribed above has been carried out, submit a Product Return Request Sheet or the Detoxification/Decontamination Certificate to SMC and await SMC's approval and further instructions before attempting to return the item.

Please refer to the International Chemical Safety Cards (ICSC) for a list of harmful substances.

If you have any further questions, please don't hesitate to contact your SMC sales representative.

JSXD and JSXZ Precautions

⚠ Warning

1. For pilot operated 2-port solenoid valves, when the valve is closed, sudden pressure resulting from the startup of the fluid supply source (pump, compressor, etc.) may cause the valve to open momentarily and leakage to occur, so please exercise caution.

2. If the product is used in the conditions in which rapid decrease in the inlet pressure of the valve and rapid increase in the outlet pressure of the valve are repeated, excessive stress will be applied to the diaphragm, which causes the diaphragm to be damaged and dropped, leading to the operation failure of the valve. Check the operating conditions before use.

3. Min. operating pressure differential (JSXD)

Be aware that even if the pressure difference is above the min. operating pressure differential when the valve is closed, the pressure difference may fall below the min. operating pressure differential when the valve opens, depending on the capacity of the supply source (pumps, compressors, etc.,) or the type of pipe restrictions (the piping is bent continuously due to elbow or tee, or narrow tube nozzle is installed in the end). If the product is used below the min. operating pressure, the operation becomes unstable, which might cause valve opening or closing failure, or oscillation, leading to failure due to insufficient pressure differential. Select an appropriate valve size with reference to the flow rate characteristics and flow rate characteristics table on pages 93 to 99.



JSX/JSX□ Series

Specific Product Precautions 9

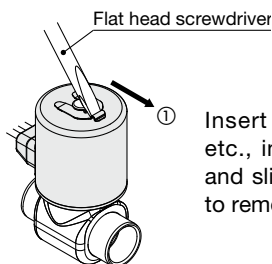
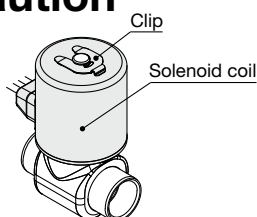
Be sure to read this before handling the products. Refer to the back cover for safety instructions. For 2-port solenoid valve for fluid control precautions, refer to the “Handling Precautions for SMC Products” and the “Operation Manual” on the SMC website: <https://www.smcworld.com>

Replacing the Solenoid Coils

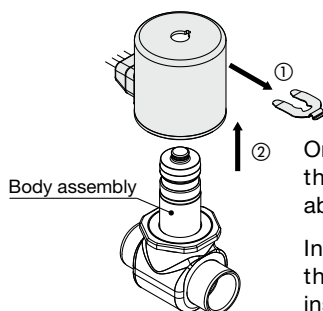
⚠ Warning

1. When replacing the solenoid coil, turn off the power supply.
2. Be careful for possible high-temperature of the solenoid coil due to the fluid temperature and operating conditions.

⚠ Caution

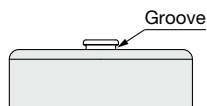


Insert a flat head screwdriver, etc., into the groove in the clip and slide it in the direction of ① to remove it.

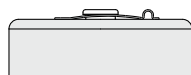


Once the clip has been removed, the coil can be removed from above (in the direction of ②).

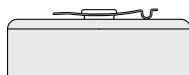
Insert the replacement coil into the body assembly, and then insert the clip by aligning it with the groove in the top of the body assembly.



Be sure to confirm the clip direction (back and front) as well as the insertion state.

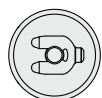


OK



Not OK

Clip direction



OK



Not OK

Inserted condition

* When inserting the coil, be sure to push it in until the groove in the body assembly is visible.

Safety Instructions

These safety instructions are intended to prevent hazardous situations and/or equipment damage. These instructions indicate the level of potential hazard with the labels of “**Caution**,” “**Warning**” or “**Danger**.” They are all important notes for safety and must be followed in addition to International Standards (ISO/IEC)*1), and other safety regulations.

⚠ Danger : **Danger** indicates a hazard with a high level of risk which, if not avoided, will result in death or serious injury.

⚠ Warning: **Warning** indicates a hazard with a medium level of risk which, if not avoided, could result in death or serious injury.

⚠ Caution: **Caution** indicates a hazard with a low level of risk which, if not avoided, could result in minor or moderate injury.

*1) ISO 4414: Pneumatic fluid power - General rules and safety requirements for systems and their components
ISO 4413: Hydraulic fluid power - General rules and safety requirements for systems and their components
IEC 60204-1: Safety of machinery - Electrical equipment of machines - Part 1: General requirements
ISO 10218-1: Robots and robotic devices - Safety requirements for industrial robots - Part 1: Robots etc.

Warning

1. The compatibility of the product is the responsibility of the person who designs the equipment or decides its specifications.

Since the product specified here is used under various operating conditions, its compatibility with specific equipment must be decided by the person who designs the equipment or decides its specifications based on necessary analysis and test results. The expected performance and safety assurance of the equipment will be the responsibility of the person who has determined its compatibility with the product. This person should also continuously review all specifications of the product referring to its latest catalog information, with a view to giving due consideration to any possibility of equipment failure when configuring the equipment.

2. Only personnel with appropriate training should operate machinery and equipment.

The product specified here may become unsafe if handled incorrectly. The assembly, operation and maintenance of machines or equipment including our products must be performed by an operator who is appropriately trained and experienced.

3. Do not service or attempt to remove product and machinery/equipment until safety is confirmed.

1. The inspection and maintenance of machinery/equipment should only be performed after measures to prevent falling or runaway of the driven objects have been confirmed.
2. When the product is to be removed, confirm that the safety measures as mentioned above are implemented and the power from any appropriate source is cut, and read and understand the specific product precautions of all relevant products carefully.
3. Before machinery/equipment is restarted, take measures to prevent unexpected operation and malfunction.

4. SMC products cannot be used beyond their specifications. They are not developed, designed, and manufactured to be used under the following conditions or environments. Use under such conditions or environments is not allowed.

1. Conditions and environments outside of the given specifications, or use outdoors or in a place exposed to direct sunlight.
2. Use for nuclear power, railways, aviation, space equipment, ships, vehicles, military application, equipment affecting human life, body, and property, combustion equipment, entertainment equipment, emergency shut-off circuits, press clutches, brake circuits, safety equipment, etc., and use for applications that do not conform to standard specifications such as catalogs and operation manuals.
3. Use for interlock circuits, except for use with double interlock such as installing a mechanical protection function in case of failure. Please periodically inspect the product to confirm that the product is operating properly.

Caution

SMC develops, designs, and manufactures products to be used for automatic control equipment, and provides them for peaceful use in manufacturing industries.

Use in non-manufacturing industries is not allowed.

Products SMC manufactures and sells cannot be used for the purpose of transactions or certification specified in the Measurement Act of each country. The new Measurement Act prohibits use of any unit other than SI units in Japan.

Limited warranty and Disclaimer/ Compliance Requirements

The product used is subject to the following “Limited warranty and Disclaimer” and “Compliance Requirements”.

Read and accept them before using the product.

Limited warranty and Disclaimer

1. The warranty period of the product is 1 year in service or 1.5 years after the product is delivered, whichever is first.*2)
Also, the product may have specified durability, running distance or replacement parts. Please consult your nearest sales branch.
2. For any failure or damage reported within the warranty period which is clearly our responsibility, a replacement product or necessary parts will be provided.
This limited warranty applies only to our product independently, and not to any other damage incurred due to the failure of the product.
3. Prior to using SMC products, please read and understand the warranty terms and disclaimers noted in the specified catalog for the particular products.

*2) **Suction cups (Vacuum pads) are excluded from this 1 year warranty.**

A suction cup (vacuum pad) is a consumable part, so it is warranted for a year after it is delivered.

Also, even within the warranty period, the wear of a product due to the use of the suction cup (vacuum pad) or failure due to the deterioration of rubber material are not allowed by the limited warranty.

Compliance Requirements

1. The use of SMC products with production equipment for the manufacture of weapons of mass destruction (WMD) or any other weapon is strictly prohibited.
2. The exports of SMC products or technology from one country to another are governed by the relevant security laws and regulations of the countries involved in the transaction. Prior to the shipment of a SMC product to another country, assure that all local rules governing that export are known and followed.

Revision History

Edition B * JSXD and JSXM types have been added.
* Brass and aluminum body materials have been added.
* An M12 connector electrical entry option has been added.
* The number of pages has been increased from 24 to 56.

Edition C * JSX□□U and JSXZ types have been added.
* The number of pages has been increased from 56 to 72.

Edition D * Vacuum, steam, and high pressure types have been added to the JSX.
* An N.O. specification has been added to the JSXD.
* An improved weather-resistant specification has been added.
* The number of pages has been increased from 72 to 92.

Edition E * A JSXR type has been added.
* A CE/UKCA-compliance table has been added.
* The number of pages has been increased from 92 to 108.

Edition F * An N.O. specification has been added to the JSXZ.
* The number of pages has been increased from 108 to 112.

⚠ Safety Instructions Be sure to read the “Handling Precautions for SMC Products” (M-E03-3) and “Operation Manual” before use.